POLITICIZED SCIENTISTS: CREDIBILITY COST OF POLITICAL EXPRESSION ON TWITTER

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As social media becomes prominent within academia, we examine its reputational costs for academics. Analyzing Twitter posts from 98,000 scientists (2016–22), we uncover substantial political expression. Online experiments with 6,000 U.S. respondents and 135 journalists, rating synthetic academic profiles with different political affiliations, reveal that politically neutral scientists are seen as the most credible. Strikingly, political expressions result in monotonic penalties: Stronger posts reduce perceived credibility of scientists and their research and audience engagement more, particularly among oppositely aligned respondents. Two surveys with scientists highlight their awareness to penalties, their perceived benefits, and a consensus on limiting political expression outside their expertise.

KEYWORDS: Twitter, Scientists' Credibility, Polarization, Online Experiment.

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1. INTRODUCTION

In a "post-truth" era, trust in science is a cornerstone of informed decision-making and effective public policy (McIntyre, 2018, Angelucci and Prat, 2024, Bursztyn et al., 2023b, Arold, 2024, Ash et al., 2025, 2024). The COVID-19 pandemic highlighted this reality, illustrating how confidence in scientific expertise shapes public health responses. Similarly, climate skepticism hinders progress toward environmental goals.[1] Skepticism extends beyond these domains, affecting economic development, education, and societal trust more broadly.

Credibility is essential to science, yet public trust is at a relative low, particularly among U.S. conservatives.[2] A key driver of distrust is the perception of political bias among scientists (Altenmüller et al., 2024). Social media amplifies these perceptions by exposing scientists' views to audiences well beyond the academic community, with nearly half of engagement coming from non-academics (Mohammadi et al., 2018, Maleki and Holmberg, 2023).[3] Such bias may be consequential rather than merely perceived: researchers' policy preferences can shape research design and reported estimates even when analyzing identical data (Borjas and Breznau, 2024).

This paper proceeds in two parts. First, we use Large Language Models (LLMs) to measure the extent and nuance of scientists' political expression on social media. Second, we assess whether their online political engagement affects public perceptions, focusing on the

[1] Higher trust in science is linked to earlier adoption of preventive measures during COVID-19 (Algan et al., 2021, Allcott et al., 2020, Bartoš et al., 2022, Bowles et al., 2023, Eichengreen et al., 2021), whereas skepticism has driven reliance on unverified remedies, such as those documented in Argentina (Calónico et al., 2023, Albornoz et al., 2024). On climate change, Druckman and McGrath (2019) highlights how differing beliefs about credible evidence influence policy support.

[2] Nichols (2017) discusses hostility toward expertise in the U.S., while Lupia et al. (2024) documents declining trust in science among U.S. respondents. Factors include misinformation (West and Bergstrom, 2021, Roozenbeek et al., 2020), historical failings (Scharff et al., 2010), the reproducibility crisis (Hendriks et al., 2020), conspiracy theories (Rutjens and Većkalov, 2022, Douglas, 2021), science-related populism (Mede and Schäfer, 2020, Mede et al., 2021), and political ideology (Cologna et al., 2025), with U.S. conservatives reporting lower trust in scientists, stronger anti-science attitudes, and less confidence in their intentions and methods (Mede, 2022, Funk et al., 2020, Li and Qian, 2022, Azevedo and Jost, 2021).

[3] Appendix Section E further documents this pattern using our data, showing that most accounts interacting with U.S. academics on Twitter cannot be identified as academics.

reputational costs for scientists as public opinion makers (Chiang and Knight, 2011, Aina, 2023) shaping individual preferences and behaviors (Algan et al., 2021, Burnitt et al., 2024, Martinez-Bravo and Stegmann, 2022, Goldstein and Kolerman, 2025). Using several online experiments, we evaluate how revealing scientists' political views on social media influences perceptions of their credibility and contributes to audience polarization.

Our findings reveal evidence of both *ideological* polarization (divergent political positions among academics) and *affective* polarization (public aversion to scientists associated with opposing partisan groups) (Lelkes, 2016, Barberá, 2020). The central questions of our study are: *To what extent do scientists express polarized political opinions on social media? And what impact does this political expression have on their perceived credibility?*

Impact is increasingly crucial for academics, with social media playing a growing role in research dissemination and public engagement with science. On the one hand, studies show that social media engagement can benefit academics (Chan et al., 2023, Klar et al., 2020, Qiu et al., 2024, Boken et al., 2023), and Altmetric data reveal a sharp rise in the online presence of scientific research on social media between 2011 and 2020.[4] On the other hand, recent cross-country evidence shows that the public primarily encounters science on social media, frequently discusses it, and often takes outspoken positions—sometimes even participating in rallies related to scientific issues, particularly in the United States (Mede et al., 2025). These trends highlight the increasing interconnection between science and politics, as well as the visibility of scientific discourse in the public sphere. Our study extends this perspective by examining how scientists use social media not just to share research, but to engage in politically salient debates—with implications for credibility and public polarization.

First, we build on Garg and Fetzer (2025b) by analyzing the extent of online political discourse and issue-based disagreement among U.S. scientists (university-based researchers) compared to general social media users. Examining political slants on social media offers three key advantages: (i) capturing a large, cross-disciplinary sample of academics, (ii) providing a nuanced view of ideological positions beyond a left-right spectrum, and (iii) fo-

[4]While this trend partly reflects overall growth in social media use, our data suggest a specific surge in the visibility of scientific research. Altmetric tracks research impact across newspapers, blogs, social media, policy briefs, patents, and academic citations.

cusing on publicly visible content, avoiding the limitations of survey-based or publication-based classifications.

Our descriptive analysis reveals that 44% of 97,737 U.S. academics on Twitter expressed political opinions by making at least one non-neutral post on any of five politically salient topics—(i) abortion rights, (ii) climate action, (iii) immigration, (iv) income redistribution, and (v) racial equality—between 2016 and 2022, compared to only 7% of the general non-academic U.S. Twitter base.[5,6] Political engagement is widespread across disciplines and topics. Notably, although 29% of academics' topical tweets reference research, only 7% of research tweets mention these topics, suggesting a selective overlap. Academics tend to be more neutral when tweeting about research, and more explicit when engaging with politically salient issues. Consistent with this, while many non-neutral tweets are grounded in verifiable facts, 51% of topical tweets contain no checkable claim. Political discourse among academics is thus predominantly *opinion-based* rather than purely informational.

Original posts are also less explicit than retweets, and high-reach academics tend to adopt more moderate stances.[7] Positions have also shifted over time, with racial equality standing out: disagreement among academics is both more pronounced and widening compared to general users.[8]

According to Morris (2001), engaging in "politically incorrect" communication can result in reputational loss, as audiences may draw adverse inferences about the speaker. Since such reputational concerns can lead to information loss, the second part of our paper quantifies this cost through an online experiment with a representative sample of 1,700 U.S.

[5]These topics are identified as the most salient in U.S. political debates by Gallup and Pew Research.

[6]A natural benchmark is Garg and Fetzer (2025b), who study the same five issues in a balanced global panel of 99,274 scholars (tweeting in both 2016:H1 and 2022:H2). In that sample, 27% of scholars ever expressed a non-neutral view. The higher U.S. share (44%) places American academics at the top of the international distribution. Sampling differs: their design excludes accounts that joined after 2016 or left before 2022 to enable within-author comparisons; our goal is to characterize the full stock of U.S. academics visible to survey respondents (Section 4).

[7]These patterns highlight the sensitivity of our classification in capturing the intensity of political expression.

[8]This trend of increased political activism among scientists has been linked to events like the March for Science (Russell and Tegelberg, 2020, Campbell et al., 2023).

respondents.[9] They assessed each scientist's personal credibility, the credibility of their research, and their willingness to engage with their content. This conjoint design allows us to isolate and precisely estimate the effects of political expression on public perceptions, offering more experimental control than a Twitter-based approach.[10]

Findings reveal a significant *credibility penalty* for scientists who engage in political discourse, regardless of political orientation. Scientists expressing either left- or right-leaning views face a monotonic decline in *personal credibility* (first outcome): Strong Republicans are rated 39% less credible than neutral scientists, Strong Democrats 11% less; Moderate Republicans and Democrats are rated 9% and 7% less credible, respectively. This penalty extends to the *credibility of their academic work* (second outcome), indicating that reputational costs go beyond personal perception and exceed standard in-group bias. It also translates into lower public *willingness to engage with their work* (third outcome): respondents are 42% less willing to read opinion pieces from Strong Republicans, 10.7% less for Strong Democrats, and 8% and 4.5% less for Moderate Republicans and Democrats, respectively, compared to neutral scientists. All effects are statistically significant at the 1% level.

Credibility penalties reflect a strong partisan bias: respondents mainly perceive scientists aligned with opposing political views as less credible. Democrat respondents rate neutral and Democrat-leaning scientists similarly but lose trust in Republican-leaning ones. Republican respondents generally prefer moderately Republican scientists to neutral, while still penalizing strongly conservative profiles, though less so than they penalize Democrat-leaning scientists. These patterns suggest that scientists' political discourse can serve as a

[9] The broader welfare implications of expert communication are complex. Chakraborty et al. (2020) distinguish between partisan endorsements (post-platform choice) and policy advocacy (pre-platform choice). While endorsements may reduce voter welfare by distorting platforms, these distortions can incentivize experts to provide informative policy advice—unless the conflict between expert and voter preferences is too large. Moreover, despite potential reputational costs, scientists can also benefit professionally from social media, including citation premiums (Chan et al., 2023, Klar et al., 2020) and improved recruitment outcomes in the academic job market (Qiu et al., 2024).

[10] Conjoint designs have been used to study medical decisions (Chan, 2022), repugnant transactions (Elías et al., 2019, Sullivan, 2021), financial choices (Macchi, 2023), dating preferences (Low, 2014), and charitable donations (List and Lucking-Reiley, 2002).

channel for polarizing public perceptions and reducing engagement with scientific content, depending on audience ideology. Both set of results can be rationalized with models of conformity (Bernheim, 1994, Orzach, 2024).

Among other tested attributes, scientists from highly ranked institutions and those in senior positions are perceived as more credible, while gender and field have no significant effect. Notably, political affiliation emerges as the most salient factor shaping public perceptions of scientists.

We conducted several robustness checks. First, we replicated our main findings with a separate sample of 2,000 U.S. respondents, confirming that results hold under a different outcome structure, as well as with an additional sample of 2,000 U.S. respondents who report being Twitter users. We also found minimal carryover effects across profiles, and results remain stable after excluding "speeders" and adjusting for multiple hypothesis testing. A permutation test and a validation task with new respondents confirmed that participants accurately identified the intended political affiliations of scientists in the vignettes. Finally, we conducted an additional robustness experiment to bound potential experimenter demand effects, in line with the approach of de Quidt et al. (2018), and found that any such influence is minimal and cannot account for our results.

To assess external validity, we tested whether the *credibility penalty* for politically engaged scientists extends to journalists—an important audience, as they regularly cover scientific research and interview scientists (Alabrese, 2022). Among 135 international journalists surveyed, we observed clear asymmetries: Republican scientists were rated 33% less credible than neutral ones, and their opinion pieces were 40% less likely to be included in newsletters. By contrast, Democrat scientists were rated only 5% less credible and 2% less likely to be featured—effects that are statistically indistinguishable from zero, likely due to the overrepresentation of liberal journalists. Partisan bias was evident: Liberal journalists were 7, 9, and 5 times less likely to rate a Republican scientist's credibility, research, and newsletter inclusion favorably compared to a Democrat. Conservative journalists showed a comparable bias in the opposite direction, being 4, 9, and 7 times less likely to favor Democrat scientists across the same dimensions.[11]

[11] This aligns with findings from Boxell and Conway (2022), who show that 16% of slant variation across outlets can be explained by journalist preferences.

According to the sender-receiver framework in Gentzkow and Shapiro (2006), a sender who fails to align their message with the receiver's prior beliefs incurs a greater credibility penalty—especially when the quality of the sender cannot be easily verified ex post. For scientists, this penalty may arise in two ways: when discussing politically charged research topics that conflict with readers' views (leading audiences to infer political bias) or when signaling a political identity cue unrelated to research (which may elicit affective polarization). To investigate these channels, we conducted an additional experimental task after the conjoint experiment. This between-subject task isolates the credibility effects of (i) tweeting about politically salient research that may implicitly signal ideological bias versus (ii) explicitly signaling political affiliation through unrelated political cues. We held the research message constant, varied the political signal in an otherwise uninformative bio, and restricted all profiles to economists to keep expertise constant.

Both sharing politically salient research and signaling political identity independently shape public perceptions. First, we find that merely tweeting about politically related research influences credibility: Democrats view economists more favorably when the research aligns with their political views, while Republicans rate economists lower when the research content is misaligned. These credibility shifts occur even in the absence of any explicit political signaling. Second, adding a political identity cue amplifies these effects. Among Democrats, a left-leaning bio further increases credibility, while a right-leaning one sharply reduces it. For Republicans, the pattern reverses: left-leaning signals exacerbate the credibility penalty for misaligned research, while right-leaning signals moderate it. These effects extend to newsletter demand—a behavioral measure of audience engagement curated following (Chopra et al., 2022, 2024)—where alignment between the audience's political views and the scientist's signal strongly predicts sign-ups. Together, these findings confirm that both inferred ideological bias (based on research content) and affective polarization (based on identity cues) contribute to audience reactions, shaping who the public trusts and engages with.

What do scientists think? We surveyed 128 international scientists on Prolific to assess whether they anticipate the credibility penalty observed in our experiment and to understand their views on political expression online. First, scientists expected a larger penalty than what we observed and considered it more acceptable to express political opinions within their area of expertise than outside it: a norm they believe to be widely shared among other academics. Second, many reported hesitating to share political views on social media, suggesting a potential for information loss in public discourse (Morris, 2001, Ottaviani and Sørensen, 2006). Third, while they anticipated mild reputational costs for commenting on topics outside their field, they expected modest reputational benefits when engaging within their expertise. However, perceptions of costs and benefits largely overlapped. To examine this further, we conducted a follow-up survey with 118 additional international scientists. Respondents perceived clear professional benefits of political expression for media appearances and policy roles, smaller benefits for network size and citations, and ambiguous or slight costs for top journal publications and academic job offers.

To probe a mechanism behind the credibility penalty, consistent with the perceived benefits scientists report, we field an additional experiment that makes a scientist's *visibility incentive* explicit. Adapting the design of Bursztyn et al. (2023a), half of the respondents, before viewing a scientist's tweet, also receive the message that *we (the researchers)* will promote the post and that the scientist knew this *ex ante*. This manipulation reveals no uniform "opportunism tax." Instead, updating is partisan. Among Democrats, the promotion information amplifies our main pattern: higher credibility for authors of aligned (left-leaning) posts and lower credibility for authors of misaligned (right-leaning) posts. Among Republicans, promotion raises credibility for authors of aligned (right-leaning) posts and attenuates the penalty for authors of misaligned (left-leaning) posts; neutral promotion is disliked. General trust in science shifts in the same direction under promotion (positive when aligned for Democrats; positive for both left- and right-leaning content among Republicans). Taken together, these results indicate that audiences recognize scientists' incentives but interpret them through partisan congruence and normative expectations about scientists' public engagement.

Overall, studying the reputational costs of scientists' online political engagement reveals a fundamental trade-off. Expressing political views can undermine scientists' credibility, limiting their ability to shape public preferences and influence decision-making while potentially deepening affective polarization. At the same time, anticipating this credibility penalty may discourage scientists from participating in policy debates, leading to information loss and diminishing the societal value of expert engagement. These effects are further mediated by scientists' perceptions of personal career benefits from engaging on social media.

Contribution to the Literature Our findings deepen our understanding of how the public perceives scientists and their communication (Altenmüller et al., 2024, Blastland et al., 2020, Norris, 2020). Prior work has explored how uncertainty and political transparency affect trust in science (Van Der Bles et al., 2019, 2020, Petersen et al., 2021). For example, Kotcher et al. (2017) found that climate scientists' advocacy generally did not reduce credibility, except when promoting nuclear power. Zhang (2023) showed that political endorsements by scientific journals, such as Nature's support for Joe Biden, can polarize perceptions of publishers and science more broadly, particularly among Republicans.

We extend Zhang (2023) in key ways. We provide the first evidence of a visibility–credibility trade-off in scientists' political expression, combining (i) observational data on scientists' political engagement on Twitter, (ii) surveys of international scientists, and (iii) online experiments with representative samples of U.S. respondents, Twitter users, and journalists. Rather than focusing on the editorial decision of a single publisher, which may reflect institutional rather than individual scientists' preferences, we provide large-scale observational evidence on scientists' own political discourse across disciplines and over time, and causally identify the reputational effects of scientists' political expression through experiments.

More broadly, we contribute to the literature on the communication (or absence) of politically salient topics in the U.S., including the "spiral of silence" (Huang and Ho, 2023), both offline (Braghieri, 2024) and online (Bursztyn et al., 2023a). This literature focuses on decisions to voice controversial opinions and their downstream effects. To our knowledge, this is the first study to examine U.S. academics' online communication on politically charged topics and its impact on public perceptions. We also contribute to work on politically motivated reasoning in the acquisition (Chopra et al., 2024, Faia et al., 2024) and interpretation (Gentzkow et al., 2023, Thaler, 2024) of information, by showing how political affiliation mediates responses to scientists' views.

We further relate to the growing literature using text-as-data to measure political ideology from sources such as official documents (Ash, 2016, Hansen et al., 2018, Grimmer, 2010), online news (Cagé et al., 2020), product descriptions (Fetzer et al., 2024), political speeches (Jensen et al., 2012, Gentzkow et al., 2019), academic publications (Garg and Fetzer, 2025a, Jelveh et al., 2024, Brown and Gupta, 2023), and surveys (Draca and Schwarz, 2024). Our study builds directly on Garg and Fetzer (2025b), who show that scientists' political expression differs from that of the general public in both topic and tone, with public narratives often shaped by highly visible but less academically impactful scientists. We extend this work by analyzing polarization in political discourse among U.S. academics and causally identifying the reputational risks of political engagement, which vary sharply by audience ideology.

Our findings contribute to broader evidence on ideological and affective polarization in the U.S., documenting how echo chambers can deepen public divides (Canen et al., 2020, 2021, Fiorina and Abrams, 2008, Alesina et al., 2020, Iyengar et al., 2019, Gentzkow and Shapiro, 2011, Levy, 2021, Chopra et al., 2024, Mosleh et al., 2021, Colleoni et al., 2014, Flaxman et al., 2016, Stewart et al., 2019, Boxell et al., 2024, Kahan et al., 2011). We find that individuals gravitate toward scientists whose views align with their own, and discount misaligned voices as less credible, highlighting the need for strategies to reduce polarization while preserving the integrity and reach of scientific discourse.

Additionally, our work contributes to the literature on social identity, originally developed by Tajfel and Turner (2003) and later incorporated into economic analysis by Akerlof and Kranton (2000) and Shayo (2009, 2020). Recent research has examined how social identity shapes economic behavior in areas such as trade policy (Grossman and Helpman, 2021), teamwork (Charness and Chen, 2020), bonus acceptance (Bursztyn et al., 2020a), policy adoption (Garcia-Hombrados et al., 2024), and everyday interactions (Braghieri et al., 2024). Party affiliation increasingly influences individual choices beyond politics, including decisions about marriage (Alford et al., 2011), dating (Huber and Malhotra, 2017), residential location (Brown et al., 2022), career path (Chinoy and Koenen, 2024), medical treatment (Kim, 2024), and even food preferences (Burnitt et al., 2024). Our findings show that scientists' social identity, specifically their political identity, reduces their perceived credibility, the credibility of their research, and the public's willingness to engage with their communication.

While our study is not a real-time Twitter experiment, it relates to recent experimental work on interventions to reduce misinformation, toxicity, and discrimination online (Guriev et al., 2023, Jiménez Durán, 2022, Beknazar-Yuzbashev et al., 2022, Angeli et al., 2022, Ajzenman et al., 2023a,b). We also contribute to the broader literature on the political effects of the internet and social media, including their role in shaping voting behavior, protests, polarization, and misinformation (see Zhuravskaya et al., 2020, for a review). Our focus, however, is on the reputational risks of the growing role of social media in academic communication.

The remainder of the paper is organized as follows. Section 2 examines the political engagement of a large sample of U.S. academics on Twitter. Section 3 presents survey evidence on scientists' own views about political expression online. Section 4 experimentally tests how political expression affects public perceptions of scientists' credibility and engagement with their content. Section 5 concludes.

2. SCIENTISTS' ONLINE POLITICAL EXPRESSION

Given the growing emphasis on academic impact, we assess whether social media is increasingly vital for scientific dissemination. We track the online diffusion of over 100,000 Scopus-indexed articles published from 2011 to 2020 in top general interest and life-science journals using Altmetric data (Alabrese, 2022, Peng et al., 2022).[12] Figure 1, Panel A, shows a steady rise in online presence, particularly on Twitter, where over 96% of articles are mentioned. Panel B further shows that the distribution of Twitter mentions has become less skewed towards zero with a thicker right tail and more high-mention outliers over time, indicating increasing variability in social media attention.

These trends highlight Twitter's growing role in scientific communication while also increasing public visibility into scientists' social media activity. Since about half of those who engage with academic content on Twitter are non-academic (Mohammadi et al., 2018, Maleki and Holmberg, 2023), and perceptions of political bias among scientists contribute to distrust in science (Altenmüller et al., 2024).

Consistent with this, Appendix Section E shows that, in our data, the majority of accounts that follow, retweet, reply to, or mention U.S. academics on Twitter cannot be identified as academics based on Mongeon et al. (2023) global registry. While interactions with other scholars remain substantial—particularly among retweeters and repliers—academics' online communication is systematically exposed to audiences beyond their professional peers. As a result, when scientists tweet, including on politically salient topics, their messages are plausibly visible to and engaged with by a broader public audience rather than being con-

[12]Of the 114,868 articles published between 2011 and 2020 in *Science*, *Nature*, *PNAS*, *Cell*, *NEJM*, and *Lancet*, 107,008 were tracked by Altmetric as of November 10, 2021. See Appendix Section C for further details.

fined to intra-academic exchange.Recent international evidence further confirms that people frequently encounter and discuss science on social media—and that in some countries, notably the United States, individuals report having participated in protests and activism around scientific issues (Mede et al., 2025).

These patterns suggest the importance of systematically tracking how scientists themselves engage in online political discourse. In this section, we therefore leverage large language models (LLM) to cost-effectively measure the extent and nuance of scientists' political expression on social networks.

2.1. *Academics on Twitter: Sample and Method*

We analyze political discourse and ideological polarization among 97,737 U.S. academics on Twitter from 2016 to 2022, a substantial subset of the 683,050 U.S. research-active academics (National Center for Education Statistics, 2020). These academics garner significant attention to their online content: over 2016–2022, the number of retweets per post averages 16.4 (SD = 1,251.4; median = 0.3), with 6.7 retweets on average for their politically salient tweets (SD = 134.9; median = 0.4).[13] These university-based researchers were identified by Mongeon et al. (2023) by matching authors' OpenAlex IDs to Twitter/X accounts via Crossref Event Data. Their method achieved high precision and moderate recall by identifying authors who tweeted their own research.[14] Building on Garg and Fetzer (2025b), we use the same Twitter data infrastructure, merging tweet-level timelines with OpenAlex records for 97,737 U.S. academics. To benchmark academics' political expres-

[13] If we instead examine the total stock of retweets accumulated by each academic during 2016 – 2022, the mean is 1,100 retweets (SD = 6,241; median = 270) overall, and 1,217 retweets (SD = 2,602; median = 414) for politically salient tweets.

[14] Mongeon et al. (2023) validate their matching against 13,208 ORCID–Twitter pairs. They recover 8,229 of those (True Positives), miss 4,979 (False Negatives), and hallucinate 361 extra pairs (False Positives), yielding a precision of 0.958, a recall of 0.623, and an F_1 of 0.755. Put differently, roughly 4% of the handles in their file are false positives, while ≈ 38 % of scholars on Twitter remain undetected. Because we analyse the resulting high-precision, moderate-recall subset without adding new accounts, our findings speak to the behaviour of the $\sim 60\%$ of research-active U.S. academics who both maintain a Twitter/X presence and occasionally self-promote their scholarship there. If the remaining non–self-tweeting scholars are, on average, less vocal or less politically engaged online, then any bias in our estimates is likely to be conservative—that is, our measures may slightly overstate how expressive (and polarised) the wider professoriate is on social media.

sion against that of the broader public, we further link these data to a comparison sample of 241,419 randomly sampled U.S. Twitter users, originally constructed by Siegel et al. (2021).[15] We then employ LLMs to detect whether tweets address specific salient topics and to classify post stances as pro, anti, or neutral toward those topics.[16]

Our dataset comprises approximately 116 million tweets (original tweets, retweets, quote retweets, and replies) from these U.S. based academics, and our analysis focuses on five politically salient issues: Abortion Rights, Climate Action, Racial Equality, Immigration, and Income Redistribution.[17]

The procedure involves two key steps. The first is *topic detection*. OpenAI's GPT-4 was used to generate dynamic keyword dictionaries designed to capture the evolving discourse on each topic. The model was prompted with:

`Provide a list of [`*`ngrams`*`] related to the topic of [`*`topic`*`] in the year [`*`year`*`]. [`*`Twitter Fine Tuning`*`].`
`Provide the [`*`ngrams`*`] as a comma-separated list.`

The `[`*`Twitter Fine Tuning`*`]` either explicitly instructed the model to `'Focus on language, phrases, or hashtags commonly used on Twitter'`, ensuring the dictionary remains contextually relevant to the platform's discourse or was left empty. This process, spanning 5 topics × 3 n-gram types × 7 years × 2 vernacular adjustments (210 prompts), yields five comprehensive keyword dictionaries applied to the full corpus of tweets. Tweets containing any keyword from a topic's dictionary were labeled as belonging to that topic (e.g., tweets mentioning "Paris Agreement" are labeled under Climate Action). The topic detection captures 5.31% (around 6.15 million) of the initial 116 million tweets. Climate Action was the most discussed topic (2.09% of all tweets), followed by Racial Equality (1.50%) and Immigration (0.86%).[18]

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[15] In Siegel et al. (2021), the authors generate user IDs uniformly at random from the Twitter user-ID space and use the API to identify accounts whose time zone or self-reported location indicates residence in the United States. We retain accounts that posted at least one tweet between 2016 and 2022.

[16] Comprehensive individual-level, time-series data are publicly available from Garg and Fetzer (2025b).

[17] We identified salient issue based on Gallup survey. Gallup's list is available here.

[18] Using GPT-4 on all tweets would be prohibitive, so we focus on using GPT-3.5 Turbo on the set of topical posts for stance detection.

The second step is *stance detection*. Using OpenAI's GPT-3.5 Turbo, topical tweets were classified into three discrete categories: "pro," "anti," and "neutral."[19] The classification used the following prompt:

`Classify this tweet's stance towards [topic] as 'pro', 'anti', 'neutral', or 'unrelated'. Tweet: [tweet].`

This process not only identifies the position of each tweet on a specified topic but also enhances precision and relevance by filtering out unrelated tweets. To reduce costs, we sampled up to three random tweets per author, month, and topic (99.55% had one tweet; 0.45% had two; 0.004% had three), ensuring adequate coverage and reliable stance estimation. See Appendix Table B.6 for examples on each topic-stance, posts-level summary statistics in Appendix Table B.3 and details on validation in Appendix Section D.

Appendix Table B.2 summarizes the key characteristics of our sample of academics. Most scientists are in STEM, followed by Medicine and Social Sciences. 54% of U.S.-based academics discussed at least one topic, and 81% of those expressed non-neutral stances. Our analysis thus focuses on 42,747 scientists who posted non-neutral topical tweets.

We call *politicized* the scientists who express a non-neutral stance on any salient topic. Our focus on non-neutral communication offers several advantages: first, it maps naturally from our methodology; second, it broadly identifies the presence of academic discourse related to political issues. Such discourse may enable the public to infer scientists' political leaning and update the perception of their credibility: a central concern addressed in the second part of this paper.

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[19] Our "pro/anti/neutral" tags do not seek to code tone (e.g. partisan invective) but rather what social-science work calls a directional policy valence: does the tweet, implicitly or explicitly, point in the policy direction of more (or less) X? A tweet that merely cites evidence ("Our LancetGH paper finds hostile-environment policies harm migrants' health") still carries a directional signal—restrictive policy $\rightarrow$ worse outcomes—which readers will reasonably interpret as evidence against such restrictions. Hence it is coded pro-immigration. Likewise, flagging that "99.9% of the literature finds human-caused warming" provides empirical support for mitigation and is therefore pro-climate-action. We argue that this does not exaggerate politicization. Our measures do not require overt partisan language, so it will indeed classify some fact-sharing as political expression. This is by design: our research question is how often academics go on record with a position that has policy implications, regardless of rhetorical style.

Appendix Table B.2 further breaks down the proportion of politicized scientists by key characteristics. The proportion remains relatively stable across categories of academic recognition (between 41% and 46%).[20] Disciplinary differences are more notable, with higher proportions in Humanities (58%) and Social Sciences (65%) versus STEM (43%) and Medicine (38%). Some gender differences also emerge, with 50% of female scientists and 40% of male scientists being vocal. More details are provided in Section 2.2.3.

2.2. *Academics Political Engagement on Twitter*

Figure 3 compares political discourse trends between academics (orange) and a random sample of U.S. Twitter users (blue) between 2016 and 2022.[21] The top-left panel ("All") shows that well over one-third of the 97,737 tracked U.S. academics express a non-neutral opinion on any specified salient topic in each month, revealing a substantial gap compared to general users.[22]

Breaking the trend down by topic, Climate Action and Racial Equality (the most frequently discussed issues) are the largest contributors to the gap. Notably, Climate Action discourse declined at the onset of the COVID-19 pandemic, while Racial Equality remained highly salient, surging in mid-2020 following the George Floyd incident and widespread protests. Abortion Rights discussions spiked in 2022 due to changes in abortion laws, and Immigration conversations peaked around the 2016 presidential election and diminished over time.

As our working definition of politically engaged academics is broad, encompassing any scientist expressing a non-neutral stance on a politically salient topic, we conduct further analysis to explore (i) the extent to which such tweets also mention research-related content and (ii) whether their statements are factual or opinion-based. These exercises help distinguish between politicization through interpretive or normative commentary and neutral communication of scientific facts.

[20] Cumulative citation categories: <100; 101–500; 501–1,000; >1,000. Categories are derived from the following distribution: 25th percentile = 36, median = 190, mean = 1205, 75th percentile = 849.

[21] See footnote 15 for further details on the benchmark sample of 241,419 random U.S. Twitter users.

[22] Monthly aggregated scatter plots are smoothed using LOESS (span=0.5) with shaded standard errors.

2.2.1. *Overlap with Politics and Research*

To better understand the interplay between scientific discourse and political engagement, we analyze how academics reference both scientific research and political figures in their tweets. This allows us to assess whether topical engagement primarily reflects the communication of evidence, political commentary, or both. Following the methodology in Section 2.1, we used GPT-4o to generate dictionaries for detecting mentions of `[scientific research papers]`. Additionally, we generated dictionaries for `[Donald Trump]`, `[Joe Biden]`, `[Politicians]`, and `[Political Candidates]`. These were combined to create a comprehensive dictionary for identifying mentions of political figures (see Appendix Table B.4 for examples).

We then examined the yearly distribution of academics' tweets across four categories: tweets mentioning politicians (blue), research papers (green), both (pink), and other content (grey). Appendix Figure A.1 shows that, on average, 10% of tweets mention politicians, 20% reference research papers, and about 1% mention both. These proportions remain relatively stable over the years, with fluctuations likely tied to significant events (e.g., larger coloured areas in 2020 may reflect discussions post-George Floyd or a COVID–19–induced surge in research tweets). Darker shades mark tweets addressing the five salient issues that largely fall in the residual grey category, namely Abortion Rights, Climate Action, Racial Equality, Immigration, and Income Redistribution.

These findings align with the cross-sectional summary statistics in Appendix Table B.3: overall, 9.8% of tweets mention politicians and 19.2% mention research papers; when restricted to the five salient topics, these rise to 16.0% and 28.9%, respectively. Notably, Climate Action tweets mention research papers 44.5% of the time versus 15% for politicians (a roughly 3:1 ratio), while Abortion Rights and Immigration tweets favor mentions of politicians (25.6% and 28%) over research papers (15% and 21.4%), with ratios of approximately 1.7:1 and 1.3:1.

Overall, the data suggest that scientists engage in discussions about research, political figures, and politically salient issues, with a moderate overlap between these domains. Given that politically salient tweets generate high engagement and research-related discourse is increasingly prevalent on Twitter (see Figure 1), it is important to examine not only whether scientists engage with political or research content, but also whether their posts rely on factual claims or reflect broader opinion-based discourse.

2.2.2. *Factual vs Non-Factual Content*

We further investigate the nature of topical engagement among scientists by assessing the factual veracity of their tweets. This analysis serves to distinguish between tweets that reflect politicized opinion and those that convey factual content, which may nevertheless align with a political stance. Although our working definition of politicized scientists is based on the expression of a *non-neutral stance* on a salient topic, this measure does not rely on tone or intent. By classifying the veracity of tweet content independently, we provide a complementary lens on the nature of topical engagement, shedding light on whether scientists express positions through opinionated commentary or through the communication of verifiable claims.

We assess whether topical tweets contain factual or non-factual statements using GPT-4o-mini. Tweets are classified into four mutually exclusive categories based on whether they include or imply a claim that is: `TRUE`, `FALSE`, `UNCLEAR`, or `NO FACTUAL CONTENT` (the default for absence of any checkable claims). The classification follows a structured prompt that also requires the extraction of any factual claims and a justification for the assigned label (see Luu et al., 2024, for a similar approach). The full prompt, together with illustrative examples and an external validation of its performance on an independent benchmark dataset, is discussed in detail in Appendix Section D.

Overall, 51% of tweets are classified as `NO FACTUAL CONTENT`, indicating that the majority of topical posts consist of opinions or general commentary rather than verifiable claims. Among the 49% of tweets that do contain factual content, 2% are labeled `FALSE`, 53% `TRUE`, and 45% `UNCLEAR` (for example, due to unverifiable forecasts or ambiguous wording).

Clear differences emerge across ideological stances (Figure 2). Tweets classified as *anti* (more conservative) are more likely to be labeled as `FALSE` or `UNCLEAR` than either *neutral* or *pro* (more progressive) posts. In contrast, *pro* tweets are most often categorized as `NO FACTUAL CONTENT`, while also showing the lowest share of `FALSE` and `UNCLEAR` classifications. To account for the possibility that retweets may introduce ambiguity (e.g., through critique or irony), we also exclude retweets from the classification. Doing so generally increases the share of `NO FACTUAL CONTENT` tweets but leaves all other patterns unchanged (available upon request).

This exercise addresses a key concern that some tweets labeled as pro or anti may not reflect politicized expression, but rather factual reporting that aligns with a given issue (e.g., new evidence on climate change or immigration policy). For example, a tweet can be labeled as "pro-climate" based on its content, while simultaneously being categorized as `TRUE` under our classification. Our results show that this is not systematically the case: while many stanced tweets (especially in the pro category) are grounded in verifiable facts, over half of non-neutral tweets contain no checkable claim at all. The factual classification thus complements the stance measure by distinguishing between opinion, ambiguous claims, and factual reporting within politicized topics.

2.2.3. *Differences by Gender and Discipline*

We examine variation in academics' online political expression by gender and discipline. Gender was determined via a binary LLM classification (Appendix Figure A.2).[23] Academics with female-labeled names express slightly more political opinions.This is especially true for Abortion Rights, Immigration, and Racial Equality, where activity increases after 2020. For Climate Action and Income Redistribution, there are no significant gender differences.

We also assess disciplinary differences using OpenAlex "Concepts." Each publication is scored on 19 root concepts; for each researcher, we average these scores over the entire period and assign the primary concept to one of three groups: Medicine, Social Sciences (Business, Economics, History, Political Science, Psychology, Sociology), or STEM (Biology, Chemistry, Engineering, Environmental Science, Geography, Geology, Materials Science, Mathematics, Physics). Figure 4 shows that, overall, U.S. academics express political views beyond their field. However, STEM scientists are most vocal about Climate Action, while Social Scientists are more active on Income Redistribution. Notably, gaps between disciplines have narrowed over time, contrasting with common stereotypes of scientist politicization (Altenmüller et al., 2024).

[23]Names were classified as "Male" or "Female" with 99% accuracy (see Appendix Section D).

2.2.4. *Scientists' Ideological Polarization on Twitter*

Our analysis thus far shows that, of the 97,737 academics sampled, 52,541 engaged in tweeting about political issues, with 81% of these expressing non-neutral stances on at least one of the five salient topics. We refer to *ideological polarization* to indicate the divergence in political views or issue positions among individuals.

For each topic and user in a given month, we calculated their slant as the difference between the number of pro-stance tweets and anti-stance tweets, divided by the total number of tweets (pro, anti, or neutral) they posted during that month on that topic.[24] This is expressed as:

$$S_{um} = \frac{pro_{um} - anti_{um}}{pro_{um} + anti_{um} + neutral_{um}}$$

In this formula, S_{um} denotes the net pro stance share of tweets by user *u* in time-period *m*, relative to all their tweets on a topic. This approach provides, for each individual, a continuous measure of slant ranging from -1 (completely anti) to 1 (completely pro) for each topic, at any point in time. Our method captures the spectrum of opinions, from strong opposition to firm support, while distinguishing between more moderate or explicit positions, a feature essential in environments where public opinions may lean toward socially desirable expressions (Bénabou and Tirole, 2006).

Figure 5 displays the net pro stance distributions S_{um} for each topic across academics and general users over the entire sampled period. Except for Racial Equality, neutral views dominate. However, multiple peaks, confirmed by Hartigan's Dip Test (Hartigan and Hartigan, 1985) (p=0), indicate strong multimodality and distinct political camps. General users cluster more at extreme conservative stances (e.g., -1) than academics, who concentrate in the moderate liberal/progressive range. Notably, for Racial Equality, with few neutral stances, users show higher consensus by clustering more at the fully pro stance compared to academics. Appendix Figure A.3 further compares early (2016–2017) and late (2021–2022) net pro stance distributions. Results suggest shifts in opinions over time, particularly on

[24]Specifically, we measured individual academics' slant following the theoretical framework of Esteban and Ray (1994), categorizing tweets into 'pro', 'anti', and a residual 'neutral' category.

topics like Immigration and Abortion Rights, for which KS is highest (Massey Jr, 1951), highlighting the dynamic nature of ideological polarization among academics.

We also examine differences in net stances between tweets mentioning scientific research or not and how differences in Twitter reach and post type influence academics' net stances. Appendix Figure A.4 shows that academic research-related tweets are significantly more moderate than non-research tweets (except for Climate Action, where differences are smaller and research tweets are slightly more progressive). The largest divergence between research and non-research-related tweets is seen in Abortion Rights: non-research tweets skew more pro-choice (+1) while research tweets cluster more near neutrality (0). Further, Appendix Figure A.5 indicates that academics with lower Twitter reach (below the median of 522 followers) exhibit more extreme stances, whereas high-reach academics appear more moderate. Finally, Appendix Figure A.6 reveals that retweets display consistently more extreme stances than original tweets, particularly for Abortion Rights and Climate Action, suggesting that original posts tend to be more measured while retweets amplify polarized views.

The pattern highlighted in Section 2.2.2 are also reflected across topics (Appendix Figure A.7). For each factual category (`TRUE`, `FALSE`, `UNCLEAR`), the distribution of tweets is generally the same as the general sample just described, indicating that most academic posts, regardless of veracity, cluster around moderate positions. Within the more ideologically extreme regions, however, we observe notable relative differences across factual categories. `FALSE` tweets are more prominent within conservative-leaning clusters, while `TRUE` or `NON FACTUAL` tweets are relatively more common in progressive-leaning clusters. Racial equality is a marked exception to this pattern. While most tweets on the topic oppose racism across the spectrum, `TRUE` posts are relatively more concentrated among conservative-leaning tweets. In contrast, `FALSE` and `UNCLEAR` posts are more evident among progressive-leaning ones, and `NON FACTUAL` tweets are somewhat in between.

Findings align with expectations: research-related tweets likely reflect professional norms of moderation, while non-research tweets allow for more direct political expression. Similarly, academics adopt a more measured tone in their original posts, potentially exercising greater caution with content directly attributable to them. Finally, with the exception of Climate Action, opinion-based content (`NON FACTUAL` tweets) tends to be more ideologically extreme (particularly in a progressive direction) compared to tweets containing verifiable factual statements. Overall, these patterns highlight the sensitivity of our measure in capturing differences between more neutral or explicit positions.

2.2.5. *Evolution of Scientists' Disagreement*

We finally examine the evolution of disagreement among scientists by computing the monthly variance of net pro stances S_{um} for each topic. Variance provides a continuous measure of opinion diversity that is insensitive to the average stance yet highly sensitive to extremes, facilitating longitudinal and cross-topic comparisons (McCarty et al., 2016). Although variance is robust and interpretable for large datasets, it does not reveal the causes of disagreement (Fiorina et al., 2005) and may overstate divergence if few hold extreme views; it also may not capture consensus well (Hopkins, 2018). Therefore, we interpret variance in relative terms—lower variances indicate greater consensus—and examine its fluctuations over time and topics to identify both divergence and emerging agreement in the evolving ideological landscape.

The top left panel in Figure 6 ("All") aggregates all topics, showing a mild increase in disagreement from 2016 to 2022, which could raise concerns about scientific consensus and public trust. While scientists appear to exhibit greater ideological distance than general users overall, a topic breakdown reveals nuances. For most topics, the general U.S. Twitter population shows higher disagreement than academics. However, for Racial Equality, academics display a growing and more pronounced disagreement compared to the public—a disparity that disproportionately affects the aggregate trend given its high prominence. For Immigration, Income Redistribution, and Abortion, public disagreement is higher, with the gap widening during the pandemic before narrowing toward the end of the period. In the case of Climate Action, disagreement between academics and the general public persists throughout, with the gap widening further by the end of 2022.

2.2.6. *Within-Academic Changes in Positions*

The variance-based measures above capture how dispersed scientists' political views are at a point in time, but they do not reveal how often individual academics actually change positions. Appendix Section F therefore provides a complementary perspective by tracking a cumulative "stock" stance for each academic and topic over time and identifying *reversal events* when this stock moves between pro, anti, and neutral positions.

Appendix Figures F.1–F.6 reveal three main patterns. First, many academics *enter* political debates from neutrality into a pro cumulative stance. This pattern is most pronounced for racial equality, where more than 50% of academics ever move from a neutral to a pro stock stance at least once, followed by climate action (around one-third) and smaller but still meaningful shares for the remaining topics. Second, entries from neutrality into anti cumulative stances are much rarer. Even on racial equality, only about 12% of academics ever move from neutral to anti, with corresponding shares for other topics generally in the low single digits. Third, true cumulative sign flips between pro and anti positions are exceedingly rare: pro-to-anti reversals are virtually absent, while anti-to-pro reversals affect at most around 2% of academics on any topic, with monthly rates typically well below 1%.

Taken together, these patterns indicate that academics' political positions are highly persistent over time. While movements from neutrality into a stance—especially a pro stance—are relatively common, complete ideological reversals are not. Most observed dynamics reflect entry into debates, occasional withdrawal back to neutrality, and stable differences across individuals, rather than frequent switching between opposing political positions.

3. SCIENTISTS' VIEWS ON ONLINE POLITICAL EXPRESSION

The most immediate advantage of social media for scientists is access to audiences beyond their immediate academic networks. Appendix Figure A.8 (left panel) shows that academics who were already active on Twitter in 2016 became significantly more visible over time, as measured by average retweets per monthly post in 2016 (blue) versus 2022 (orange), based on their initial posting activity. This pattern is even more pronounced among those who occasionally engaged with political topics (right panel), who received higher engagement than their peers, both at the beginning and at the end of the period.

This visibility, however, may come at a cost. As scientists become more public-facing, particularly on politically sensitive topics, they risk being perceived as biased or partisan. Such perceptions can erode their most valuable asset: credibility. The remainder of this section investigates whether scientists anticipate this trade-off and how they evaluate the professional costs and benefits of political expression online.

To address these questions, we examine scientists' self-reported views using two surveys. We first recruited 128 scientists worldwide via Prolific.[25] Appendix Table B.7 describes the sample: 94% are employed (over 60% in universities), including 43% postdoctoral researchers, 28% faculty, and 29% industry professionals; 34% work in life sciences/biomedicine, 34% in social sciences, and the rest in physical sciences or technology.

First, we quantified whether scientists expect a credibility penalty for expressing political opinions on social media—that is, whether they effectively anticipate the reputational cost that our core experiment in Section 4 uncovers. Second, following Bursztyn and Yang (2022), we elicited both the scientists' own views and their perceptions of the opinions of peers, that is, their first- and second-order beliefs on the acceptability of political expression within and beyond their expertise. Third, we asked about personal experiences with publicly expressing political opinions.

To gauge the expected credibility penalty, we told respondents that scientists who **do not** express political opinions receive a trust score of 7.2/10 based on our main experiment (see next Section 4), then asked: "*What is the reported trust level for scientists who **do** express political opinions on social media?*" (participants earned a £0.50 bonus for a correct response). As shown in Figure 7, respondents expect a 30% loss in public trust, nearly double the experimentally observed loss of 16.6%.

While the perception of a reputational loss aligns with cheap talk models under reputational concerns (Morris, 2001, Ottaviani and Sørensen, 2006), the overestimation of the penalty stands in contrast to these models, which assume that reputational costs are common knowledge and that beliefs are correct in equilibrium. Exploring this discrepancy is beyond the scope of this paper, but we offer two possible explanations. One is that a more prominent audience for scientists' reputational concerns may be fellow scientists rather than the U.S. public or journalists, as in our experiment. Another is that scientists may simply misperceive the beliefs of the U.S. public, consistent with findings in Bursztyn et al. (2020b).

Relative to professional norms, Panel A of Appendix Figure A.9 shows that scientists consider political expression acceptable when related to their expertise, but not beyond

[25] Prolific respondents were screened for employment in "Research," PhD-level education, and English fluency.

it.[26] Panel B shows they believe peers share this norm. Finally, Panel A of Appendix Figure A.10 indicates that scientists are significantly more hesitant to comment on topics outside their field,[27] while Panel B shows they expect net negative repercussions for out-of-field comments and net benefits for in-field ones, though these net differences are small.

To further examine the perceived professional costs and benefits of expressing political views online, we again surveyed 118 international scientists via Prolific. This survey disentangles the perceived impact of political expression on specific professional outcomes such as network size, citations, academic job offers, media appearances, policy roles, and top publications.

Figure 8 compares scientists' perceived outcomes for three hypothetical profiles: Scientist X (grey), who expresses political views related to their research; Scientist Z (red), who expresses views on unrelated topics; and Scientist Y, identical to either X or Z but refraining from political expression. Outcome distributions shifted toward X or Z (green areas) indicate perceived benefits, while a denser left tail for Y (blue areas) signals perceived costs. The results reveal clear perceived benefits of political expression for media appearances and policy roles, milder benefits for network size and citations, and ambiguous or slight costs for top publications and academic job offers.

Finally, we posed three open-ended questions to capture scientists' views on what constitutes "politicized" behavior, the boundaries of expertise for political commentary, and personal experiences with its costs and benefits. Appendix Table B.8 summarizes these responses, highlighting recurring concerns about credibility, expertise boundaries, and the trade-offs of political engagement. While our Twitter data cannot fully capture scientists' motivations, survey responses consistently suggest that academics perceive greater benefits when political opinions align with their research expertise.

[26] This aligns with Garg and Fetzer (2025b), who find that academics adopt more pro-social views on topics where they hold expertise.

[27] This complements evidence of a 'spiral of silence' in academia (e.g., Norris, 2020).

4. SCIENTISTS' POLITICAL EXPRESSION AND PUBLIC PERCEPTIONS

Having examined the scope of online political discourse and issue-based disagreement among scientists, as well as scientists' perceptions of costs and benefits of political expression, we now assess the risks of such expression.[28]

Specifically, we study how scientists' political expression affects public perceptions of their credibility and contributes to audience polarization. We ran a conjoint experiment with 1,704 respondents from a broadly representative U.S. sample, recruited via Prolific: a platform widely used for experimental research (Bursztyn et al., 2023a, Enke et al., 2023).

Our sample broadly reflects the U.S. population in terms of key dimensions, including political affiliation, region, ethnicity, and gender. As is typical with Prolific, respondents tend to be slightly younger, more educated, and have higher incomes than the general population. Importantly, prior work shows that Prolific's Republican sample is representative of the broader Republican population (Braghieri et al., 2024, Kashner and Stalinski, 2024). Appendix Table B.9 provides full demographic comparisons with the population census. To ensure data quality, all participants passed an attention check before beginning the main task.

We opted for an online experiment over a field experiment on X due to several advantages, despite a minor conceptual mismatch with Section 2. First, the conjoint design on Prolific lets us systematically vary scientists' attributes—something hard to achieve on X. Second, a field experiment would require creating fake scientist profiles to engage with users, a method both impractical and likely to raise authenticity concerns. Third, typical field outcomes like follow-backs and retweets (Ajzenman et al., 2023a,b) are too narrow for capturing perceived credibility.

4.1. *Experimental Design*

Following best practices for conjoint experiments (Hainmueller et al., 2015), we created five hypothetical scientist profiles, randomly varying key attributes: gender (male/female),

[28] An overview of all experiments and surveys conducted for this paper can be found in Appendix Table B.1. All studies and surveys were pre-registered on AsPredicted with numbers 166935, 179009, 181452, 186295, 206927, and 235820.

research field (Social Sciences, STEM, Medicine, Humanities), seniority (junior/senior), university prestige (high/low), and political affiliation.

The latter (our main attribute) is conveyed through a Twitter-style bio (as suggested in Rogers and Jones (2021))and a high-engagement tweet written by real scientists,[29] spanning five categories: Strong Democrat, Moderate Democrat, Neutral (baseline), Moderate Republican, and Strong Republican.

The five political profiles were created by randomizing the remaining attributes, ensuring each had a unique combination of characteristics. Vignettes followed a consistent format and were shown in random order to avoid order effects (see Appendix Table B.11 and Figure A.11 for a visual representation).

> *The profile you are seeing is a [**Gender**] scientist.*
> *This scientist works in the field of [**Research Field**]*
> *Currently, this scientist is a [**Seniority**] at the [**University Affiliation**].*
> *The scientist is active on X (formerly known as Twitter).*
> *The Twitter bio of the scientist is: "[**Twitter Bio**]".*
> *A recent selected Tweet reads: "[**Twitter Post**]".*

Participants rated the credibility of the scientist and their research separately on a 0–10 scale, and indicated their willingness to read an opinion piece by a similar scientist (0–10).[30] To encourage truthful responses, participants were told they would receive an article from a real scientist matching their ratings.[31] The design included 160 unique pro-

[29]High-engagement tweets (those with many likes, retweets, and comments) are used for their greater visibility. Minor modifications were made to the wording of the posts to preserve the anonymity of the academic author.

[30]Hainmueller et al. (2015) shows that single- and paired-vignette designs yield similar estimates; we use the simpler single-vignette approach.

[31]The same incentivization that we employ has been widely used in experiments in labor economics (Kessler et al., 2019, Cullen et al., 2023, Kessler et al., 2024), gender economics (Low, 2014), finance decision in development countries (Macchi, 2023, Colonnelli et al., 2024), and in education (Exley et al., 2024). While the elicitation procedure might not be fully incentive compatible, prior work finds no significant difference between hypothetical and incentivized responses (Hainmueller et al., 2015, Brañas-Garza et al., 2021, 2023, Enke et al., 2022).

files, generated from all combinations of 2 (gender) × 4 (field) × 2 (seniority) × 2 (university affiliation) × 5 (political profile).

Our within-subject conjoint design covers the full political spectrum and provides greater power to detect the effect of each attribute, even though the tweets are not identical in content and style. A between-subject design, by contrast, would allow only limited variation: for example, varying the Twitter bio while holding the tweet constant. However, to rule out any potential confounder, we also ran a between-subject task in which respondents rated the credibility of an economist (discipline held constant). In this additional task, detailed in Section 4.4, the tweet (promoting a research paper) was fixed, while political leaning was signaled only through the bio (left or right). This design allowed us to isolate whether credibility penalties stem from politically salient research topics or from identity cues unrelated to scientific content.

Our design addresses common concerns in conjoint experiments (Hainmueller et al., 2015), particularly attribute-order effects and experimenter demand effects. To mitigate order effects, we placed the primary attribute—political affiliation—at the bottom of the profile. To reduce demand effects, we highlighted multiple salient attributes and informed respondents that they would receive an opinion piece from a real scientist matching their top-rated profile. We explicitly discuss and bound potential demand effects by conducting a robustness experiment where we explicitly induce demand effects, as discussed in Section 4.2.2. To enhance profile plausibility, we informed participants that all tweets were from real scientists and stated in the consent form that no false information was used. Full experimental instructions are provided in the supplementary Instructions Material.

4.1.1. *Validating Twitter Political Signals*

To validate our political affiliation signals, we surveyed 98 new Prolific participants who categorized each combination of Twitter bios and tweets from the main experiment into one of five labels: Strong Republican, Moderate Republican, Neutral, Moderate Democrat, or Strong Democrat. Appendix Figure A.12 shows that most responses matched the intended labels, with misclassifications largely limited to adjacent categories. This high accuracy confirms that our signals were reliably perceived, supporting the credibility of our design.

4.2. *Impact on Public Perceptions*

Figure 9 shows that scientists' online political expression leads to a monotonic penalty: neutral profiles receive the highest ratings for personal credibility, research credibility, and willingness to read opinion pieces, with outcomes declining significantly as political affiliations become more extreme on either the left or right.

Scientists expressing any political stance are viewed as less credible than neutral ones (Panel A), supporting the stereotype that scientists should remain impartial (Altenmüller et al., 2024). Relative to neutral scientists, Strong Republicans are rated 39% less credible and Strong Democrats 11% less. Moderate Republicans and Moderate Democrats incur smaller penalties: 9% and 7%, respectively. An identical pattern holds for research credibility, indicating that political discourse impacts both personal and scientific credibility.

Similarly, willingness to read declines monotonically: respondents are 42% less willing to read from Strong Republicans and 10.7% less willing to read from Strong Democrats. Moderate Republicans and Moderate Democrats see more limited declines of 8% and 4.5%, respectively. All results remain robust when controlling for respondent characteristics (Appendix Figure A.13, Panels C and D).

A model of conformity like Bernheim (1994) proposes a mechanism that rationalizes why the scientists who engage in political communication experience a credibility penalty. In Bernheim (1994)'s model of conformity, individuals care directly about social status, which depends on public perceptions of their underlying predispositions, and small departures from a homogeneous norm incur smoothly increasing but endogenous status penalties. Interpreting scientific neutrality as such a norm, the expression of partisan stances constitutes a departure that signals bias and thus lowers status.

While political affiliation is the most salient attribute, a few additional characteristics also marginally affect public perceptions. Scientists from prestigious institutions (e.g., Harvard, UC Berkeley, UChicago) are rated as more credible and attract higher readership than those from less prestigious ones (e.g., University of Arkansas, University of Connecticut), as shown in Appendix Figure A.13 and Appendix Tables B.12 to B.14. When examining each political profile separately, the benefit of a prestigious institutional affiliation holds for all except the neutral academic, and it even results in a penalty for Strong Democrat profiles.

Moreover, Full Professors are generally seen as more credible and are more likely to be read than Assistant Professors.

Other attributes are never statistically significant. Specifically, gender does not appear to affect credibility—though neutral male scientists are (non-significantly but consistently) less likely to be read than their female counterparts, partly consistent with findings in Sievertsen and Smith (2025).

4.2.1. *Heterogeneity by Respondent Leaning*

To determine whether scientists' online political expression may contribute to polarizing audience views, we analyzed how perceptions of different scientist profiles vary based on respondents' political leanings. Panel B of Figure 9 reveals significant heterogeneity tied to political identity.

Respondents identifying as *Democrat* or *leaning Democrat* assign substantially lower credibility to the Republican-leaning scientists: the Moderate Republican scientist faces an 18.3% credibility penalty and the Strong Republican a 60% penalty (Panel B, dark blue), with willingness-to-read dropping by 23.2% and 69%, respectively (light blue). These respondents view the Neutral and Democrat-leaning scientists similarly.

Conversely, respondents identifying as *Republican* or *leaning Republican* rate the Democrat-leaning scientists as 16% (Moderate) to 26% (Strong) less credible (Panel B, dark red), and are 18%–30% less willing to read their opinions (light red). Notably, they view the Moderate Republican scientist as 3.1% more credible than the Neutral one and are 11.6% more willing to read them. While the Strong Republican scientist is rated 12% lower in credibility than the Neutral one, they remain more credible than the Democrat-leaning scientists, with only a 7% drop in willingness-to-read.

Overall, the gap in willingness to engage with scientists holding opposing political identities ranges from 18%–23% for moderate stances and increases to 30%–69% for extreme ones. These results, consistent with Ajzenman et al. (2023b) and Rathje et al. (2021), show a monotonic pattern: stronger political signals lead to larger penalties among politically opposed respondents. Crucially, the drop in personal credibility extends to perceived research credibility and translates into lower willingness to engage with their content. Barrios et al. (2024) documents a comparable effect: a 30% decline in trust in an economics paper due to a conflict of interest, aligning closely with our findings.

Although the scientists' tweets themselves contain no content that would undermine their expertise or research, the starkly different reactions by Democrats versus Republicans point to in-group/out-group dynamics—where group identity drives belief updating independent of informational value (Tajfel et al., 1971, Tajfel and Turner, 2003)—as the engine of affective polarization.[32] To capture this formally, we draw on Orzach (2024)'s dynamic conformity model, in which each observer trades off learning from a sender's private signal against a "conformity" cost that rises with the squared distance between the sender's public stance and the observer's own in-group norm. In our context, Democrats treat neutrality or mild Democratic cues as the normative benchmark, and Republicans treat neutrality or mild Republican cues likewise; any stronger partisan expression thus incurs a conformity penalty. Once the conformity cost lies above the threshold for pooling equilibria, observers ignore private information entirely and "pool" on their in-group norm, resulting in monotonically larger discounts in both credibility and willingness-to-read as scientists' signals deviate further from that norm. This mechanism exactly reproduces the Panel B heterogeneity: strong out-group signals (e.g., Strong Republicans for Democratic respondents) trigger the largest penalties, moderate out-group signals incur intermediate penalties, and neutral or in-group signals face little or no discount.

4.2.2. *Robustness Checks*

We conducted several robustness checks to address potential concerns: (1) sample and procedure robustness, (2) variation in perceived expertise, (3) differential respondent attention, (4) misperception of political profiles, (5) multiple hypothesis testing, and (6) experimenter demand effects.

Replication among Twitter users. A potential concern is that the credibility penalties observed in our general-population samples might differ among active Twitter users, who are more accustomed to political content online. We therefore ran a replication with 2,000 U.S. respondents recruited on Prolific who reported using Twitter. Of these, 85% had used Twitter at least once in the past week and 36% used it daily, indicating high platform en-

[32] In-group favoritism enhances positive affect toward one's own group while fostering disdain for perceived out-groups, and polarization intensifies when group identities become central to self-concept (Iyengar and Westwood, 2015, Mason, 2018).

gagement.[33] Appendix Figure A.20 shows that the estimates are virtually unchanged relative to those in the main experiment with a general-population sample, confirming that our core findings generalize to a population more accustomed to political communication on social media.

Replication with Modified Outcomes We also replicated the main experiment with 2,000 new Prolific respondents, eliciting only two outcomes: credibility and willingness to read. Results are virtually identical, confirming robustness to procedural changes and to a new, representative U.S. sample (available upon request).

Role of Expertise As discipline is randomized across profiles, we test whether matching a scientist's expertise to the topic of their post affects perceptions (e.g., the Strong Republican may be a medical expert; the Moderate Democrat a STEM expert). Appendix Tables B.12–B.14 show no significant effect of such alignment on credibility or willingness to engage.

Carryover Effects A crucial assumption for our design is the absence of carryover effects across profiles. Such effects would imply that a respondent's evaluation of one profile is influenced by those seen before or after, introducing bias. Although unlikely due to randomization, we explicitly test this by analyzing responses within each round. Appendix Figure A.16 confirms that results are consistent across rounds, ruling out order effects.

Excluding 'Speeders' A common concern in online experiments is that some respondents rush through tasks without paying adequate attention, introducing noise. We therefore exclude respondents who completed the survey in under one minute (median time: 7.2 minutes). Appendix Figure A.17 shows results remain stable, indicating inattentive respondents are not driving our findings.

Permutation Test We further test whether the baseline 'neutral' scientist inadvertently signals political identity. We randomly reassigned labels across the five profiles seen by

[33] Because it is difficult to define representativeness for Twitter users, we set quotas for gender and political affiliation to mirror the U.S. population. As shown in Appendix Table B.10, the resulting sample is younger and more educated than the U.S. average—patterns consistent with a social media user base.

each respondent and re-estimated our main result, repeating the procedure 100 times. Appendix Figure A.18 shows the resulting coefficients cluster around zero, supporting the neutrality of the baseline.

Multiple Hypothesis Testing One advantage of a conjoint experiment is the ability to simultaneously test multiple attributes, though this requires correction for multiple hypothesis testing. We adjust for multiple comparisons using the false discovery rate method (Benjamini et al., 2006), correcting for 11 treatments (16 attribute categories minus five baselines). Appendix Table B.15 shows that estimates remain statistically significant after correction.

Experimenter Demand Our design minimizes demand effects by placing political affiliation last among randomized attributes and incentivizing truthful responses via a tailored newsletter. Yet, we bound potential residual effects by conducting a separate experiment with 354 U.S. respondents. After rating a neutral profile, subjects were shown either a Strong Republican or Strong Democrat profile. Before rating the second, they were nudged to rate it either higher or lower to induce demand (de Quidt et al., 2018). Appendix Figure A.14 shows that these nudges had little effect: neutral scientists remain the most credible and readable, while left- and right-leaning profiles incur penalties.

4.3. *Impact on Journalists Perceptions*

To assess external validity, we conducted a simplified experiment with 135 international journalists recruited via Prolific. The sample includes experienced professionals employed in various roles and outlets (50% with over 5 years of experience, 78% as reporters or editors, and over half based in the U.S. or the UK), with roughly 60% affiliated with politically oriented outlets (see detailed summary statistics in Appendix Table B.16).

In this streamlined experiment, journalists viewed three scientist profiles—signaling Strong Republican, Strong Democrat, and Neutral ideologies via Twitter biographies (without tweets).[34] They evaluated the credibility of the scientists and their research and indicated their willingness to feature an opinion piece from a scientist with such characteristics. To incentivize thoughtful responses, journalists were informed that an opinion piece from

[34] We reduced the number of profiles to account for the limited sample size while maintaining statistical power.

their highest-rated scientist would appear in a newsletter shown to 100 readers, with a monetary bonus awarded if their selected piece ranked among the top five preferred by readers.

Appendix Figure A.15 replicates our main findings in this new sample of international journalists: neutral scientists are rated highest, while ideologically non-neutral profiles incur significant penalties. Specifically, journalists perceive the Strong Republican scientist and their research as 33% and 32% less credible, respectively, and are 40% less willing to feature their opinion piece, relative to a neutral scientist (Panel A). By contrast, the Strong Democrat scientist is rated 5% less credible (4% for their research) and elicits a 2% lower willingness to feature their opinion piece; however, these smaller effects are not statistically significant. The more pronounced effects for Republican profiles are mainly driven by the liberal majority among journalists (62%): among liberal respondents, penalties for Republican scientists range from 42% to 56% (Panel B), while conservative journalists impose penalties of 18% to 28% on Democrat scientists.

We further explored factors shaping journalists' preferences by surveying their views on disclosing scientists' political leanings (when known), expectations of reader reactions to featuring politically engaged scientists, and willingness to engage with such scientists who are politically active on social media. Appendix Table B.17 shows that over half of the journalists believe a scientist's political leaning should be disclosed and that featuring politically active scientists can affect a newspaper's credibility. While they anticipate mixed reader reactions, most remain inclined to reach out to politically active scientists, suggesting that journalists may play a role in amplifying scientists' political opinions and shaping public perceptions of scientific credibility.

4.4. *Separating the Effect of Tweeting about Research on Salient Topics from Signaling Political Identity*

Scientists may share political views while discussing their research on politically charged issues or when expressing opinions unrelated to their work. We aim to disentangle whether these two types of communication have different effects on public perception. Specifically, we test whether perceived credibility is influenced by (1) tweeting about research on politically salient topics, which may signal ideological bias (Altenmüller et al., 2024), or (2) signaling political identity through content unrelated to research (Ajzenman et al., 2023b, Rathje et al., 2021, Levy, 2021). While other mechanisms may matter, our design isolates these two core, policy-relevant channels.

Experimental Design To disentangle the two channels, we added a final task immediately after the main experiment in Section 4.1. Respondents were randomly assigned to one of four conditions that varied whether an economist (who regularly discusses politically salient issues) tweets about recent research on a politically sensitive topic (vs. a non-sensitive one), and whether their Twitter bio includes a clear political signal (vs. a neutral one). Expertise is held constant across all profiles.

In the *passive control*, the economist discusses game theory research on a non-political issue. In the *active control*, the economist tweets about research on the negative impact of policies on migrants' health: a topic with a potentially left-leaning narrative.[35] The difference between active and passive control isolates the effect of politically sensitive research content on perceived ideological bias. Both profiles feature neutral Twitter bios. Two treatment arms add a political signal in the Twitter bio—either *Left* ("advocate for equality") or *Right* ("proud patriot")—while keeping the same politically salient research as in the active control (see Appendix Figure A.19). This isolates the effect of explicit political identity.

After viewing one of the four profiles, respondents rated the economist's personal and research credibility and their willingness to read an opinion piece from a similar economist. They were also invited to subscribe to a free newsletter on U.S. socio-economic issues featuring similar economists, delivered via Prolific message (see screenshot in Instructions Material). Finally, we measured respondents' general trust in scientists using three Likert items, averaged into a composite index.

Results Figure 10 shows that both channels, research content and political identity cues, influence audience perceptions, with effects varying by respondents' political leanings. Among Democrats, perceptions remain stable when the economist discusses non-political research content. However, politically aligned research increases perceived credibility by 8.6% (economist) and 6.3% (research), and raises willingness to read by 33%. Adding a left-leaning bio further increases credibility by 1.7% and willingness to read by 7.1%.

[35] The research topic was chosen to clearly align with one political side and oppose the other, facilitating interpretation. Symmetrical effects are expected with right-leaning aligned research.

By contrast, a right-leaning bio reduces credibility by 8.6%, bringing it back to the non-political baseline, and lowering willingness to read by 4.5%. Republican respondents show the opposite pattern: when the economist discusses politically misaligned research, credibility drops by 12% relative to the non-salient benchmark. This penalty increases by 5% with a left-leaning signal and is reduced by 5% with a right-leaning one. Willingness to read follows a similar trajectory.

Newsletter sign-ups and overall trust in science show similar, though less pronounced, patterns. Correlating newsletter demand and willingness to read validates the latter outcome ($\beta = 0.064$, $p < 0.001$). Recontacting respondents a few weeks later further confirms engagement: of 595 participants who expressed interest in the newsletter, 440 were successfully recontacted, and 86% clicked the newsletter link to join.

The bottom of Figure 10 presents average outcomes for treated respondents relative to either the active or passive control, allowing a clearer comparison between in-group and out-group conditions. In-group respondents (those whose political leanings match the political signal) rate scientists more favorably across all outcomes. Regression results in Appendix Tables B.18 and B.19 confirm these findings.

These results highlight two main insights. First, audience responses vary sharply by political alignment: for the same content, credibility and engagement increase when the political signal aligns with the respondent's views—clear evidence of affective polarization. Second, both channels independently shape perceptions of scientists' credibility.

4.5. *Isolating the Effect of Strategic Considerations behind Scientists' Political Expression*

Our descriptive and experimental evidence establishes a credibility penalty for scientists who express political views, but it does not identify *why* audiences penalize them. In sender–receiver settings, observers can attribute a message either to sincere beliefs or to *strategic* motives (e.g., visibility or audience-building); the attribution shapes how they update about the sender's type.

Guided by this logic, and by our survey evidence that scientists perceive visibility as a salient benefit of online communication, we designed an additional experiment to isolate whether perceived *visibility incentives* drive part of the credibility loss. The key idea is to shift perceptions of strategic intent without altering the message content or the scientist's identity. We implement this in the replication sample of Twitter users, where exposure to political content is more common, and evaluate how making visibility incentives explicit shifts credibility and related outcomes.

4.5.1. *Experimental Design*

In the second task of the Twitter-user replication we specifically implemented a 2×3 between-subjects design to test whether credibility penalties arise because audiences interpret scientists' political expression as *strategically motivated*, adapting Bursztyn et al. (2023a) to academics' political expression.

We varied the slant of the tweet shared by an economist with mid-sized reach whose expertise was described as being in immigration. All tweets reported factual statistics but were framed to suggest three ideological leanings: (i) neutral, (ii) left-leaning, or (iii) right-leaning. Crossed with this, we randomized whether respondents were told *before* viewing the tweet that the economist had been informed *ex ante* that we (the researchers) would promote the post on X/Twitter to increase its visibility ("Promotion"), or received no such information ("No Promotion"). This manipulation directly links political posting to perceived strategic motives.[36]

Consistent with the survey evidence in Section 3, the *Promotion* manipulation made salient that the economist's behavior could be interpreted as visibility-seeking, potentially altering respondents' evaluations of the scientist. Specifically, outcomes included: (i) credibility of the economist (0–10), (ii) credibility of the economist's research (0–10), (iii) willingness to read a related opinion piece (0–10), (iv) perceived political bias of the economist

[36] In the *Promotion* condition, subjects read: "*You are matched with a (migration expert) economist who posted the following factual tweet on their personal social media account. Before writing and sharing the post, the economist was informed that we (the researchers) would promote the post to increase its visibility and reach. Promoted posts on X/Twitter are displayed to a broader audience beyond the economist's current followers. Visibility typically extends to users with similar interests (such as other academics, policy professionals, or journalists) based on engagement patterns and topical relevance. Tweet: . . .* ".

In the *No Promotion* condition, respondents read only: "*You are matched with a (migration expert) economist who posted the following factual tweet on their personal social media account. Tweet: . . .* ", with all sentences about promotion omitted. Further details in the Instructions Material.

(-5 = strongly left to +5 = strongly right), (v) willingness to subscribe to a newsletter of similar economists (Yes/No), and (vi) an index of trust in science, computed as the average across three Likert-scale items. In addition, we elicited second-order beliefs about perceived reputational costs: respondents estimated how many out of 100 Twitter users would rate the economist as highly credible, somewhat credible, neutral, somewhat not credible, or not at all credible.

We preregistered three possible patterns to identify the operative mechanism:

1. *No promotion effect.* If promotion has no effect across promotion conditions, respondents already account for strategic motives when evaluating political expression.
2. *Motivated reasoning.* If promotion *selectively* affects credibility—reducing it when the tweet's leaning is misaligned with respondents' partisanship and increasing it when aligned—this indicates partisan interpretation of strategic motives.
3. *Opportunism.* If promotion reduces credibility *uniformly* across leanings, this suggests a general aversion to perceived opportunism.

Several design choices deserve mention. To balance realism and ethics, we asked three economists to post the tweets on Twitter and then promptly deleted them after promotion, while ensuring that their identities were never revealed to respondents. The tweets contained only *verifiable factual information*, so that participants were not exposed to misinformation; the wording differed only in framing, and respondents were explicitly told that the content was truthful.[37] We omitted explicit source references in the survey to avoid confounding reactions, but separately elicited respondents' judgments of the *credibility of the information* itself.

4.5.2. *Results*

We report four main findings: (i) the slant manipulation worked and promotion did not change perceived ideology; (ii) promotion alters perceived credibility depending on partisan alignment; (iii) the Trust in Science Index shifts in the same direction; and (iv) second-order beliefs about how others would judge the scientist show no systematic change under promotion.

[37] See the exact posts in Appendix Figure A.21. The links to the three facts used to portray a neutral narrative versus a left- or right-leaning one can be found here: neutral, left-wing, and right-wing.

Manipulation check. We first verify that the *slant* manipulation worked as intended. Respondents who viewed the left- (right-) leaning tweet rated the economist as more left- (right-) biased than those who viewed the neutral tweet. Importantly, adding the information on *Promotion* did not shift perceived ideological bias, indicating that the promotion treatment does not confound the slant manipulation (see Bias columns in Figure 11 and Appendix Table B.20).

Promotion effects on credibility. We next test whether making *Promotion* salient (i.e., explicitly informing respondents that the economist had been told *before* posting that *we, the researchers* would promote their tweet) alters perceptions of the scientist. Overall, we find no evidence of a uniform penalty from promotion across conditions, providing no support for the opportunism hypothesis. Effects hinge on partisan alignment. Among Democrats, the promotion information magnifies partisan reactions: relative to the corresponding non-promotion cells, the scientist's perceived credibility is higher when the tweet is left-leaning and lower when it is right-leaning. Neutral promotion is essentially null among Democrats. Among Republicans, the promotion information raises the scientist's perceived credibility for right-leaning tweets and attenuates the penalty for left-leaning tweets; by contrast, making promotion salient for the neutral tweet is associated with a credibility penalty. Overall, Democrats' responses are consistent with motivated reasoning, whereas Republicans' responses contradict a generalized opportunism story and suggest selective approval of promotion for ideological messages but a dislike of promotion when the message is neutral (see Credibility columns in Figure 11 and Appendix Table B.20).

Promotion effects on trust in science. Making *Promotion* salient also shifts the Trust in Science Index, but asymmetrically across respondents' political leanings. Among Democrats, general trust increases when the economist posts a left-leaning tweet under promotion, with a smaller positive effect for the neutral message; promotion of right-leaning tweets has essentially no effect. Among Republicans, promotion increases general trust for both left- and right-leaning tweets, while neutral promotion has no impact (see Trust in Science Idx columns in Figure 11 and Appendix Table B.20). These patterns suggest that making our intent to promote the message salient can be read by some audiences as constructive scientific engagement, broadly so among Republicans, and selectively (when aligned) among Democrats, consistent with evidence on divergent normative views of scientists' roles in society and policymaking (Cologna et al., 2025, Capozza et al., 2025).

Second-order beliefs about reputational costs. We also tested whether making *Promotion* salient changes respondents' *second-order* beliefs about how others would judge the economist. After viewing the tweet, respondents allocated 100 hypothetical Twitter users across credibility categories; we analyze the expected share who would find the economist "not credible." Across slant and party cells, differences between promotion and no-promotion are small and imprecise, with confidence intervals overlapping zero (Appendix Table B.20 columns "Other find Not Credible"). We therefore find no systematic evidence that the promotion information increases the perceived reputational costs of posting, alleviating concerns that our credibility effects are confounded by beliefs about how others would react.

In summary, this second task isolates the role of strategic considerations by making the scientists' *visibility incentive* to promote their tweets explicit. Receivers update on this information, but not uniformly. Among Democrats, the promotion information amplifies the experiment pattern, raising the credibility gap between authors of aligned and misaligned messages, consistent with a selective attribution of penalty to strategic motives for out-group signals. Among Republicans, the promotion information closes the credibility gap from the main experiment: it boosts credibility for authors of aligned messages and attenuates the penalty for authors of misaligned ones, contradicting a generalized "opportunism tax." In parallel, trust in science moves positively under promotion (aligned among Democrats; broadly among Republicans). Overall, audiences understand and incorporate scientists' incentives, but updating depends on partisan congruence and normative expectations about scientists' public engagement rather than a universal penalty for strategic behavior.

5. CONCLUSION

This study demonstrates that scientists' public expression of political views on social media significantly shapes public perceptions of their credibility.

First, we show that political discourse is common among U.S. academics: between 2016 and 2022, roughly 44% of 97,737 scientists actively discussed political issues on Twitter—over six times the rate observed among a random sample of U.S. Twitter users. Topics like climate change and racial equality are frequently discussed, often with divergent viewpoints. While some of these posts convey verifiable facts, a majority contain no checkable claims and are best characterized as opinion-based commentary. This distinction is important for understanding how such discourse may be perceived by the public and contribute to assessments of scientists' credibility.

Second, using conjoint experiments with 6,000 U.S. representative respondents and 135 international journalists rating synthetic academic profiles with varied political affiliations, we identify a monotonic credibility penalty for scientists who express political views. The more extreme the position, on either left or right, the less credible the scientist and their research are perceived, and the lower the willingness to engage with their content, especially among ideologically opposed respondents. A follow-on mechanism test indicates that perceived *visibility incentives* help explain the credibility penalty.

Complementary surveys of 128 international scientists reveal that they expect an even larger credibility penalty than what we observe experimentally. Scientists broadly view political expression as acceptable when related to their expertise, but not beyond it—reflecting a prevailing norm within academia. At the same time, they recognize potential professional benefits from political engagement, particularly when aligned with their research.

Our findings highlight a significant challenge to the Mertonian norm of Universalism (Merton, 1973), which holds that scientific work should be judged on its merits rather than on the personal attributes or beliefs of its producers. In contemporary digital environments, however, political cues—especially when salient and easily interpretable—systematically shape public perceptions of credibility and engagement, independently of their substantive quality.

These results carry important policy implications. Public trust in science is central to evidence-based policymaking, yet political expression by scientists can undermine this trust, particularly in polarized settings. At the same time, discouraging public engagement would risk narrowing the flow of expertise into public debate. The relevant policy challenge is therefore not whether scientists should speak, but how the institutional and digital contexts in which they speak shape the trade-off between visibility and credibility.

Institutions can play a constructive role by clarifying norms around public communication, investing in media and digital literacy, and supporting scientists in distinguishing between evidence-based communication and low-information political signaling. While negative reactions to research communication on politically charged topics may be unavoidable, explicit partisan cues—such as political identifiers in online profiles—appear especially costly, offering little informational value while narrowing audiences and intensifying affective polarization. More broadly, platform design matters: engagement-driven algorithms may amplify precisely those political signals that carry the largest credibility penalties. Preserving trust in science thus requires attention not only to individual behavior, but also to the digital environments in which scientific communication is disseminated and interpreted.

REFERENCES

AINA, CHIARA (2023): "Tailored Stories," *Mimeo*. [2]

AJZENMAN, NICOLÁS, BRUNO FERMAN, AND PEDRO C. SANT'ANNA (2023a): "Discrimination in the Formation of Academic Networks: a Field Experiment on Econtwitter," *IZA Discussion Paper No. 15878*. [9, 24]

AJZENMAN, NICOLAS, BRUNO FERMAN, AND PEDRO C. SANT'ANNA (2023b): "Rooting for the Same Team: Shared Social Identities in a Polarized Context," *SSRN Working Paper No. 4326148*. [9, 24, 28, 32]

AKERLOF, GEORGE A. AND RACHEL E. KRANTON (2000): "Economics and Identity," *The Quarterly Journal of Economics*, 115 (3), 715–753. [9]

ALABRESE, ELEONORA (2022): "Bad Science: Retractions and Media Coverage," *CESifo Working Paper No. 10195*. [5, 10]

ALBORNOZ, FACUNDO, NICOLAS BOTTAN, GUILLERMO CRUCES, BRIDGET HOFFMANN, AND MARÍA LOMBARDI (2024): "Backlash Against Expert Recommendations: Reactions to COVID-19 Advice in Latin America," *Journal of Economic Behavior & Organization*, 228, 106752. [1]

ALESINA, ALBERTO, ARMANDO MIANO, AND STEFANIE STANTCHEVA (2020): "The Polarization of Reality," *AEA Papers and Proceedings*, 110, 324–328. [9]

ALFORD, JOHN R., PETER K. HATEMI, JOHN R. HIBBING, NICHOLAS G. MARTIN, AND LINDON J. EAVES (2011): "The Politics of Mate Choice," *The Journal of Politics*, 73 (2), 362–379. [9]

ALGAN, YANN, DANIEL COHEN, EVA DAVOINE, MARTIAL FOUCAULT, AND STEFANIE STANTCHEVA (2021): "Trust in scientists in times of pandemic: Panel evidence from 12 countries," *Proceedings of the Academy of Sciences*, 118 (40), 1–8. [1, 2]

ALLCOTT, HUNT, LEVI BOXELL, JACOB CONWAY, MATTHEW GENTZKOW, MICHAEL THALER, AND DAVID YANG (2020): "Polarization and public health: Partisan differences in social distancing during the coronavirus pandemic," *Journal of public economics*, 191, 104254. [1]

ALTENMÜLLER, MARLENE S., TOBIAS WINGEN, AND ANNA SCHULTE (2024): "Explaining Polarized Trust in Scientists: a Political Stereotype-approach," *Science Communication*, 46 (1), 92–115. [1, 8, 10, 17, 27, 32]

Angeli, Deivis, Matt Lowe, et al. (2022): "Virtue Signals," *Working Paper*. [9]

Angelucci, Charles and Andrea Prat (2024): "Is Journalistic Truth Dead? Measuring How Informed Voters Are About Political News," *American Economic Review*, 114 (4), 887–925. [1]

Arold, Benjamin W. (2024): "Evolution Vs. Creationism in the Classroom: the Lasting Effects of Science Education," *The Quarterly Journal of Economics*, 2331–2375. [1]

Ash, Elliott (2016): "The Political Economy of Tax Laws in the US States," *Working Paper*. [8]

Ash, Elliott, Daniel L Chen, and Suresh Naidu (2025): "Ideas Have Consequences: The Impact of Law and Economics on American Justice," *The Quarterly Journal of Economics*, qjaf042. [1]

Ash, Elliott, Sergio Galletta, Dominik Hangartner, Yotam Margalit, and Matteo Pinna (2024): "The Effect of Fox News on Health Behavior During Covid-19," *Political Analysis*, 32 (2), 275–284. [1]

Azevedo, Flávio and John T. Jost (2021): "The Ideological Basis of Antiscientific Attitudes: Effects of Authoritarianism, Conservatism, Religiosity, Social Dominance, and System Justification," *Group Processes & Intergroup Relations*, 24 (4), 518–549. [1]

Barberá, Pablo (2020): "Social Media, Echo Chambers, and Political Polarization," *Social Media and Democracy: the State of the Field, Prospects for Reform*, 34. [2]

Barrios, John, Filippo Maria Lancieri, Joshua Levy, Shashank Singh, Tommaso M Valletti, and Luigi Zingales (2024): "The conflict-of-interest discount in the marketplace of ideas," Tech. rep., New Working Paper Series. [28]

Bartoš, Vojtěch, Michal Bauer, Jana Cahlíková, and Julie Chytilová (2022): "Communicating Doctors' Consensus Persistently Increases Covid-19 Vaccinations," *Nature*, 606 (7914), 542–549. [1]

Beknazar-Yuzbashev, George, Rafael Jiménez Durán, Jesse McCrosky, and Mateusz Stalinski (2022): "Toxic Content and User Engagement on Social Media: Evidence From a Field Experiment," *SSRN Working Paper No. 4307346*. [9]

Bénabou, Roland and Jean Tirole (2006): "Incentives and Prosocial Behavior," *American Economic Review*, 96 (5), 1652–1678. [18]

Benjamini, Yoav, Abba M. Krieger, and Daniel Yekutieli (2006): "Adaptive Linear Step-up Procedures That Control the False Discovery Rate," *Biometrika*, 93 (3), 491–507. [31]

Bernheim, B Douglas (1994): "A theory of conformity," *Journal of Political Economy*, 102 (5), 841–877. [5, 27]

Blastland, Michael, Alexandra LJ Freeman, Sander van der Linden, Theresa M. Marteau, and David Spiegelhalter (2020): "Five Rules for Evidence Communication," *Nature*, 587 (7834), 362–364. [8]

Boken, Johannes, Mirko Draca, Nicola Mastrorocco, and Arianna Ornaghi (2023): "The Returns to Viral Media: the Case of Us Campaign Contributions," *Warwick Economics Research Papers Series No. 1472*. [2]

Borjas, George J and Nate Breznau (2024): "Ideological Bias in Estimates of the Impact of Immigration," *National Bureau of Economic Research*, (w33274). [1]

BOWLES, JEREMY, KEVIN CROKE, HORACIO LARREGUY, JOHN MARSHALL, AND SHELLEY LIU (2023): "Sustaining Exposure to Fact-checks: Misinformation Discernment, Media Consumption, and Its Political Implications," *SSRN Working Paper No. 4582703*. [1]

BOXELL, LEVI AND JACOB CONWAY (2022): "Journalist ideology and the production of news: Evidence from movers," *Available at SSRN 4295587*. [5]

BOXELL, LEVI, MATTHEW GENTZKOW, AND JESSE M. SHAPIRO (2024): "Cross-country Trends in Affective Polarization," *The Review of Economics and Statistics*, 106 (2), 557–565. [9]

BRAGHIERI, LUCA (2024): "Political Correctness, Social Image, and Information Transmission," *American Economic Review*, 114 (12), 3877–3904. [8]

BRAGHIERI, LUCA, PETER SCHWARDMANN, AND EGON TRIPODI (2024): "Talking Across the Aisle," *Mimeo*. [9, 24]

BRAÑAS-GARZA, PABLO, LORENZO ESTEPA-MOHEDANO, DIEGO JORRAT, VICTOR OROZCO, AND ERICKA RASCÓN-RAMÍREZ (2021): "To Pay or Not to Pay: Measuring Risk Preferences in Lab and Field," *Judgment and Decision Making*, 16 (5), 1290–1313. [25]

BRAÑAS-GARZA, PABLO, DIEGO JORRAT, ANTONIO M. ESPÍN, AND ANGEL SÁNCHEZ (2023): "Paid and Hypothetical Time Preferences Are the Same: Lab, Field and Online Evidence," *Experimental Economics*, 26 (2), 412–434. [25]

BROWN, JACOB R., ENRICO CANTONI, RYAN D. ENOS, VINCENT PONS, AND EMILIE SARTRE (2022): "The Increase in Partisan Segregation in the United States," *NICEP Research Discussion paper 2023-09*. [9]

BROWN, MATTHEW AND HARSH GUPTA (2023): "The role of political ideology in media coverage of science," *Working Paper*. [8]

BURNITT, CHRISTOPHER, JARRETT GARS, AND MATEUSZ STALINSKI (2024): "The Politics of Food," *Mimeo*. [2, 9]

BURSZTYN, LEONARDO, MICHAEL CALLEN, BRUNO FERMAN, SAAD GULZAR, ALI HASANAIN, AND NOAM YUCHTMAN (2020a): "Political Identity: Experimental Evidence on Anti-americanism in Pakistan," *Journal of the European Economic Association*, 18 (5), 2532–2560. [9]

BURSZTYN, LEONARDO, GEORGY EGOROV, INGAR HAALAND, AAKAASH RAO, AND CHRISTOPHER ROTH (2023a): "Justifying Dissent," *The Quarterly Journal of Economics*, 138 (3), 1403–1451. [7, 8, 24, 35]

BURSZTYN, LEONARDO, ALESSANDRA L. GONZÁLEZ, AND DAVID YANAGIZAWA-DROTT (2020b): "Misperceived Social Norms: Women Working Outside the Home in Saudi Arabia," *American Economic Review*, 110 (10), 2997–3029. [22]

BURSZTYN, LEONARDO, AAKAASH RAO, CHRISTOPHER ROTH, AND DAVID YANAGIZAWA-DROTT (2023b): "Opinions As Facts," *The Review of Economic Studies*, 90 (4), 1832–1864. [1]

BURSZTYN, LEONARDO AND DAVID Y. YANG (2022): "Misperceptions About Others," *Annual Review of Economics*, 14 (1), 425–452. [22]

CAGÉ, JULIA, NICOLAS HERVÉ, AND MARIE-LUCE VIAUD (2020): "The Production of Information in an Online World," *The Review of Economic Studies*, 87 (5), 2126–2164. [8]

CALÓNICO, SEBASTIAN, RAFAEL DI TELLA, AND JUAN CRUZ LOPEZ DEL VALLE (2023): "The Political Economy of a "Miracle Cure": the Case of Nebulized Ibuprofen and Its Diffusion in Argentina," *National Bureau of Economic Research No. w31781*. [1]

CAMPBELL, ERYN, JOHN KOTCHER, TERESA MYERS, JOHN COOK, AMANDA C. BORTH, AND EDWARD MAIBACH (2023): "What Motivates People to Defend Science: Evidence From the 2017 March for Science," *Plos One*, 18 (11), e0290222. [3]

CANEN, NATHAN, CHAD KENDALL, AND FRANCESCO TREBBI (2020): "Unbundling polarization," *Econometrica*, 88 (3), 1197–1233. [9]

CANEN, NATHAN J., CHAD KENDALL, AND FRANCESCO TREBBI (2021): "Political Parties As Drivers of Us Polarization: 1927-2018," Tech. rep., National Bureau of Economic Research. [9]

CAPOZZA, FRANCESCO, KRISHNA SRINIVASAN, AND MATTIE TOMA (2025): "Science by Consensus: Eliciting Citizens' and Experts' R&D Spending Priorities," *Working Paper*. [38]

CHAKRABORTY, ARCHISHMAN, PARIKSHIT GHOSH, AND JAIDEEP ROY (2020): "Expert-captured Democracies," *American Economic Review*, 110 (6), 1713–1751. [4]

CHAN, ALEX (2022): "Discrimination Against Doctors: a Field Experiment," *Working Paper*. [4]

CHAN, HO FAI, ALI SINA ÖNDER, SASCHA SCHWEITZER, AND BENNO TORGLER (2023): "Twitter and Citations," *Economics Letters*, 231, 111270. [2, 4]

CHARNESS, GARY AND YAN CHEN (2020): "Social Identity, Group Behavior, and Teams," *Annual Review of Economics*, 12 (1), 691–713. [9]

CHIANG, CHUN-FANG AND BRIAN KNIGHT (2011): "Media bias and influence: Evidence from newspaper endorsements," *The Review of economic studies*, 78 (3), 795–820. [2]

CHINOY, SAHIL AND MARTIN KOENEN (2024): "Political Sorting in the US Labor Market: Evidence and Explanations," *Working Paper*. [9]

CHOPRA, FELIX, INGAR HAALAND, AND CHRISTOPHER ROTH (2022): "Do People Demand Fact-checked News? Evidence From US Democrats," *Journal of Public Economics*, 205, 104549. [6]

——— (2024): "The demand for news: Accuracy concerns versus belief confirmation motives," *The Economic Journal*, 134 (661), 1806–1834. [6, 8, 9]

COLLEONI, ELANOR, ALESSANDRO ROZZA, AND ADAM ARVIDSSON (2014): "Echo Chamber or Public Sphere? Predicting Political Orientation and Measuring Political Homophily in Twitter Using Big Data," *Journal of Communication*, 64 (2), 317–332. [9]

COLOGNA, VIKTORIA, NIELS MEDE, SEBASTIAN BERGER, JOHN BESLEY, CAMERON BRICK, MARINA JOUBERT, EDWARD MAIBACH, SABINA MIHELJ, NAOMI ORESKES, MIKE SCHÄFER, ET AL. (2025): "Trust in Scientists and Their Role in Society Across 68 Countries," *Nature Human Behaviour*. [1, 38]

COLONNELLI, EMANUELE, BO LI, AND ERNEST LIU (2024): "Investing with the government: A field experiment in China," *Journal of Political Economy*, 132 (1), 248–294. [25]

CULLEN, ZOË, WILL DOBBIE, AND MITCHELL HOFFMAN (2023): "Increasing the demand for workers with a criminal record," *The Quarterly Journal of Economics*, 138 (1), 103–150. [25]

DE QUIDT, JONATHAN, JOHANNES HAUSHOFER, AND CHRISTOPHER ROTH (2018): "Measuring and Bounding Experimenter Demand," *American Economic Review*, 108 (11), 3266–3302. [5, 31]

DOUGLAS, KAREN M. (2021): "Are Conspiracy Theories Harmless?" *The Spanish Journal of Psychology*, 24, e13. [1]

DRACA, MIRKO AND CARLO SCHWARZ (2024): "How Polarised Are Citizens? Measuring Ideology From the Ground Up," *The Economic Journal*, 134 (661), 1950–1984. [8]

DRUCKMAN, JAMES N. AND MARY C. MCGRATH (2019): "The Evidence for Motivated Reasoning in Climate Change Preference Formation," *Nature Climate Change*, 9 (2), 111–119. [1]

EICHENGREEN, BARRY, CEVAT GIRAY AKSOY, AND ORKUN SAKA (2021): "Revenge of the Experts: Will Covid-19 Renew or Diminish Public Trust in Science?" *Journal of Public Economics*, 193, 104343. [1]

ELÍAS, JULIO J., NICOLA LACETERA, AND MARIO MACIS (2019): "Paying for Kidneys? a Randomized Survey and Choice Experiment," *American Economic Review*, 109 (8), 2855–2888. [4]

ENKE, BENJAMIN, THOMAS GRAEBER, AND RYAN OPREA (2023): "Confidence, Self-selection, and Bias in the Aggregate," *American Economic Review*, 113 (7), 1933–1966. [24]

ENKE, BENJAMIN, RICARDO RODRIGUEZ-PADILLA, AND FLORIAN ZIMMERMANN (2022): "Moral Universalism: Measurement and Economic Relevance," *Management Science*, 68 (5), 3590–3603. [25]

ESTEBAN, JOAN-MARIA AND DEBRAJ RAY (1994): "On the Measurement of Polarization," *Econometrica*, 62 (4), 819–851. [18]

EXLEY, CHRISTINE L, RAYMOND FISMAN, JUDD B KESSLER, LOUIS-PIERRE LEPAGE, XIAOMENG LI, CORINNE LOW, XIAOYUE SHAN, MATTIE TOMA, AND BASIT ZAFAR (2024): "Information-optional policies and the gender concealment gap," Tech. rep., National Bureau of Economic Research. [25]

FAIA, ESTER, ANDREAS FUSTER, VINCENZO PEZONE, AND BASIT ZAFAR (2024): "Biases in information selection and processing: Survey evidence from the pandemic," *Review of Economics and Statistics*, 106 (3), 829–847. [8]

FETZER, THIEMO, JOHN PETER LAMBERT, BENNET FELD, AND PRASHANT GARG (2024): "AI-Generated Production Networks: Measurement and Applications to Global Trade," Discussion Paper 19708, CEPR Press, Paris & London. [8]

FIORINA, MORRIS P. AND SAMUEL J. ABRAMS (2008): "Political Polarization in the American Public," *Annual Review Political Science*, 11, 563–588. [9]

FIORINA, MORRIS P., J. ABRAMS SAMUEL, AND C. POPE JEREMY (2005): *Culture War? the Myth of a Polarized America*, New York: Pearson Longman. [20]

FLAXMAN, SETH, SHARAD GOEL, AND JUSTIN M. RAO (2016): "Filter Bubbles, Echo Chambers, and Online News Consumption," *Public Opinion Quarterly*, 80 (S1), 298–320. [9]

FUNK, CARY, ALEC TYSON, BRIAN KENNEDY, AND COURTNEY JOHNSON (2020): "Science and Scientists Held in High Esteem Across Global Publics," *Pew Research Center*, 29. [1]

GARCIA-HOMBRADOS, JORGE, MARCEL JANSEN, Á.NGEL MARTÍNEZ, BERKAY ÖZCAN, PEDRO REY-BIEL, AND ANTONIO ROLDÁN-MONÉS (2024): "Ideological Alignment and Evidence-based Policy Adoption," *IZA Discussion Paper No. 17007*. [9]

GARG, PRASHANT AND THIEMO FETZER (2025a): "Causal claims in economics," *arXiv preprint arXiv:2501.06873*. [8]

——— (2025b): "Political expression of academics on Twitter," *Nature Human Behaviour*. [2, 3, 8, 11, 12, 23]

GENTZKOW, MATTHEW AND JESSE M. SHAPIRO (2006): "Media Bias and Reputation," *Journal of Political Economy*, 114 (2), 280–316. [6]

——— (2011): "Ideological Segregation Online and Offline," *The Quarterly Journal of Economics*, 126 (4), 1799–1839. [9]

GENTZKOW, MATTHEW, JESSE M. SHAPIRO, AND MATT TADDY (2019): "Measuring Group Differences in High-dimensional Choices: Method and Application to Congressional Speech," *Econometrica*, 87 (4), 1307–1340. [8]

GENTZKOW, MATTHEW, MICHAEL B WONG, AND ALLEN T ZHANG (2023): "Ideological Bias and Trust in Information Sources," *Working Paper*. [8]

GOLDSTEIN, YOAV AND MATAN KOLERMAN (2025): "Changing Minds: How Academic Fields Shape Political Attitudes," *Available at SSRN 5196889*. [2]

GRIMMER, JUSTIN (2010): "A Bayesian Hierarchical Topic Model for Political Texts: Measuring Expressed Agendas in Senate Press Releases," *Political Analysis*, 18 (1), 1–35. [8]

GROSSMAN, GENE M. AND ELHANAN HELPMAN (2021): "Identity Politics and Trade Policy," *The Review of Economic Studies*, 88 (3), 1101–1126. [9]

GURIEV, SERGEI, EMERIC HENRY, THÉO MARQUIS, AND EKATERINA ZHURAVSKAYA (2023): "Curtailing False News, Amplifying Truth," *SSRN Working Paper No. 4616553*. [9]

HAINMUELLER, JENS, DOMINIK HANGARTNER, AND TEPPEI YAMAMOTO (2015): "Validating Vignette and Conjoint Survey Experiments Against Real-world Behavior," *Proceedings of the National Academy of Sciences*, 112 (8), 2395–2400. [24, 25, 26]

HANSEN, STEPHEN, MICHAEL MCMAHON, AND ANDREA PRAT (2018): "Transparency and Deliberation Within the Fomc: a Computational Linguistics Approach," *The Quarterly Journal of Economics*, 133 (2), 801–870. [8]

HARTIGAN, JOHN A. AND PAMELA M. HARTIGAN (1985): "The Dip Test of Unimodality," *The Annals of Statistics*, 13 (1), 70–84. [18]

HENDRIKS, FRIEDERIKE, DOROTHE KIENHUES, AND RAINER BROMME (2020): "Replication Crisis= Trust Crisis? the Effect of Successful vs Failed Replications on Laypeople's Trust in Researchers and Research," *Public Understanding of Science*, 29 (3), 270–288. [1]

HOPKINS, DANIEL J. (2018): *The Increasingly United States: How and Why American Political Behavior Nationalized*, University of Chicago Press. [20]

HUANG, YIHONG AND YUEN HO (2023): "Breaking the Spiral of Silence," *Mimeo*. [8]

HUBER, GREGORY A. AND NEIL MALHOTRA (2017): "Political Homophily in Social Relationships: Evidence From Online Dating Behavior," *The Journal of Politics*, 79 (1), 269–283. [9]

IYENGAR, SHANTO, YPHTACH LELKES, MATTHEW LEVENDUSKY, NEIL MALHOTRA, AND SEAN J. WESTWOOD (2019): "The Origins and Consequences of Affective Polarization in the United States," *Annual Review of Political Science*, 22, 129–146. [9]

IYENGAR, SHANTO AND SEAN J WESTWOOD (2015): "Fear and loathing across party lines: New evidence on group polarization," *American Journal of Political Science*, 59 (3), 690–707. [29]

JELVEH, ZUBIN, BRUCE KOGUT, AND SURESH NAIDU (2024): "Political Language in Economics," *The Economic Journal*, 134 (662), 2439–2469. [8]

JENSEN, JACOB, SURESH NAIDU, ETHAN KAPLAN, LAURENCE WILSE-SAMSON, DAVID GERGEN, MICHAEL ZUCKERMAN, AND ARTHUR SPIRLING (2012): "Political Polarization and the Dynamics of Political Language: Evidence From 130 Years of Partisan Speech," *Brookings Papers on Economic Activity*, 1–81. [8]

JIMÉNEZ DURÁN, RAFAEL (2022): "The Economics of Content Moderation: Evidence From Hate Speech on Twitter," *Stigler Center for the Study of the Economy the State Working Paper No. 324*. [9]

KAHAN, DAN M., HANK JENKINS-SMITH, AND DONALD BRAMAN (2011): "Cultural Cognition of Scientific Consensus," *Journal of Risk Research*, 14 (2), 147–174. [9]

KASHNER, DANIEL AND MATEUSZ STALINSKI (2024): "Preempting polarization: An experiment on opinion formation," *Journal of Public Economics*, 234, 105122. [24]

KESSLER, JUDD B, CORINNE LOW, AND XIAOYUE SHAN (2024): "Lowering the playing field: Discrimination through sequential spillover effects," *Review of Economics and Statistics*, 1–28. [25]

KESSLER, JUDD B., CORINNE LOW, AND COLIN D. SULLIVAN (2019): "Incentivized Resume Rating: Eliciting Employer Preferences Without Deception," *American Economic Review*, 109 (11), 3713–3744. [25]

KIM, WOOJIN (2024): "Political polarization in medicine," *Working Paper*. [9]

KLAR, SAMARA, YANNA KRUPNIKOV, JOHN BARRY RYAN, KATHLEEN SEARLES, AND YOTAM SHMARGAD (2020): "Using Social Media to Promote Academic Research: Identifying the Benefits of Twitter for Sharing Academic Work," *PLoS One*, 15 (4), e0229446. [2, 4]

KOTCHER, JOHN E., TERESA A. MYERS, EMILY K. VRAGA, NEIL STENHOUSE, AND EDWARD W. MAIBACH (2017): "Does Engagement in Advocacy Hurt the Credibility of Scientists? Results from a Randomized National Survey Experiment," *Environmental Communication*, 11 (3), 415–429. [8]

LELKES, YPHTACH (2016): "Mass Polarization: Manifestations and Measurements," *Public Opinion Quarterly*, 80 (S1), 392–410. [2]

LEVY, RO'EE (2021): "Social Media, News Consumption, and Polarization: Evidence From a Field Experiment," *American Economic Review*, 111 (3), 831–870. [9, 33]

LI, NAN AND YACHAO QIAN (2022): "Polarization of Public Trust in Scientists Between 1978 and 2018: Insights From a Cross-decade Comparison Using Interpretable Machine Learning," *Politics and the Life Sciences*, 41 (1), 45–54. [1]

LIST, JOHN A. AND DAVID LUCKING-REILEY (2002): "The Effects of Seed Money and Refunds on Charitable Giving: Experimental Evidence From a University Capital Campaign," *Journal of Political Economy*, 110 (1), 215–233. [4]

LOW, CORINNE (2014): "Pricing the Biological Clock: Reproductive Capital on the US Marriage Market," *Wharton School of Business 17*. [4, 25]

LUPIA, ARTHUR, DAVID B. ALLISON, KATHLEEN HALL JAMIESON, JENNIFER HEIMBERG, MAGDALENA SKIPPER, AND SUSAN M. WOLF (2024): "Trends in US public confidence in science and opportunities for progress," *Proceedings of the National Academy of Sciences*, 121 (11), e2319488121. [1]

LUU, SON T, HIEP NGUYEN, TRUNG VO, AND LE-MINH NGUYEN (2024): "ZeFaV: Boosting Large Language Models for Zero-Shot Fact Verification," in *Pacific Rim International Conference on Artificial Intelligence*, Springer, 288–295. [16]

MACCHI, ELISA (2023): "Worth your weight: experimental evidence on the benefits of obesity in low-income countries," *American Economic Review*, 113 (9), 2287–2322. [4, 25]

MALEKI, ASHRAF AND KIM HOLMBERG (2023): "Who Are Tweeting About Academic Publications? A Cochrane Systematic Review and Meta-Analysis of Altmetric Studies," *arXiv preprint arXiv:2312.06399*. [1, 10]

MARTINEZ-BRAVO, MONICA AND ANDREAS STEGMANN (2022): "In Vaccines We Trust? the Effects of the CIA's Vaccine Ruse on Immunization in Pakistan," *Journal of the European Economic Association*, 20 (1), 150–186. [2]

MASON, LILLIANA (2018): "Ideologues without issues: The polarizing consequences of ideological identities," *Public Opinion Quarterly*, 82 (S1), 866–887. [29]

MASSEY JR, FRANK J. (1951): "The Kolmogorov-Smirnov Test for Goodness of Fit," *Journal of the American Statistical Association*, 46 (253), 68–78. [19]

MCCARTY, NOLAN, KEITH T. POOLE, AND HOWARD ROSENTHAL (2016): *Polarized America: the Dance of Ideology and Unequal Riches*, MIT Press. [20]

MCINTYRE, LEE (2018): *Post-truth*, MIT Press. [1]

MEDE, NIELS G. (2022): "Legacy Media As Inhibitors and Drivers of Public Reservations Against Science: Global Survey Evidence on the Link Between Media Use and Anti-science Attitudes," *Humanities and Social Sciences Communications*, 9 (1), 1–11. [1]

MEDE, NIELS G, VIKTORIA COLOGNA, SEBASTIAN BERGER, JOHN BESLEY, CAMERON BRICK, MARINA JOUBERT, EDWARD MAIBACH, SABINA MIHELJ, NAOMI ORESKES, MIKE S SCHÄFER, ET AL. (2025): "Public communication about science in 68 countries: Global evidence on how people encounter information about science and engage with it," *OSF Preprints*. [2, 11]

MEDE, NIELS G. AND MIKE S. SCHÄFER (2020): "Science-related Populism: Conceptualizing Populist Demands Toward Science," *Public Understanding of Science*, 29 (5), 473–491. [1]

MEDE, NIELS G., MIKE S. SCHÄFER, AND TOBIAS F.ÜCHSLIN (2021): "The Scipop Scale for Measuring Science-related Populist Attitudes in Surveys: Development, Test, and Validation," *International Journal of Public Opinion Research*, 33 (2), 273–293. [1]

MERTON, ROBERT K. (1973): *The Sociology of Science: Theoretical and Empirical Investigations*, University of Chicago Press. [39]

MOHAMMADI, EHSAN, MIKE THELWALL, MARY KWASNY, AND KRISTI L. HOLMES (2018): "Academic information on Twitter: A user survey," *PLOS ONE*, 13 (5), 1–18. [1, 10]

MONGEON, PHILIPPE, TIMOTHY D. BOWMAN, AND RODRIGO COSTAS (2023): "An Open Data Set of Scholars on Twitter," *Quantitative Science Studies*, 4 (2), 314–324. [10, 11]

MORRIS, STEPHEN (2001): "Political Correctness," *Journal of Political Economy*, 109 (2), 231–265. [3, 7, 22]

MOSLEH, MOHSEN, CAMERON MARTEL, DEAN ECKLES, AND DAVID G. RAND (2021): "Shared Partisanship Dramatically Increases Social Tie Formation in a Twitter Field Experiment," *Proceedings of the National Academy of Sciences*, 118 (7), e2022761118. [9]

NICHOLS, TOM (2017): *The Death of Expertise: the Campaign Against Established Knowledge and Why It Matters*, Oxford University Press. [1]

NORRIS, PIPPA (2020): "Closed Minds? Is a 'Cancel Culture'stifling Academic Freedom and Intellectual Debate in Political Science?" *HKS Working Paper No. RWP20-025*. [8, 23]

ORZACH, ROI (2024): "Conformity Concerns: A Dynamic Perspective," *Available at SSRN 4992030*. [5, 29]

OTTAVIANI, MARCO AND PETER NORMAN SØRENSEN (2006): "Reputational Cheap Talk," *The Rand Journal of Economics*, 37 (1), 155–175. [7, 22]

PENG, HAO, DANIEL M. ROMERO, AND EMOKE-Á.GNES HORVÁT (2022): "Dynamics of Cross-platform Attention to Retracted Papers," *Proceedings of the National Academy of Sciences*, 119 (25), e2119086119. [10]

PETERSEN, MICHAEL BANG, ALEXANDER BOR, FREDERIK J.ØRGENSEN, AND MARIE FLY LINDHOLT (2021): "Transparent Communication About Negative Features of Covid-19 Vaccines Decreases Acceptance But Increases Trust," *Proceedings of the National Academy of Sciences*, 118 (29), e2024597118. [8]

QIU, JINGYI, YAN CHEN, ALAIN COHN, AND ALVIN E. ROTH (2024): "Social Media and Job Market Success: a Field Experiment on Twitter," *SSRN Working Paper No. 4778120*. [2, 4]

RATHJE, STEVE, JAY J. VAN BAVEL, AND SANDER VAN DER LINDEN (2021): "Out-group Animosity Drives Engagement on Social Media," *Proceedings of the National Academy of Sciences*, 118 (26), e2024292118. [28, 33]

ROGERS, NICK AND JASON J JONES (2021): "Using twitter bios to measure changes in self-identity: Are Americans defining themselves more politically over time?" *Journal of Social Computing*, 2 (1), 1–13. [25]

ROOZENBEEK, JON, CLAUDIA R. SCHNEIDER, SARAH DRYHURST, JOHN KERR, ALEXANDRA LJ FREEMAN, GABRIEL RECCHIA, ANNE MARTHE VAN DER BLES, AND SANDER VAN DER LINDEN (2020): "Susceptibility to Misinformation About Covid-19 Around the World," *Royal Society Open Science*, 7 (10), 201199. [1]

RUSSELL, ADRIENNE AND MATTHEW TEGELBERG (2020): "Beyond the Boundaries of Science: Resistance to Misinformation by Scientist Citizens," *Journalism*, 21 (3), 327–344. [3]

RUTJENS, BASTIAAN T. AND BOJANA VEĆKALOV (2022): "Conspiracy Beliefs and Science Rejection," *Current Opinion in Psychology*, 46, 101392. [1]

SCHARFF, DARCELL P., KATHERINE J. MATHEWS, PAMELA JACKSON, JONATHAN HOFFSUEMMER, EMEOBONG MARTIN, AND DOROTHY EDWARDS (2010): "More Than Tuskegee: Understanding Mistrust About Research Participation," *Journal of Health Care for the Poor and Underserved*, 21 (3), 879. [1]

SHAYO, MOSES (2009): "A Model of Social Identity With an Application to Political Economy: Nation, Class, and Redistribution," *American Political Science Review*, 103 (2), 147–174. [9]

——— (2020): "Social Identity and Economic Policy," *Annual Review of Economics*, 12 (1), 355–389. [9]

SIEGEL, ALEXANDRA A, EVGENII NIKITIN, PABLO BARBERÁ, JOANNA STERLING, BETHANY PULLEN, RICHARD BONNEAU, JONATHAN NAGLER, JOSHUA A TUCKER, ET AL. (2021): "Trumping hate on Twitter? Online hate speech in the 2016 US election campaign and its aftermath," *Quarterly Journal of Political Science*, 16 (1), 71–104. [12]

SIEVERTSEN, HANS H. AND SARAH SMITH (2025): "Do female experts face an authority gap? Evidence from economics," *Journal of Economic Behavior Organization*, 231, 106912. [28]

STEWART, ALEXANDER J., MOHSEN MOSLEH, MARINA DIAKONOVA, ANTONIO A. ARECHAR, DAVID G. RAND, AND JOSHUA B. PLOTKIN (2019): "Information Gerrymandering and Undemocratic Decisions," *Nature*, 573 (7772), 117–121. [9]

SULLIVAN, COLIN D. (2021): "Eliciting Preferences Over Life and Death: Experimental Evidence From Organ Transplantation," . [4]

TAJFEL, HENRI, MICHAEL G. BILLIG, ROBERT P. BUNDY, AND CLAUDE FLAMENT (1971): "Social categorization and intergroup behaviour," *European Journal of Social Psychology*, 1 (2), 149–178. [29]

TAJFEL, HENRI AND JOHN C. TURNER (2003): "The Social Identity Theory of Intergroup Behavior," *Social Psychology*, 4, 73–98. [9, 29]

THALER, MICHAEL (2024): "The fake news effect: Experimentally identifying motivated reasoning using trust in news," *American Economic Journal: Microeconomics*, 16 (2), 1–38. [8]

VAN DER BLES, ANNE MARTHE, SANDER VAN DER LINDEN, ALEXANDRA LJ FREEMAN, JAMES MITCHELL, ANA B. GALVAO, LISA ZAVAL, AND DAVID J. SPIEGELHALTER (2019): "Communicating Uncertainty About Facts, Numbers and Science," *Royal Society Open Science*, 6 (5), 181870. [8]

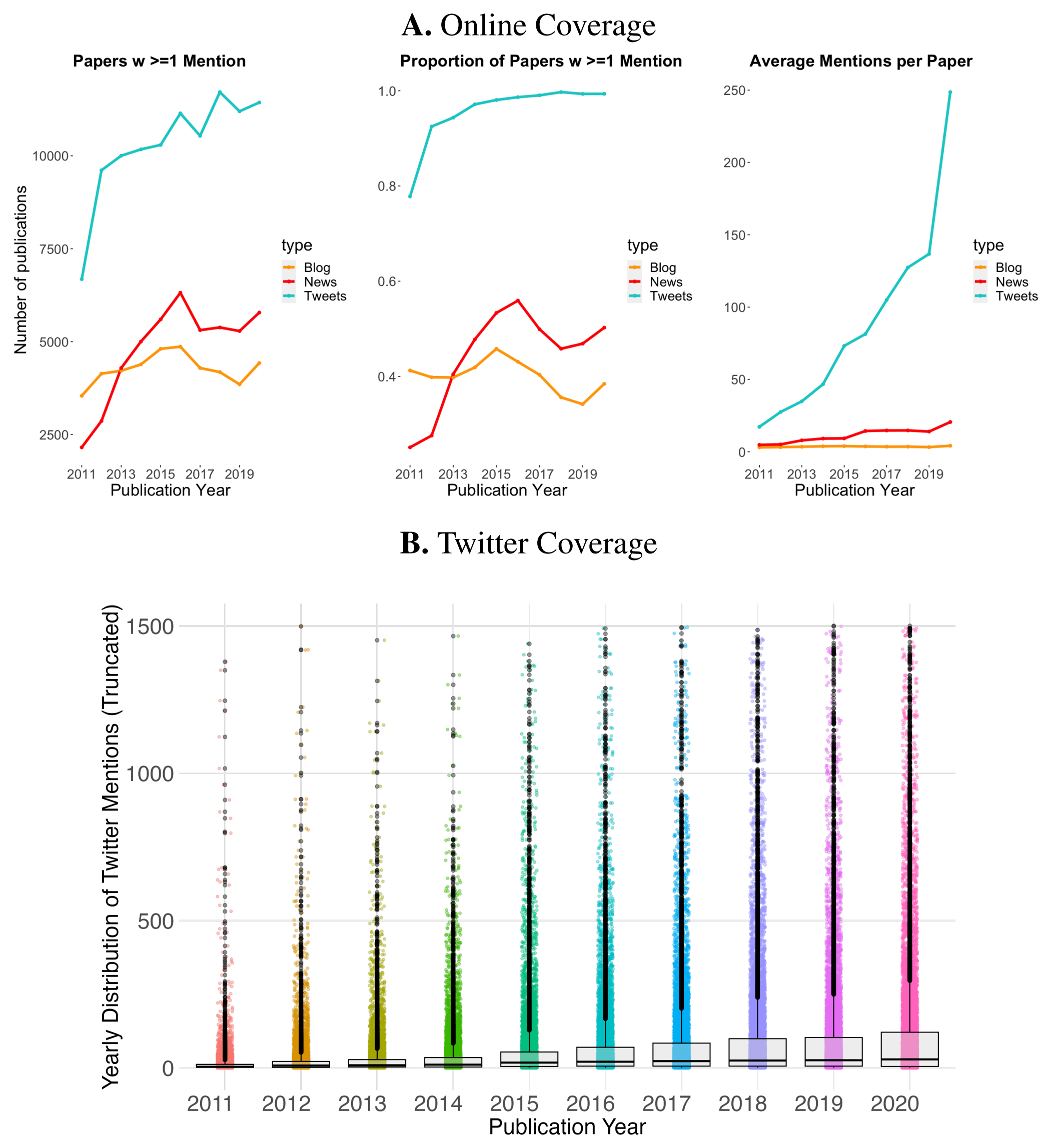
VAN DER BLES, ANNE MARTHE, SANDER VAN DER LINDEN, ALEXANDRA LJ FREEMAN, AND DAVID J. SPIEGELHALTER (2020): "The Effects of Communicating Uncertainty on Public Trust in Facts and Numbers," *Proceedings of the National Academy of Sciences*, 117 (14), 7672–7683. [8]

WEST, JEVIN D. AND CARL T. BERGSTROM (2021): "Misinformation in and About Science," *Proceedings of the National Academy of Sciences*, 118 (15), e1912444117. [1]

ZHANG, FLOYD JIUYUN (2023): "Political Endorsement by Nature and Trust in Scientific Expertise During Covid-19," *Nature Human Behaviour*, 7 (5), 696–706. [8]

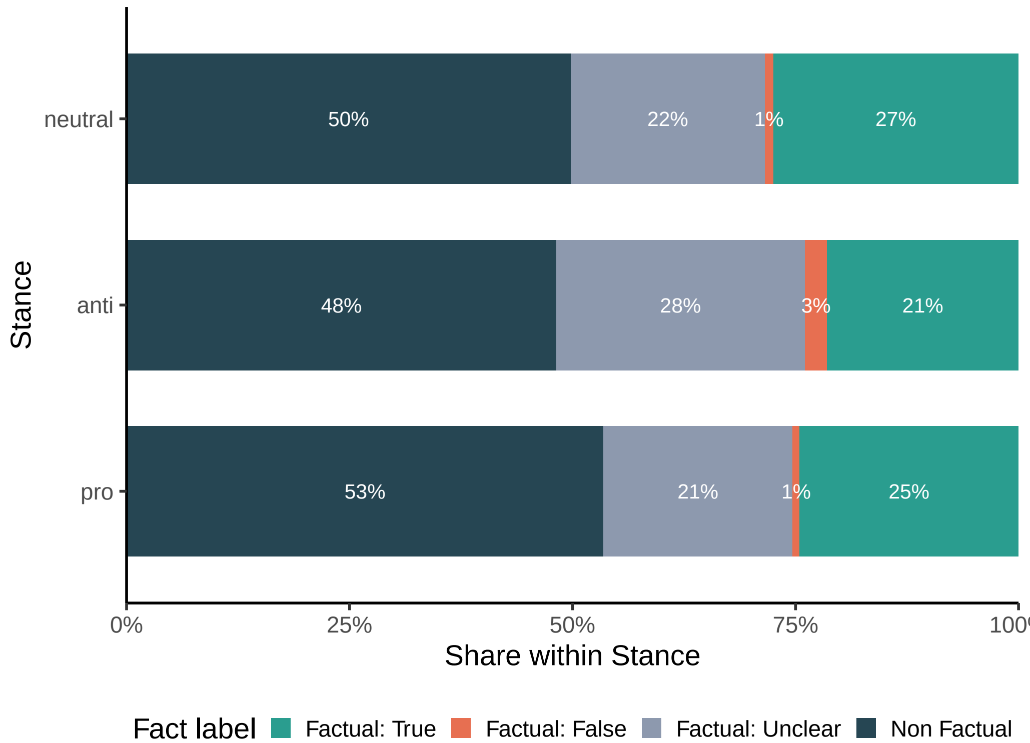
ZHURAVSKAYA, EKATERINA, MARIA PETROVA, AND RUBEN ENIKOLOPOV (2020): "Political Effects of the Internet and Social Media," *Annual Review of Economics*, 12, 415–438. [9]

FIGURE 1.—Online presence of research articles published in general interest journals between 2011 and 2020



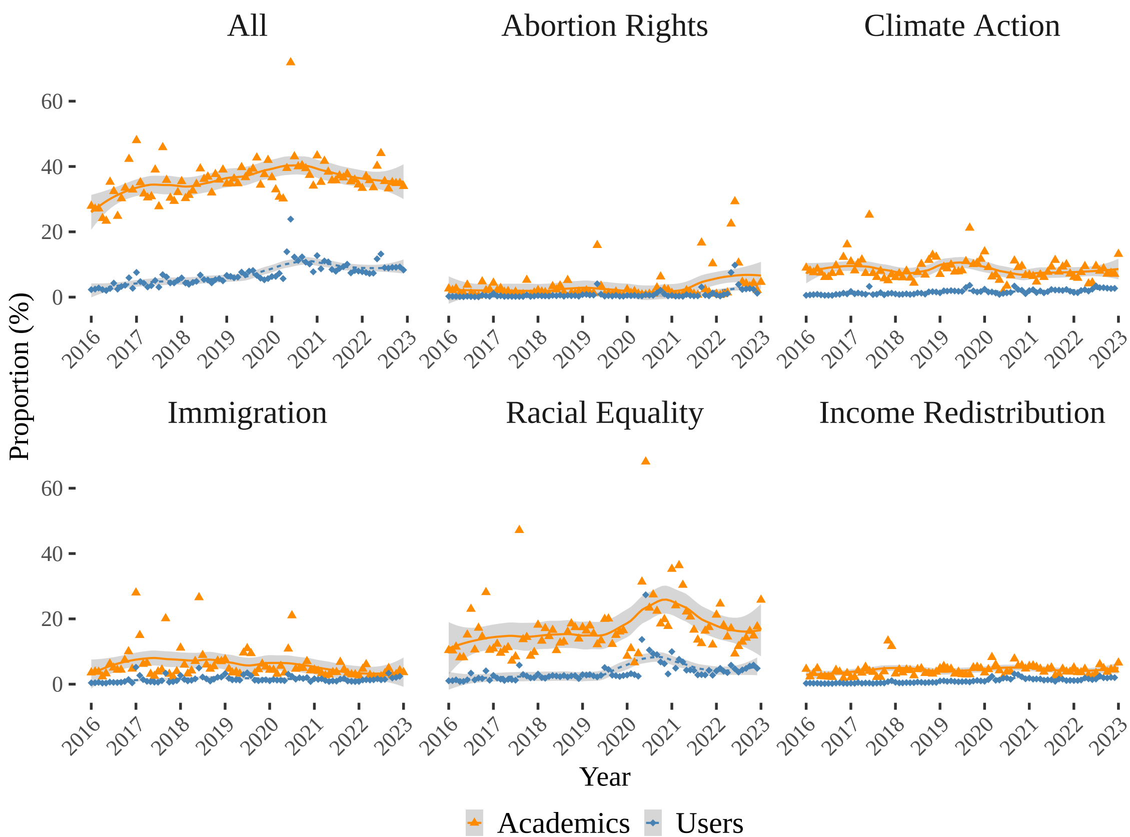
Note: Figure A provides trends in online coverage of scientific articles published in *Science*, *Nature*, *PNAS*, *Cell*, *NEJM*, and *Lancet* between 2011 and 2020. Online appearances across blog posts, newspaper articles, or Twitter are retrieved from Altmetric (accessed on November 10th, 2021). The figure suggests that scientific articles with any online appearances have increased over time, in absolute number (first row), as a proportion of all articles published (second row), and per average number of appearances per published paper (third row). Figure B provides trends in Twitter coverage of scientific articles published in *Science*, *Nature*, *PNAS*, *Cell*, *NEJM*, and *Lancet* between 2011 and 2020. The data present the distribution of Twitter mentions per article, retrieved from Altmetric (accessed on November 10th, 2021). The figure indicates growing online presence, with the distribution of Twitter mentions becoming less skewed towards zero, with a ticker right tale and a rise in high-mention outliers over time.

FIGURE 2.—Distribution of Factual vs Non-Factual Content by Ideological Stance.



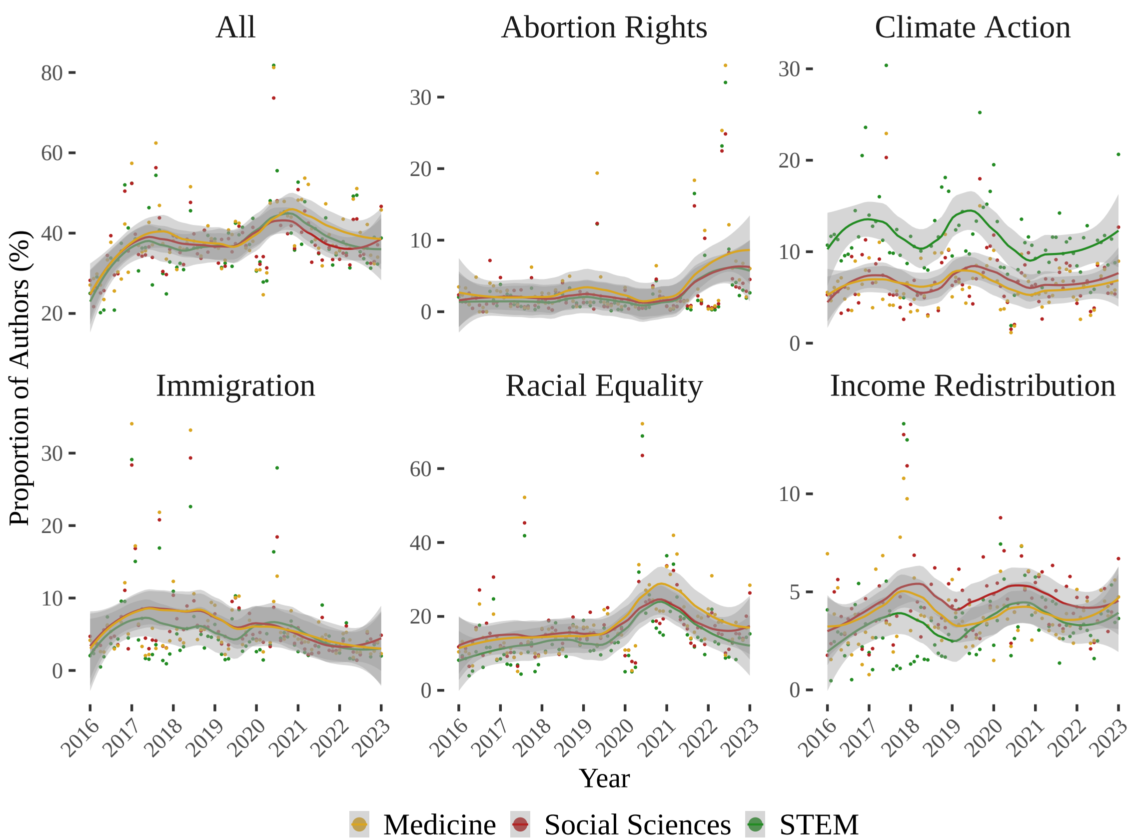
Note: This figure shows the distribution of tweets by academics (2016–2022), highlighting the share of factual (TRUE, FALSE, UNCLEAR) vs non-factual content across ideological stances. Over half of topical tweets (51%) are non-factual, i.e., opinion or commentary which do not contain any verifiable claim. Among factual tweets, 53% are TRUE, 45% UNCLEAR, and 2% FALSE. Conservative-leaning (anti) tweets show relatively more FALSE and UNCLEAR content; progressive-leaning (pro) tweets show a higher share of non-factual content and fewer FALSE classifications. This classification was conducted using GPT-4o-mini, following a structured prompt to label tweets and extract factual claims (full prompt in Appendix Section D) and helps clarify that a non-neutral stance does not necessarily indicate politicized expression, but may reflect factual reporting aligned with a given issue (e.g., new evidence on climate change). We show this is not systematically the case: while many stanced tweets (particularly in the *pro* category) are grounded in verifiable facts, over half of non-neutral tweets contain no checkable claim at all. Our factual classification thus complements the stance measure by distinguishing between opinion, ambiguous claims, and factual reporting within salient topics.

FIGURE 3.—Proportion of US Users and Academics with a Political Opinion Over Time



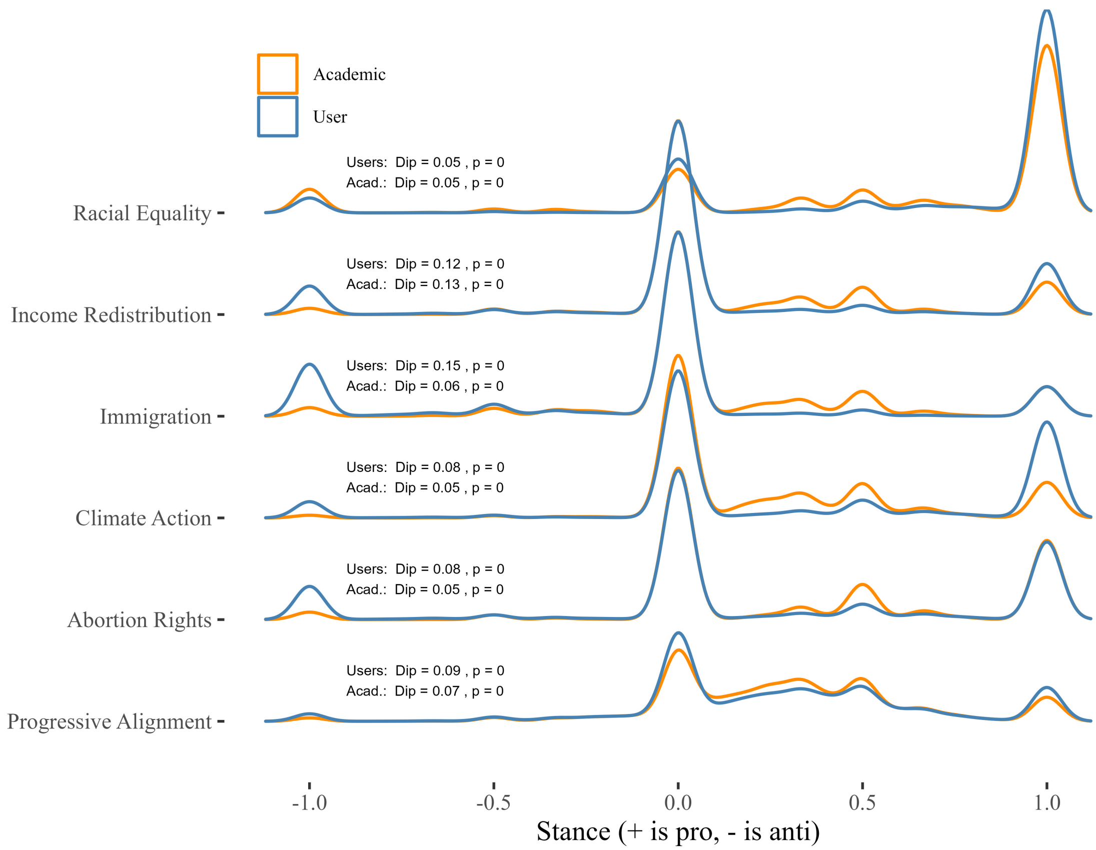
Note: The figure illustrates trends in politicization of conversations by academics and general users on Twitter from 2016 to 2022. Monthly aggregated scatter plots display expressed stance for each topic, with a LOESS applied for trend visualization. Standard errors are depicted in the shaded region. In the "All" panel, around 40% of tracked US academics expressed opinions on predefined political issues, compared to 5-10% of general users. Variations and spikes are observed across topics, with Climate Action and Racial Equality showing the largest disparities. Climate Action witnessed significant declines during the COVID-19 pandemic onset. Mid-2020 saw a surge in attention to Racial Equality, reflecting the outcry after the George Floyd incident. Other topics exhibit stable increasing trends, with occasional short-lived spikes, notably in Abortion Rights around changes in laws in 2022. Immigration discussions, while less frequent, maintained regularity, with heightened attention during the 2016 presidential election.

FIGURE 4.—Proportion of Academics with Political Opinions Over Time by Topic and Discipline



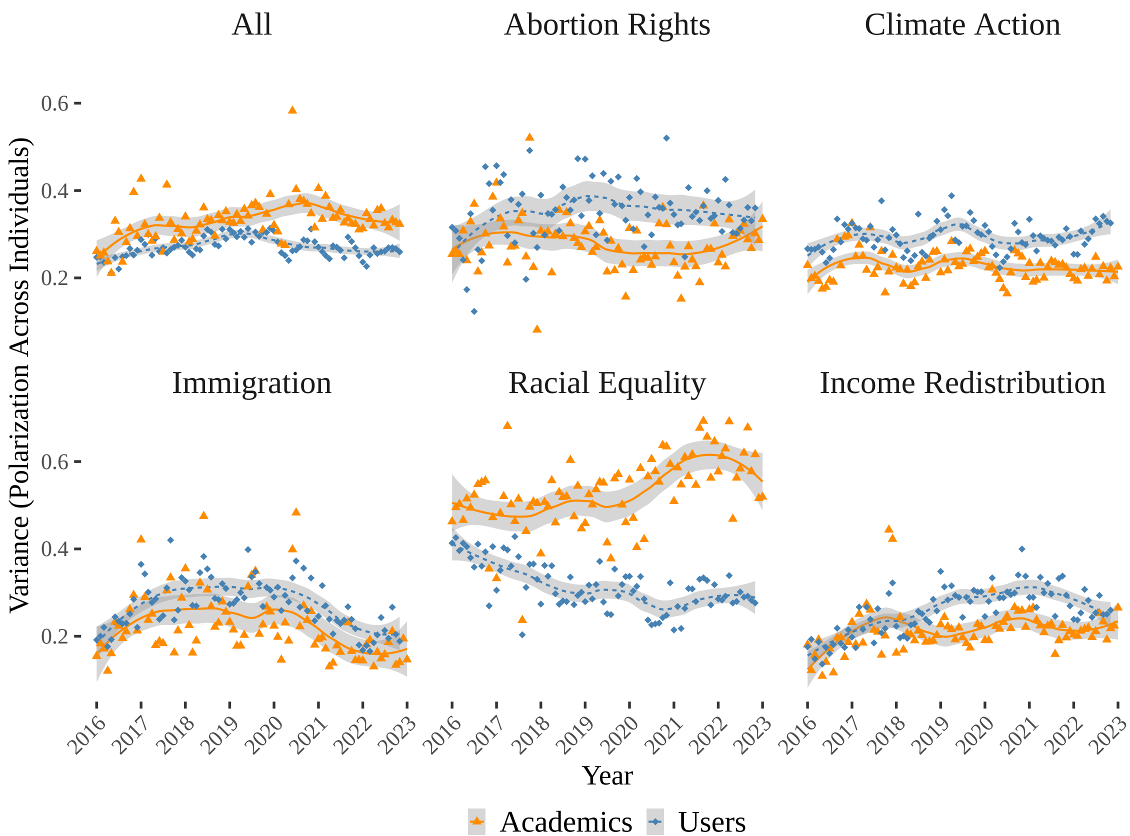
Note: OpenAlex describes "Concepts" as abstract ideas that a work is about: "Concepts are hierarchical, like a tree. There are 19 root-level concepts, and six layers of descendants branching out from them, containing about 65 thousand concepts all told". OpenAlex classifies each work with a high level of accuracy. We combine the 19 root-level concepts into 3 broad categories for ease of comparison: (1) Medicine, (2) Social Sciences, and (3) STEM. We omit Humanities from the time-series depiction given we have only around 100 humanities' academics, which would create unreliable trends. Each work by an author can belong to multiple concepts and a score from 0 to 1 is given to each concept, where values closer to 1 reflect the likelihood the work belongs to that concept. For each academic, we take the average score for all the root-level concepts across all their works from 2016-2022. We then pick the primary concept of that academic as the root level concept with the highest average score. This depicts the proportion of academics within each concept category expressing an opinion about a political topic. Our analysis reveals distinct patterns and spikes across concept categories for two issues: Climate Action and Income Redistribution. STEM academics consistently show significantly higher engagement with Climate Action, almost doubling the frequency of expressions from other fields in any given period. On Income Redistribution, however, STEM academics were notably less vocal than those in Medicine or Social Sciences before 2020, after which the differences narrowed.

FIGURE 5.—Cross-sectional Ideological Polarization across US Users and Academics, 2016-2022



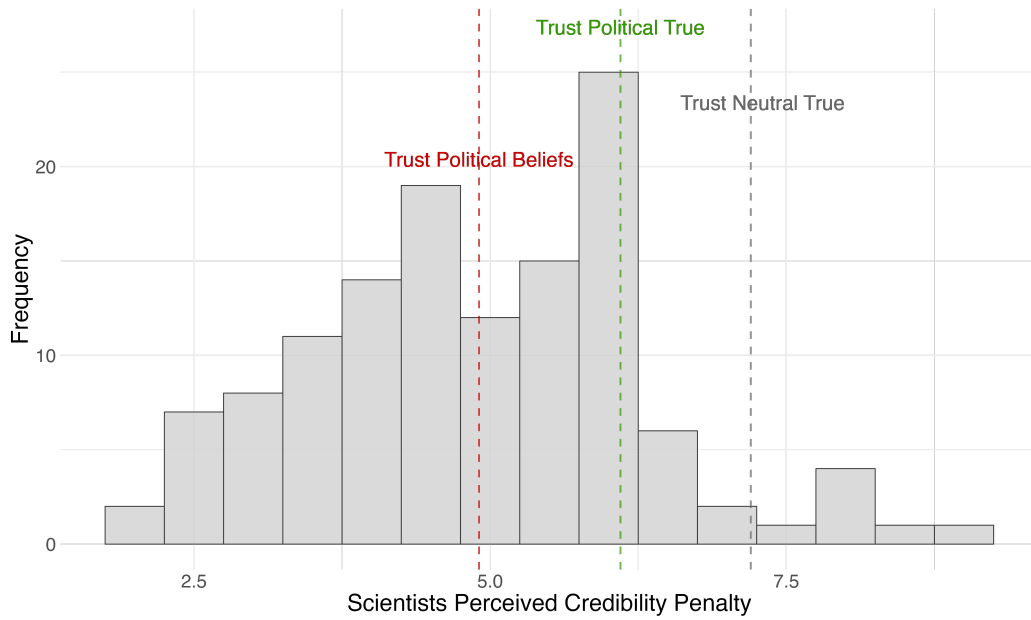
Note: The figure illustrates the density distribution of net stance across various political topics among U.S. academics (in orange) and general users (in blue) from 2016 to 2022. Topics include Income Redistribution, Climate Action, Immigration, Abortion Rights, and Racial Equality, with "Progressive Alignment" representing the average stance across these topics. The x-axis represents the net stance, where positive values indicate a pro-stance and negative values indicate an anti-stance. The y-axis indicates different topics, with density distributions shown as ridgelines. Each ridgeline highlights where individuals tend to cluster in their expressed opinions. Black vertical lines within each distribution represent the mean net stance for each topic. Hartigan's Dip Test identifies multimodal distributions, suggesting distinct ideological camps. The dip statistic and corresponding p-value are annotated for each topic, demonstrating statistically significant multimodalities for both groups. Compared to academics, general users tend to cluster more around extreme viewpoints, especially towards the conservative (-1) side on most issues. Academics, on the other hand, have a larger mass in the moderate liberal/progressive range. An exception to this is the issue of race, where general users show more consensus, with a larger mass towards pro-racial equality. Both groups exhibit statistically significant multimodal distributions (p-values of 0), with users showing slightly higher dip statistics on average, indicating more pronounced ideological polarization.

FIGURE 6.—Issue Disagreement Across Academics By Topics Over Time



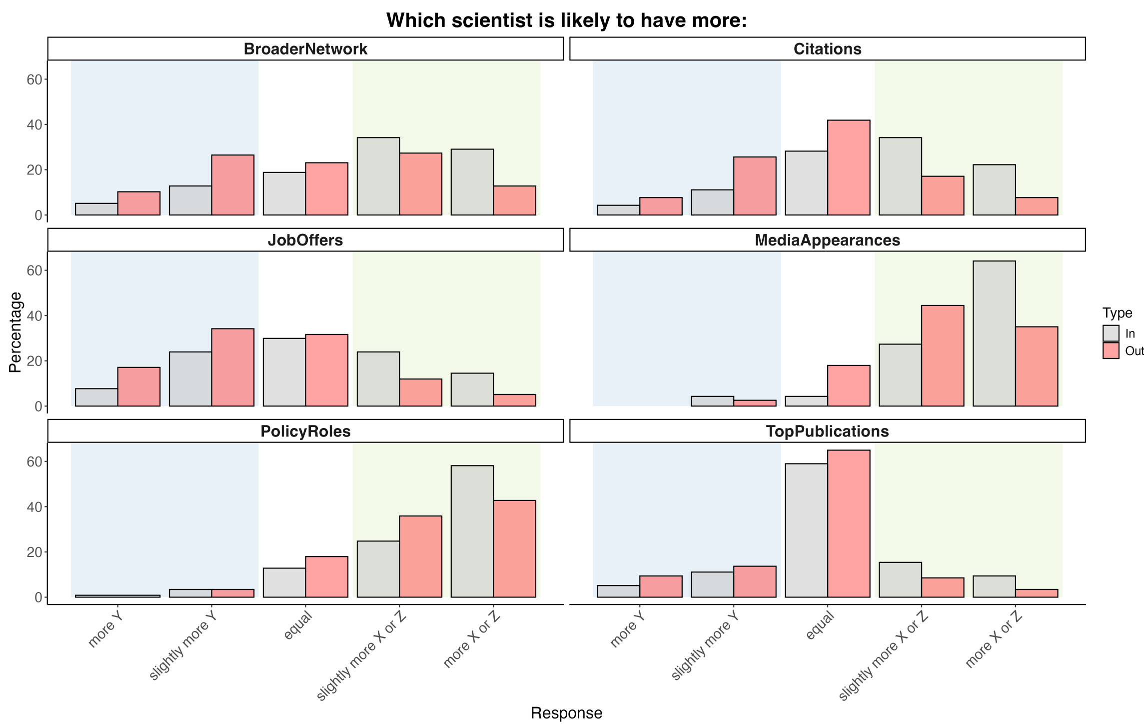
Note: The figure measures ideological polarization by computing the variance of net stance held by a population on predefined political topics. Variance across users reflects the dispersion of political opinion on each topic, providing a continuous measure of ideological disagreement. Aggregating across topics yields an overall measure of political variance. Similar to the increasing trend in expression of opinion, polarization increases between 2016 and 2022, raising concerns about its impact on scientific consensus-building and public trust in scientific expertise. Race exhibits wider debate among academics relative to the general population. Disparities in ideological disagreement are observed across topics, with gaps widening, particularly around the pandemic period for Immigration, Income Redistribution, and Abortion, but also closing by the end of 2022. The gap persists for Climate Action throughout the sample period and grows at the end of 2022.

FIGURE 7.—Scientists' Beliefs about Credibility Penalty



Note: This figure reports the distribution of responses from 128 scientists recruited on Prolific to the following question: *"What do you think is the reported level of trust for scientists who do express political opinions on social media?"*. Prior to asking the question, we informed the academic respondents that we had surveyed a representative sample of the U.S. on their perceived credibility of scientists, distinguishing between those who had expressed political opinions online and those who had not. We anchored our scientists' beliefs on the public perceived credibility for scientists who *do not* express political opinions online (indicated by the grey dashed line). The average answer of our sample of scientists is shown by the red dashed line and is significantly lower than the true value obtained in the main survey of a representative sample of U.S. respondents, which is indicated by the green dashed line.

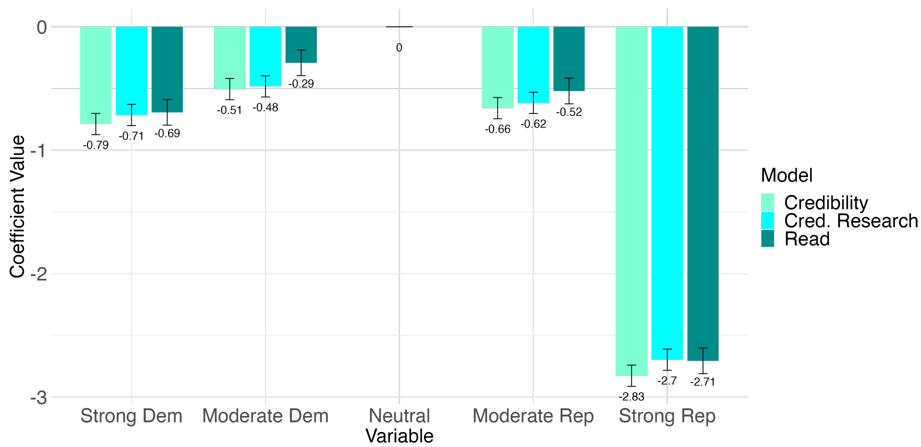
FIGURE 8.—Scientists' Perceived Costs and Benefits from Public Political Expression



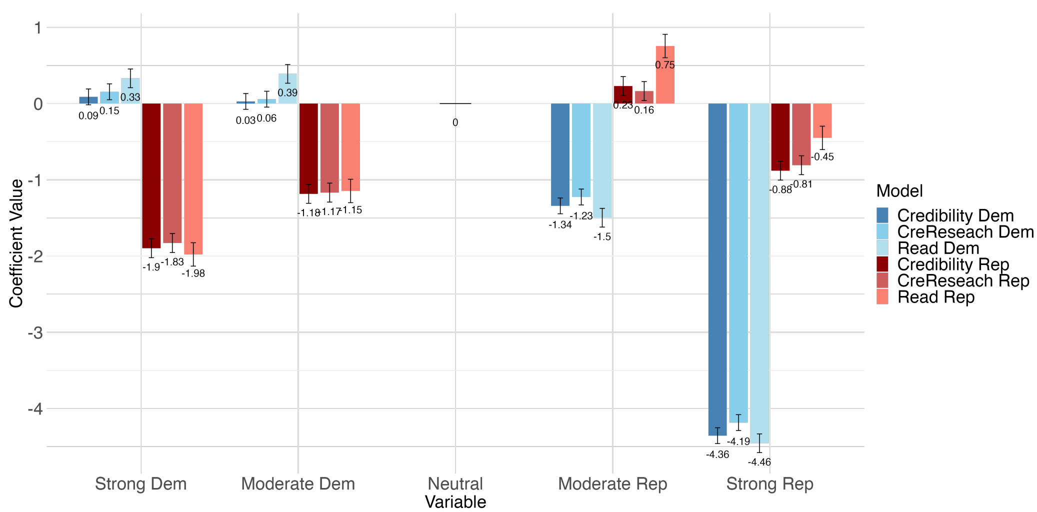
Note: The figure illustrates scientists' perceptions of professional benefits based on public political expression. Respondents compared the outcomes for Scientist X (grey), who shares opinions on politically charged topics related to their research, and Scientist Z (red), who shares opinions on politically charged topics unrelated to their research, against Scientist Y, who refrains from such expression (respectively, either related or unrelated to their research). The evaluated benefits include network size, citations, academic job offers, media appearances, policy roles, and top publications. A distribution skewed towards X or Z (green area) reflects perceived benefits from political expression, while a denser left tail towards Y (blue area) indicates perceived costs. Results suggest clear benefits for media appearances and policy roles, milder benefits for network size and citations, and ambiguous or slight perceived costs for top publications and academic job offers.

FIGURE 9.—Impact of Scientists' Political Expression on Perceptions from the General Public. Credibility and public Willingness to Read Peak at Neutral, with a Monotonic Penalty for Scientists Political Expression to the 'Left' and 'Right' of Neutral.

A. Base Model

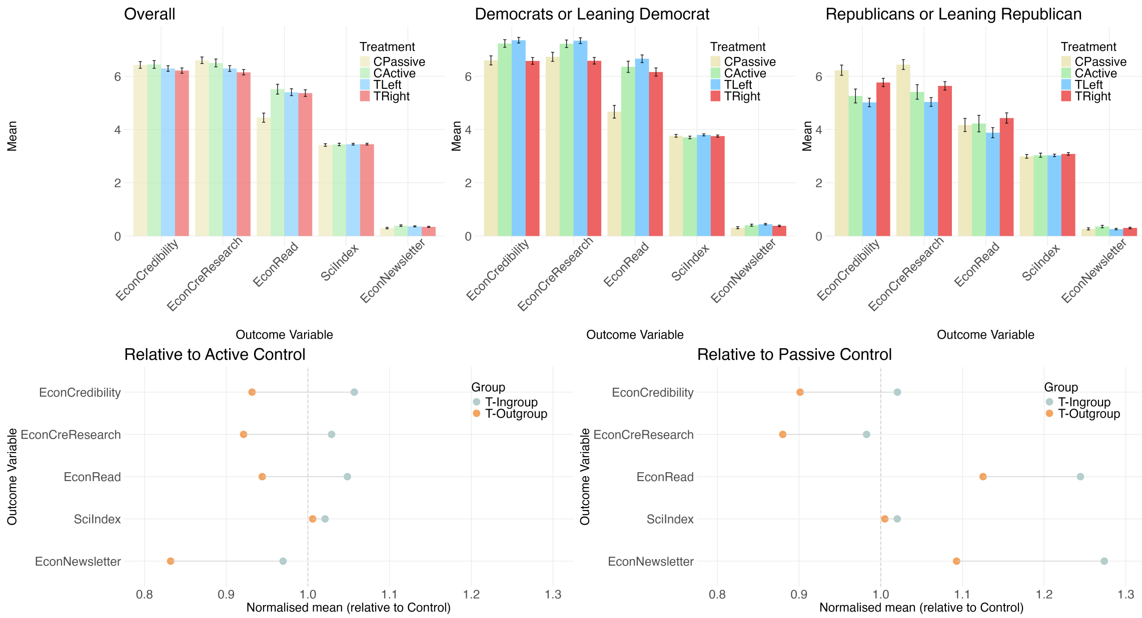


B. Heterogeneity by Respondents' Partisanship



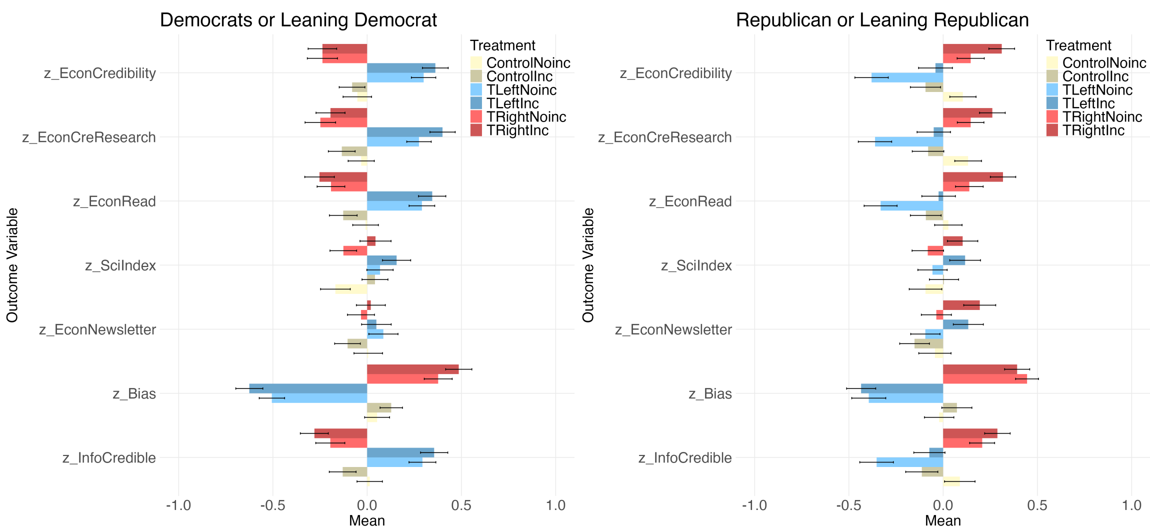
Note: Coefficients are obtained by regressing scientists' characteristics on respondents' perceived credibility or willingness to read content from scientists. The x-axis represents different political affiliations of scientists, estimated by indicator variables for "Strong Republican", "Moderate Republican", "Strong Democrat", or "Moderate Democrat", with "Neutral" as the excluded category. The y-axis shows the coefficient values indicating the impact on credibility and willingness to read. The data reveals a peak in credibility for neutral scientists, with a decline for both left- and right-leaning scientists. Standard errors are clustered at the individual level. Additional regressors include indicator variables to control for other scientist characteristics: institutional affiliation (Harvard, UC Berkeley, or Chicago, versus Arkansas or Connecticut), field of research (Medicine, Mathematics, Engineering, Economics, or Literature), seniority of role (Full professor or Assistant Professor), and gender (male or female). (N = 1704, 940 Dem. or Lean Dem., 745 Rep. or Lean Rep., 19 Other leaning.)

FIGURE 10.—Separating the Effect of Communicating Salient Research from a Scientists' Pure Political Signal on Respondents' Perceived Credibility. Any Signal Impacts Scientists' Credibility. Effects are Moderated by the Congruence between Participants' Leaning and Scientists' Signal.



Note: Figures show results of our second experimental task where respondents are divided into four groups: in the *passive control* group, respondents are exposed to an economist who neither advertises own research in a politically salient issue nor signals any political affiliation; respondents in the *active control* group are exposed to the profile of an economist advertising own research in a politically salient issue with no political signal; respondents in the *treatment left (right)* group are exposed to an economist advertising their research in a politically salient issue together with a left (right) political signal. The politically salient research is favourable to a democrat leaning narrative. After viewing one of the four profiles, we collect the following outcome variables for each respondent: their perceived credibility of the economist, their perceived credibility of the economist's research, their willingness to read an opinion from the economist, their intention to sign up for a newsletter containing opinions from a similar profile, and a composite index of general trust in science. *Democrats* show higher credibility when exposed to politically aligned research. The left political signal increases perceived credibility, while the right signal reduces it. Similarly, willingness to read is higher with politically aligned research; the left signal increases it, while the right signal decreases it. *Republicans* exhibit significantly reduced perceived credibility when exposed to misaligned politically salient research, especially with a left signal, though less so with a right signal. Willingness to read is highest with a congruent right signal and lowest with a left signal. For both sub-samples, newsletter sign-up and overall trust in science move similarly, but changes are less pronounced. Additionally, normalising our group averages relative to the active or passive control, at the bottom, we observe that in-group respondents (when scientist signal and respondents leaning align) perceive significantly higher credibility of scientists and their research, are more willing to read their opinion, and are more likely to sign up to the newsletter, relative to out-group respondents. (N = 1704, 940 Dem. or Lean Dem., 745 Rep. or Lean Rep., 19 Other leaning.)

FIGURE 11.—Isolating the Effect of Strategic Considerations behind Scientists' Political Expression on Respondents' Perceived Credibility: Making *Visibility Incentives* Explicit. Subjects Respond to Scientists' Incentives, but Updating Depends on Partisan Congruence.



Note: Figure shows results of the experimental task isolating the effect of explicit strategic motives behind scientists' political expression. Respondents, selected among Twitter users, were randomly assigned to a 2×3 design: Varying by *Tweet leaning* (Neutral, Left, Right) and *Promotion information* (No Promotion vs Promotion). In the Promotion condition, respondents were told that *we (the researchers)* would promote an economist's tweet and that the economist knew this *before* posting; no such information was provided in the No-Promotion condition. Outcomes include credibility of the economist, credibility of the economist's research, willingness to read related content, a Trust in Science Index, perceived ideological bias of the economist, and credibility of the information in the tweet. Bars plot conditional means of standardized outcomes; whiskers show 95% confidence intervals. Promotion information does not produce a uniform penalty. Among Democrats, it amplifies partisan reactions. *Relative to No-Promotion*, perceived credibility is higher for *promoted left-leaning* tweets and lower for *promoted right-leaning* tweets, while neutral promotion has no impact. Among Republicans, promotion raises perceived credibility for *promoted right-leaning* tweets and attenuates the penalty for *promoted left-leaning* tweets; promotion of a neutral tweet is penalized. General trust in science increases for Democrats when facing left (and neutral) promoted content, and for Republicans when facing either left- or right-leaning promoted content. Perceived ideological bias of the economist shifts with tweet slant as intended, confirming the manipulation. (N = 2,000; 1,081 Dem./Lean Dem.; 906 Rep./Lean Rep.; 13 Other.)

SUPPLEMENT TO "POLITICIZED SCIENTISTS:
CREDIBILITY COST OF POLITICAL EXPRESSION ON TWITTER"

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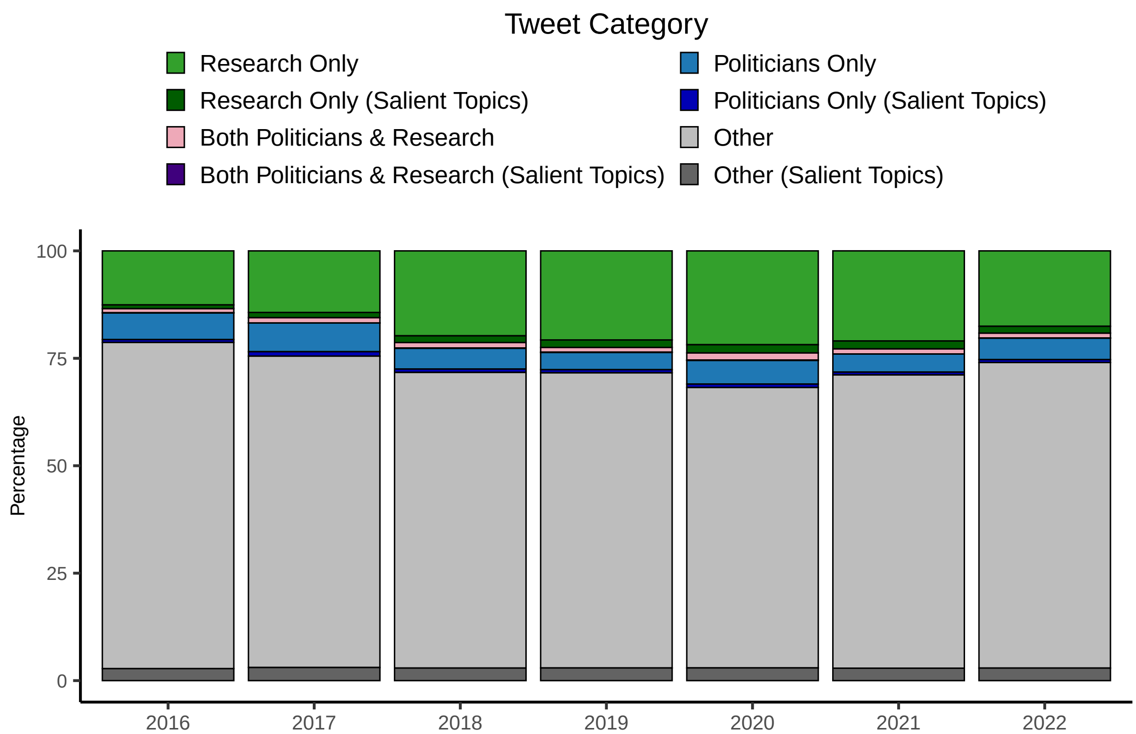
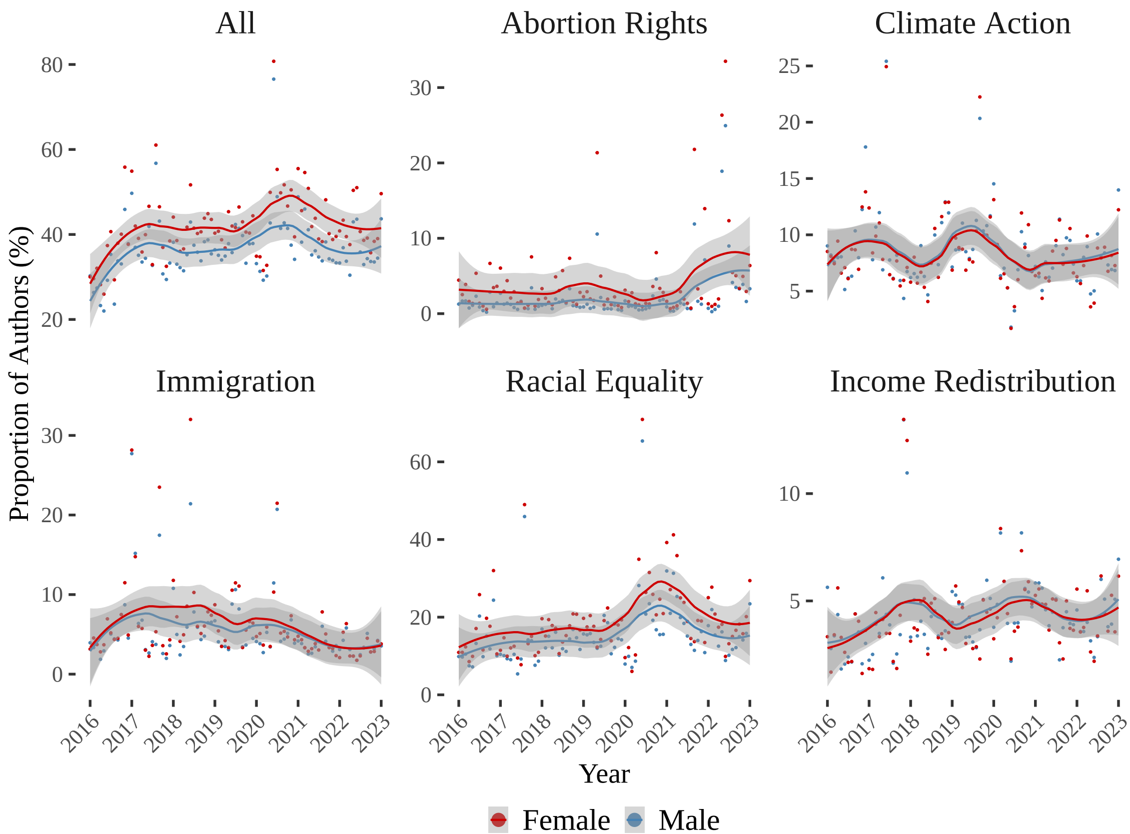
APPENDIX A: ONLINE FIGURES

FIGURE A.1.—Yearly Distribution of Tweets by Academics Mentioning Politicians, Research Papers, and Salient Topics



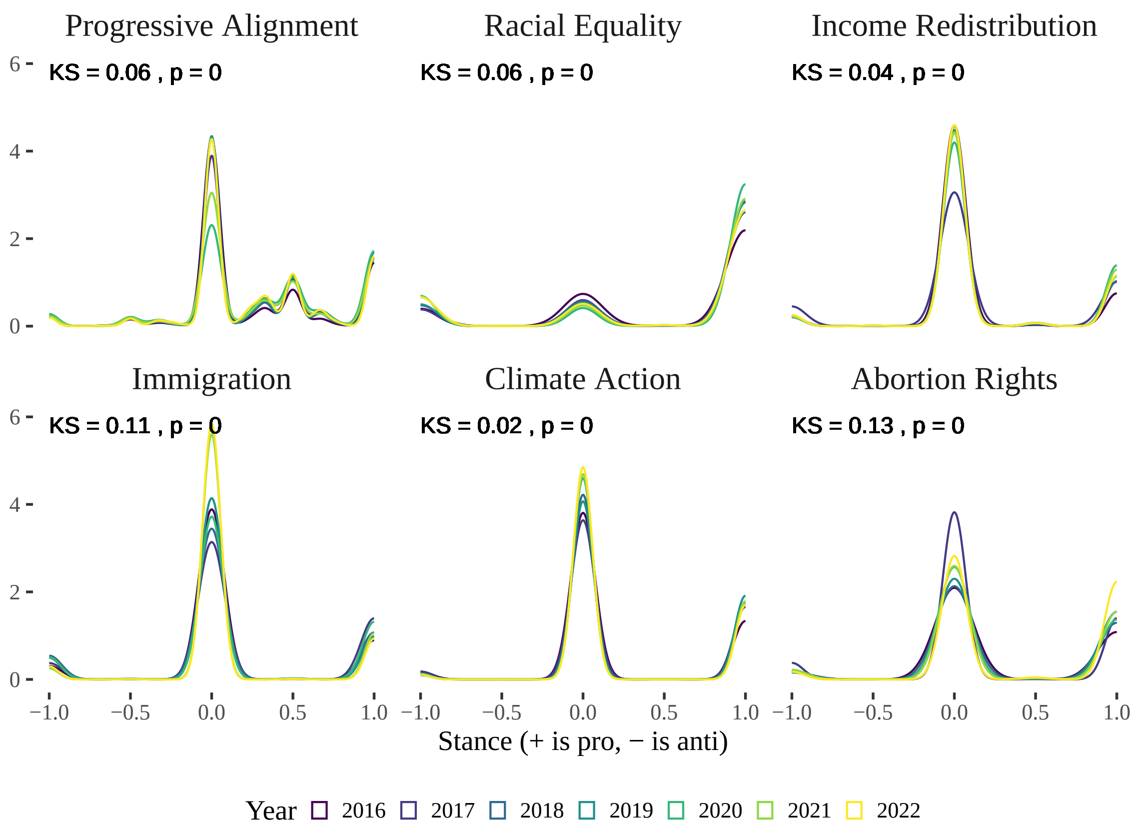
Note: This figure depicts the yearly distribution of tweets by academics from 2016 to 2022, highlighting the proportion of tweets that mention politicians, research papers, both, and other content. It helps to understand the interaction between political discourse and academic content over time. Each bar represents the total percentage of tweets for a given year, with colors indicating the categories: green for tweets mentioning research papers, blue for tweets mentioning politicians, purple for tweets mentioning both politicians and research papers, and grey for other tweets. Darker shades within each color represent tweets related to the five salient topics (Abortion Rights, Climate Action, Racial Equality, Immigration, and Income Redistribution). The "*Politicians Only (Salient Topics)*" category includes tweets that mention politicians and one of the salient topics, the "*Research Only (Salient Topics)*" category includes tweets that mention research papers and one of the salient topics, and the "*Both Politicians & Research (Salient Topics)*" category includes tweets that mention both politicians and research papers within the salient topics. The overlaps between categories, represented by the pink sections, are relatively small across all years, typically around 1% (with the salient topics subset being even smaller, at 0.01%). The overall proportion of tweets mentioning politicians remains around 10%, while approximately 20% of tweets mention research papers. Notable spikes in tweets mentioning politicians and research papers are observed in certain years, reflecting significant political or scientific events such as the 2016 and 2020 US presidential elections, the George Floyd incident in 2020, and the COVID-19 pandemic.

FIGURE A.2.—Proportion of Academics with Political Opinions Over Time by Topic and Gender



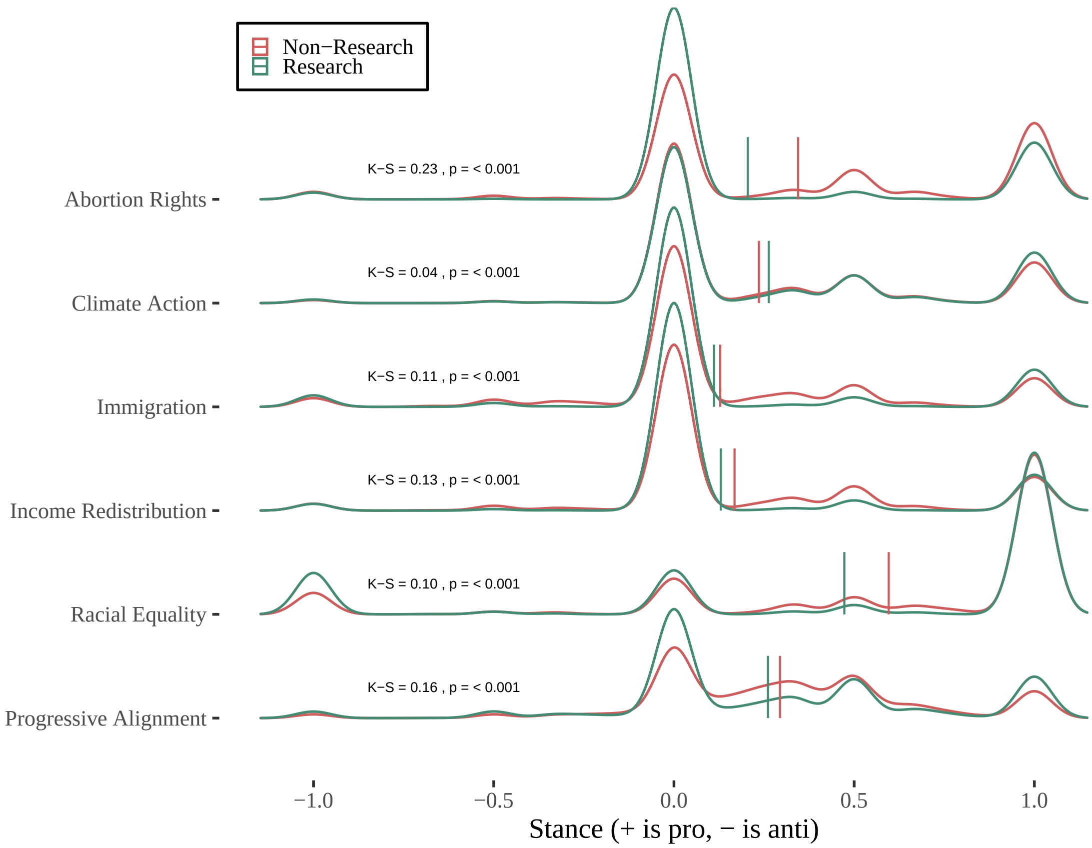
Note: We passed the full OpenAlex academic name as it appears on their papers to an LLM to classify the gender as 'Male', 'Female', or 'Unclear'. Close to 99% of names were labeled Male or Female. Using this binary classification of gender, we can explore sub-population differences in political expression. This is displayed in the figure broken down by political topics. In general, we find academics with names classified as female to express slightly more political opinions overall, especially on topics Abortion Rights, Immigration, and Racial Equality. Most statistically significant differences occur post-2020 (when confidence intervals overlap the least). The topics of Climate Action and Income Redistribution display the least differences.

FIGURE A.3.—Evolution of Ideological Polarization among Academics



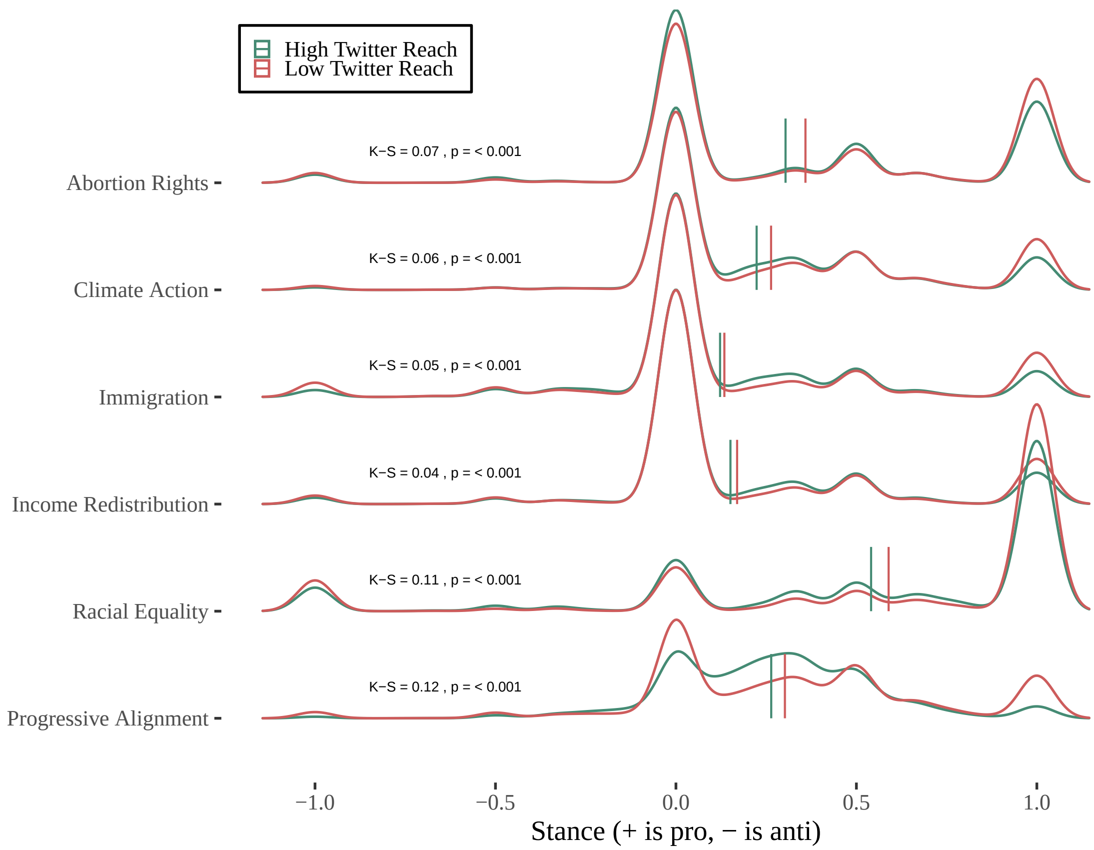
Note: This figure investigates the dynamics of ideological polarization among academics over time, focusing on key political issues. Each panel presents yearly distributions of stances of U.S. academics on various topics, testing for distribution equality between early (2016-2017) and late (2021-2022) years. Topics include Income Redistribution, Climate Action, Immigration, Abortion Rights, and Racial Equality, with "Progressive Alignment" representing the average stance across all topics. The x-axis shows the net stance, with positive values indicating a pro-stance and negative values an anti-stance. Density distributions are depicted as overlapping yearly ridgelines, with different colors representing different years. Each plot includes the Kolmogorov-Smirnov (KS) test results, testing distributions equality between 2016-2017 and 2021-2022, the KS statistic and corresponding p-value are provided. The figure demonstrates significant shifts in ideological stances over time, particularly on Immigration and Abortion Rights, which show larger KS statistics, indicating substantial distributional changes, whereas Climate Action shows smaller shifts.

FIGURE A.4.—Cross-sectional Ideological Polarization based on whether Academic Tweets are Research Related



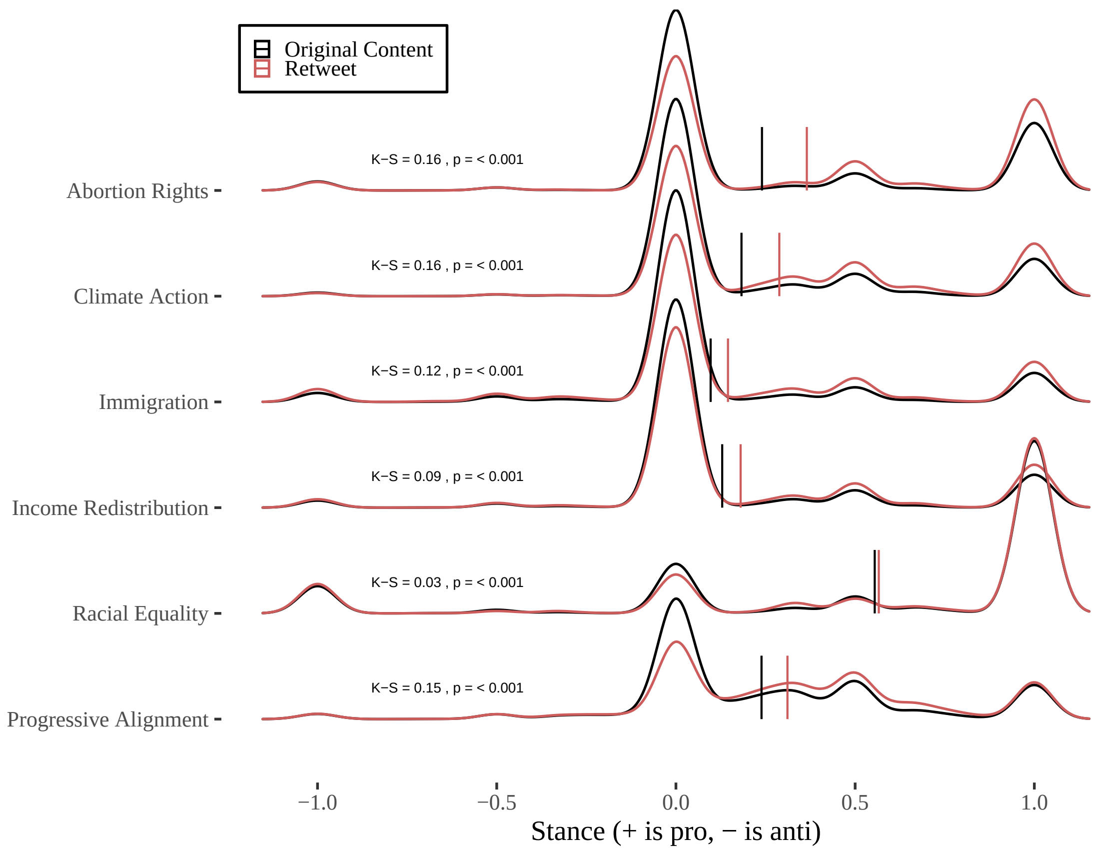
Note: The figure presents the density distributions of net stances across various political topics among U.S. academics from 2016 to 2022, comparing tweets that mention research papers (in green) with those that do not (in red). Topics include Income Redistribution, Climate Action, Immigration, Abortion Rights, and Racial Equality, with "Progressive Alignment" representing the average stance across these topics. The x-axis represents the net stance, where positive values indicate a pro stance and negative values indicate an anti stance. The y-axis lists the different topics, with density distributions shown as ridgelines. Vertical mean lines are included for each distribution, colored according to the tweet type. Kolmogorov-Smirnov (K-S) test results are annotated for each topic, indicating statistically significant differences between the distributions of research-related and non-research-related tweets (p-values < 0.001). The largest divergence is observed in Abortion Rights, where non-research tweets show a larger mass toward the pro stance (+1), while research-related tweets are more centered around neutrality (0). Generally, non-research tweets tend to be more liberal across most topics, except for Climate Action, where research-related tweets are more progressive. This suggests that the context of discussion influences the expression of political stances among academics.

FIGURE A.5.—Cross-sectional Ideological Polarization based on High vs. Low Twitter Reach



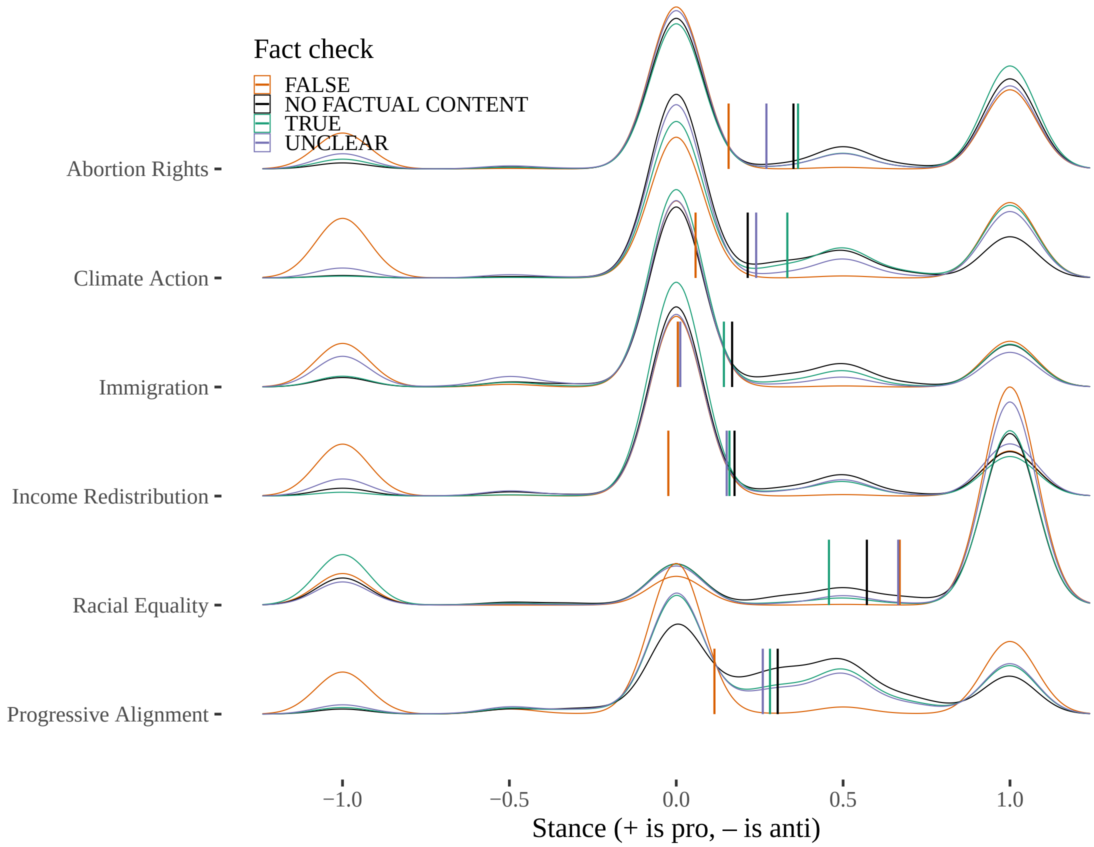
Note: The figure presents the density distributions of net stances across various political topics among U.S. academics from 2016 to 2022, comparing academics with high Twitter reach (in green) and low Twitter reach (in red). High and low Twitter reach is defined based on whether the follower count is above or below the median of the filtered dataset (median = 522; 1st quartile = 219; 3rd quartile = 1,302; max = 4,587,745). Topics include Income Redistribution, Climate Action, Immigration, Abortion Rights, and Racial Equality, with "Progressive Alignment" representing the average stance across these topics. The x-axis represents the net stance, where positive values indicate a pro stance and negative values indicate an anti stance. The y-axis lists the different topics, with density distributions shown as ridgelines. Vertical mean lines are included for each distribution, colored according to the Twitter reach category. Kolmogorov-Smirnov (K-S) test results are annotated for each topic, indicating statistically significant differences between the distributions of high and low Twitter reach (p-values < 0.001). Low reach tweets tend to exhibit more polarized stances compared to high reach tweets, with the largest divergence observed in Abortion Rights and Racial Equality.

FIGURE A.6.—Cross-sectional Ideological Polarization based on Retweets vs. Original Content



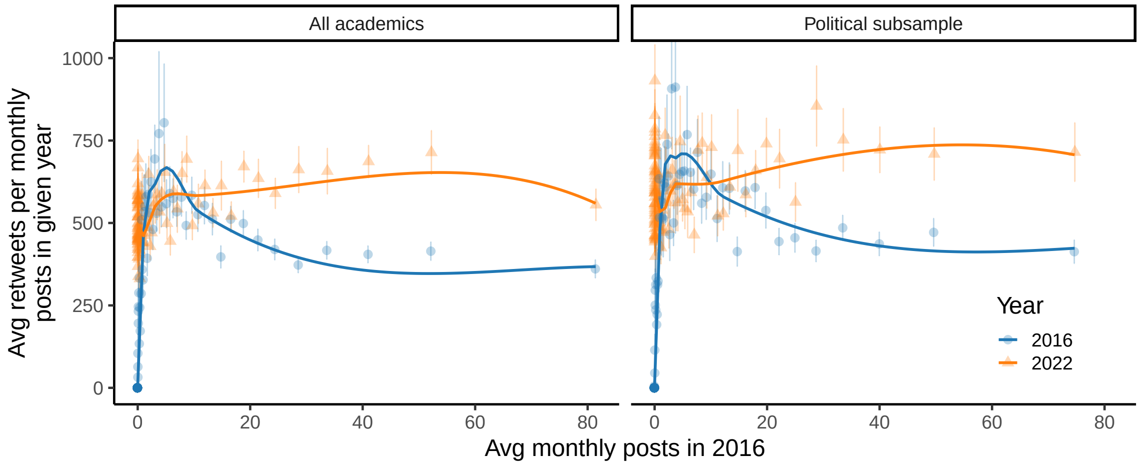
Note: The figure presents the density distributions of net stances across various political topics among U.S. academics from 2016 to 2022, comparing retweets (in red) with original content (in black). Original content includes original tweets, replies and any retweets with additional comments (i.e., quoted retweets). Topics include Income Redistribution, Climate Action, Immigration, Abortion Rights, and Racial Equality, with "Progressive Alignment" representing the average stance across these topics. The x-axis represents the net stance, where positive values indicate a pro stance and negative values indicate an anti stance. The y-axis lists the different topics, with density distributions shown as ridgelines. Vertical mean lines are included for each distribution, colored according to the content type. Kolmogorov-Smirnov (K-S) test results are annotated for each topic, indicating statistically significant differences between the distributions of retweets and original content (p-values < 0.001). Retweets tend to exhibit more polarized stances compared to original content, with the largest divergence observed in Abortion Rights and Climate Action.

FIGURE A.7.—Cross-sectional Ideological Polarization based on Factual vs. Non-factual Content



Note: The figure presents the density distributions of net stances across various political topics among U.S. academics from 2016 to 2022, factual content classification (TRUE, FALSE, UNCLEAR) and non-factual content (default, no verifiable claims). Most tweets cluster around moderate positions (stance ≈0), with slightly more density on the progressive end (+) and less on the conservative end (−). Within more extreme stance regions, `FALSE` tweets are relatively more common among conservative-leaning posts, while `TRUE` tweets are more concentrated among progressive-leaning ones. Racial equality is an exception: `TRUE` tweets are relatively more common on the conservative end, and `FALSE` ones more on the progressive end. Tweet classification was performed using GPT-4o-mini, based on a structured prompt requiring both labeling and extraction of factual claims. Full prompt in Appendix Section D.

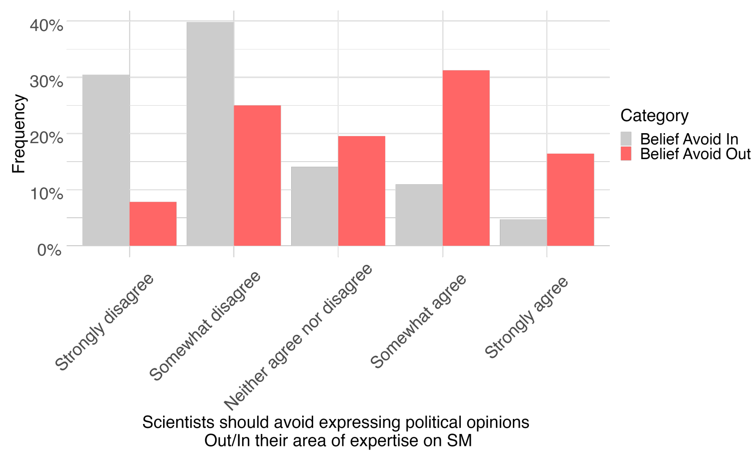
FIGURE A.8.—Engagement vs. early activity: full sample (left) vs. political subsample (right)



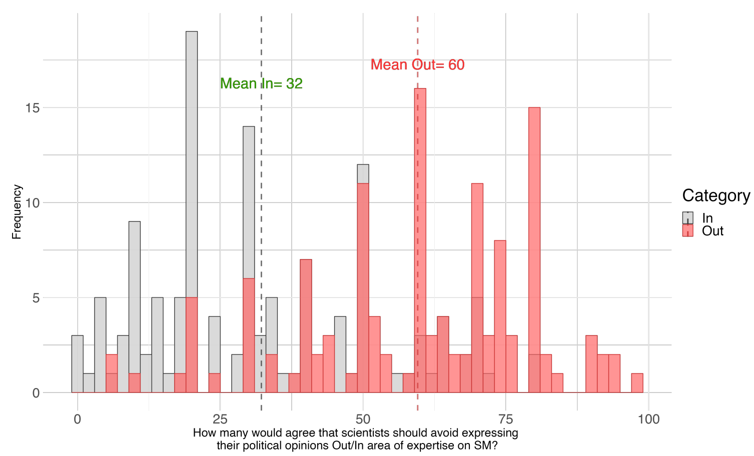
Note: This figure shows average retweets per monthly post in 2016 (blue circles) and 2022 (orange triangles), for academics grouped into 100 equal-sized bins based on their average monthly posting frequency in 2016. In the full sample (left panel), monthly tweet counts in 2016 range from Min = 0.00 to Max = 381.17 (Mean = 4.76; Median = 0.00; 1st Quartile = 0.00; 3rd Quartile = 2.92). In the politically active subsample (right panel)—academics who posted at least once between 2016 and 2022 on a politically salient topic—tweet counts range from Min = 0.00 to Max = 289.33 (Mean = 4.97; Median = 0.08; 1st Quartile = 0.00; 3rd Quartile = 3.92). Both panels display a "hump-shaped" pattern in 2016, where engagement initially increases with baseline posting frequency and then tapers off. Nonetheless, the figure overall shows that academics who were already active in 2016 became significantly more visible over time, with the trend even more pronounced among those who engaged in political topics, who received consistently higher engagement than their peers, both at the beginning and at the end of the period.

FIGURE A.9.—Scientists' Own Beliefs and Belief's on Other Scientists Beliefs around Academics Publicly Expressing Political Views

A. Scientists Should Avoid Expressing Political Opinions



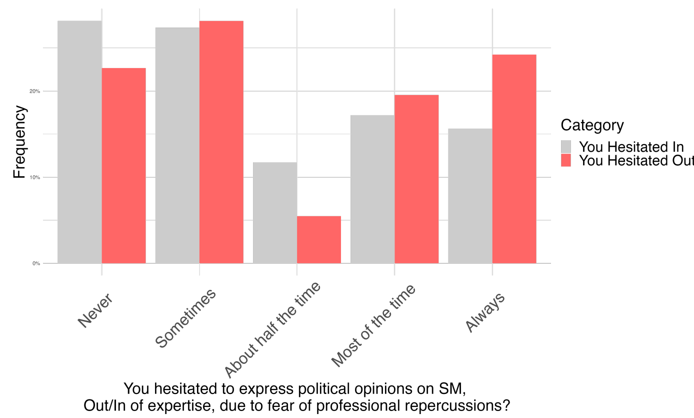
B. How many Scientists Agree on Avoiding to Express Political Opinions



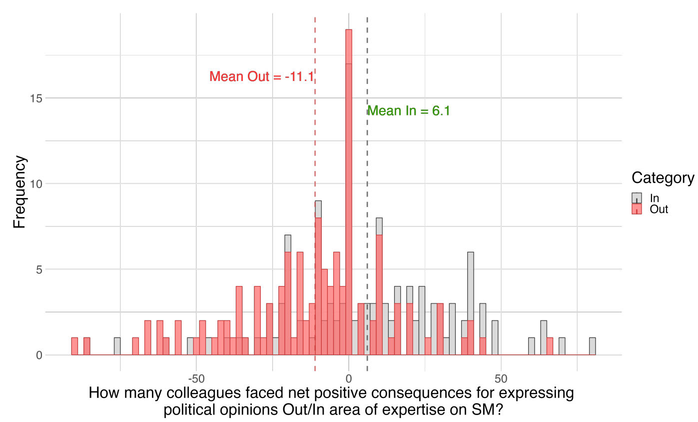
Note: The figure illustrates scientists' own beliefs and their beliefs about other scientists' views on the public expression of political opinions on social media, based on a sample of 128 scientists recruited on Prolific. Panel A shows scientists' responses to their level of agreement with the following statements: *"Scientists and researchers should avoid expressing their political opinions* ***outside*** *their area of expertise on social media."* (red bars) and *"Scientists and researchers should avoid expressing their political opinions* ***within*** *their area of expertise on social media."* (grey bars). For either statement, scientists could choose one of five options: "Strongly disagree," "Somewhat disagree," "Neither agree nor disagree," "Somewhat agree," and "Strongly agree." Our respondents think that expressing political views on social media is more acceptable within their own area of expertise than outside it. Panel B illustrates scientists' responses to the following questions: *Out of 100 scientists and researchers, how many do you think would agree with the statement: "Scientists and researchers should avoid expressing their political opinions* ***outside*** *their area of expertise on social media?"* (red bars) and *Out of 100 scientists and researchers, how many do you think would agree with the statement: "Scientists and researchers should avoid expressing their political opinions* ***about*** *their area of expertise on social media?"* (grey bars). Our respondents believe that other scientists also think that publicly expressing political views outside their own area of expertise is less acceptable than within their own area of expertise.

FIGURE A.10.—Scientists' Experience Hesitating to Express Political Opinions and their Perception around Positive and Negative Consequences from Public Political Expression

A. Have Hesitated to Express Political Opinions



B. Perceived Consequences of Expressing Political Opinions



Note: The figure illustrates scientists' hesitation to express political views online and their perceptions of colleagues facing consequences from public political expression, based on a sample of 128 scientists recruited on Prolific. Panel A shows responses to whether they hesitated to express political opinions on social media due to professional concerns, either *"outside* their area of expertise (red bars) or *"within* their area of expertise (grey bars). Respondents chose from: "Always," "Most of the time," "About half the time," "Sometimes," and "Never." Panel B illustrates responses to the perceived consequences for colleagues expressing political opinions. Scientists estimated how many out of 100 colleagues faced *negative consequences* and *positive consequences* for opinions expressed outside (about) their area of expertise. Red bars represent the difference in responses for areas outside one's expertise, and grey bars represent the same for areas within one's expertise. Scientists anticipate net reputational costs for expressing political opinions outside their field compared to net benefits within their field. On average, scientists believe more colleagues faced *negative* consequences for expressing political opinions outside their area of expertise and more colleagues benefited from expressing views within their area of expertise. However, there is significant variation in responses, with perceptions of costs and benefits largely overlapping for opinions expressed outside versus inside one's area of expertise.

FIGURE A.11.—Vignettes of scientists profiles

Strong Democrat

The profile you are seeing is a **Female** scientist. This scientist works in the field of **American Literature**.

Currently, this scientist is **Assistant Professor** at the **University of Connecticut**.

The scientist is active on X (formerly known as Twitter). The twitter bio of the scientist is: "**Academic. Human rights advocate 🌈✊🏿**"

A recent selected Tweet reads: "**Research compellingly underscores a grave injustice: African American infants and mothers in the socio-economic apex face markedly poorer health outcomes compared to their Caucasian counterparts at the economic base. This stark disparity demands urgent systemic reforms to address deep-rooted inequities.**"

How credible do you think this scientist is?

Not credible at all — Very Credible
0 1 2 3 4 5 6 7 8 9 10

Credible

How credible do you think the scientist's own research is?

Not credible at all — Very Credible
0 1 2 3 4 5 6 7 8 9 10

Credible

How willing you are to read an opinion piece from this scientist?

Not willing at all — Very willing
0 1 2 3 4 5 6 7 8 9 10

Willing to read

Strong Republican

The scientist is active on X (formerly known as Twitter). The twitter bio of the scientist is: "**Academic, Republican, #biblebelieve 🇺🇸**"

A recent selected Tweet reads: "**For those advocating for civil rights and pro-life values (which are inherently linked), take note. There are individuals who have courageously highlighted the inhumane procedures that proponents of abortion, such as @JoeBiden, are pushing for nationwide acceptance and funding. This is unequivocally unacceptable.**

Moderate Republican

The scientist is active on X (formerly known as Twitter). The twitter bio of the scientist is: "**Academic. American. Sharing research, family and community stories 🏡🤝**"

A recent selected Tweet reads: "**Maintaining law and order is critical for the stability of any community. Initiatives to reduce police funding compromise public safety and put our neighborhoods at risk. The pursuit of safety and justice should transcend political boundaries.**"

Neutral (excluded category)

The scientist is active on X (formerly known as Twitter). The twitter bio of the scientist is: "**Academic. Discovering truths of the world. 📚**"

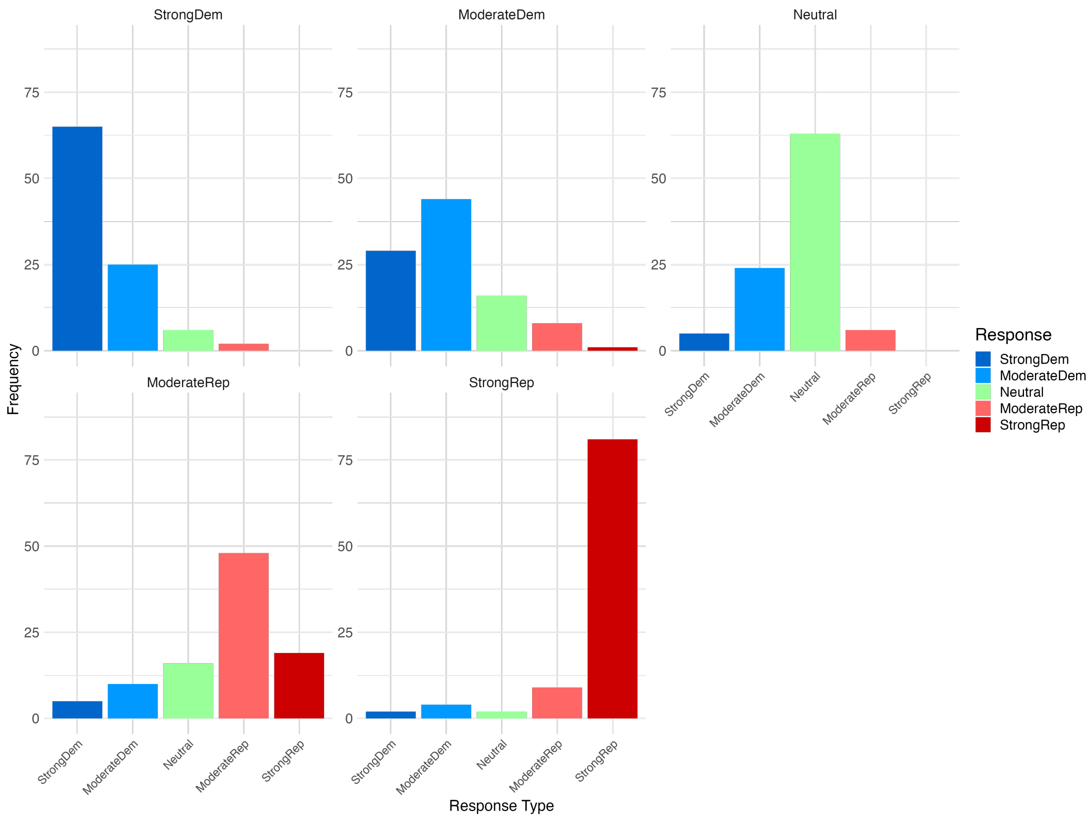
A recent selected Tweet reads: "**On December 5, 1932, eminent physicist Albert Einstein was granted a visa, facilitating his pivotal relocation to the United States, a move that significantly influenced the trajectory of theoretical physics research in the 20th century. #OnThisDay**

Moderate Democrat

The scientist is active on X (formerly known as Twitter). The twitter bio of the scientist is: "**Climber and friend of the environment ⛰️🌱**"

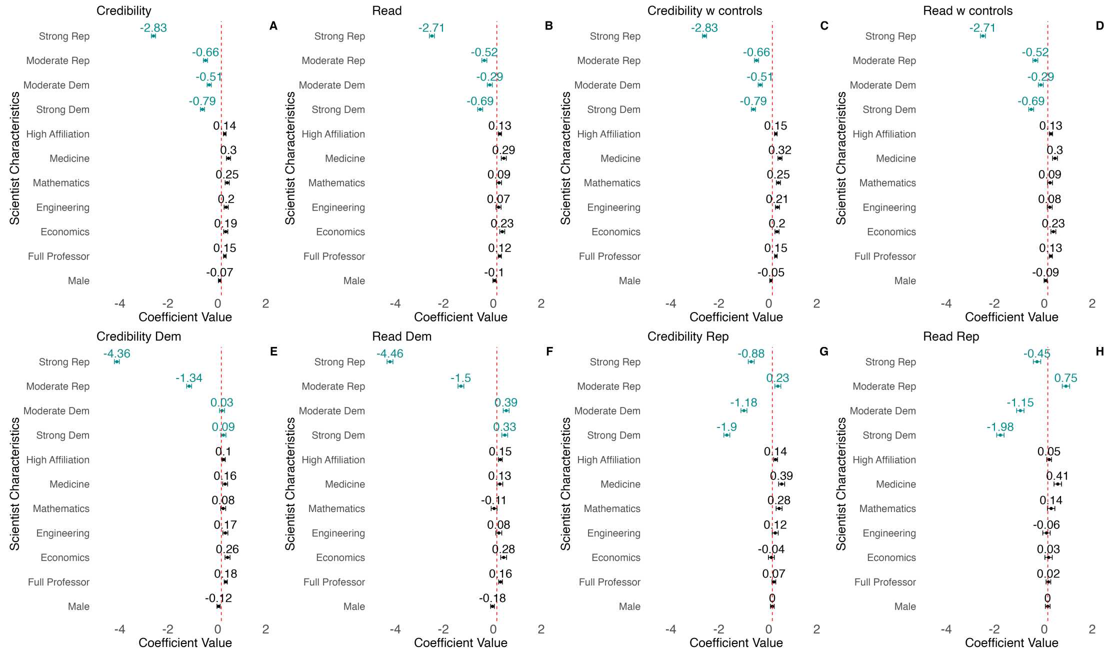
A recent selected Tweet reads: "**Researchers at Exxon precisely forecasted the extent of global warming resulting from fossil fuel combustion in studies starting in 1970s, according to a research paper. Despite this, the company cast skepticism on the findings, contributing to a postponement of government climate initiatives.**

FIGURE A.12.—Frequency of Responses by Intended Political Leaning of Twitter Signal



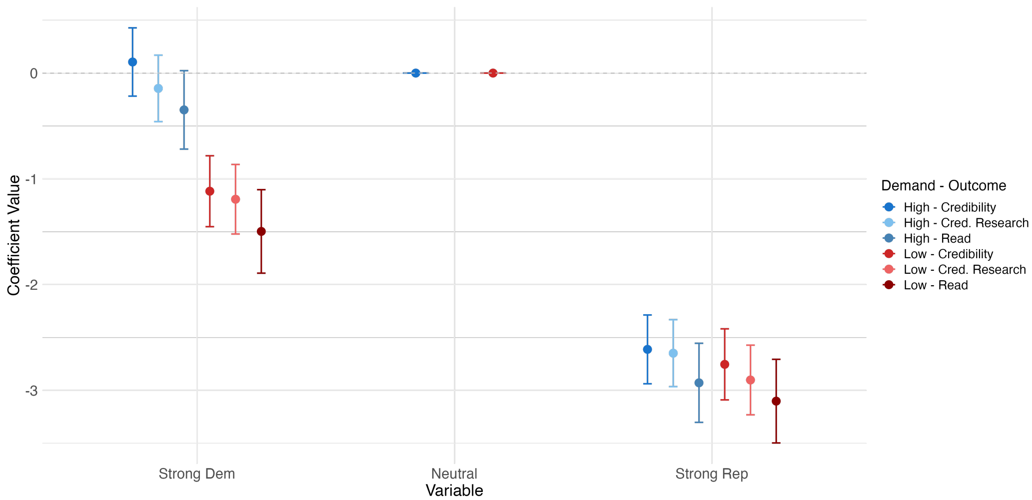
Note: The figure presents the results of a validation of the political signals used in the main experiment. In the validation, we asked 98 respondents, recruited on Prolific, to classify the political leaning of five vignettes. Each vignette displayed one of five different Twitter bios and Twitter posts combinations, which were used in the main experiment. Respondents were presented with each of the five vignettes in random order and asked to classify each into one of five categories: "Strongly Republican," "Moderately Republican," "Strongly Democrat," "Moderately Democrat," and "Neutral." Each plot displays a histogram of responses for each political signal (vignette) used in the main experiment. For each histogram, the mode answer correctly identifies the political leaning of the vignette profile, thereby validating our main exercise.

FIGURE A.13.—Effect of scientists' characteristics on respondents' perceived credibility. Any perceived political leaning of scientists reduces their credibility. Effects are heterogeneous, with Democrats showing reduced credibility for Republican scientists, and vice versa.



Note: Coefficients are obtained by regressing scientists' characteristics on respondents' perceived credibility or willingness to read from scientists. All the standard errors are clustered at the individual level. Political leaning is indicated by "Strong Republican," "Moderate Republican," "Strong Democrat," or "Moderate Democrat," with "Neutral" as the excluded category. High Affiliation signifies institutions such as Harvard, UC Berkeley, or Chicago, versus Arkansas or Connecticut. Research fields include Medicine, Mathematics, Engineering, and Economics, with Literature excluded. Full professor indicates full professors versus assistant professors. Male is coded as one for male scientists. Controls encompass respondents' age, gender, income, ethnicity, education, employment status, religion, region, and political leaning. (N = 1704, 940 Dem. or Lean Dem., 745 Rep. or Lean Rep., 19 Other leaning.)

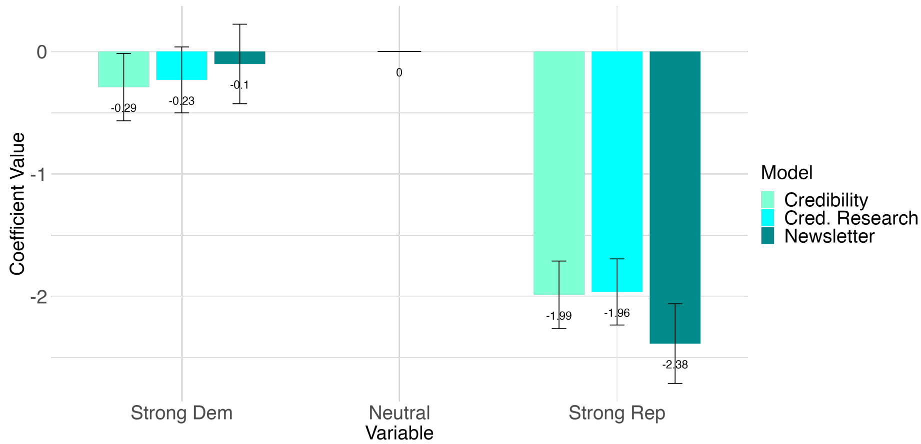
FIGURE A.14.—Bounding the impact of scientists' political expression on perceived credibility and willingness to read: Credibility and public willingness to read peak at neutral, while both left- and right-leaning scientists face a credibility penalty regardless of the demand condition.



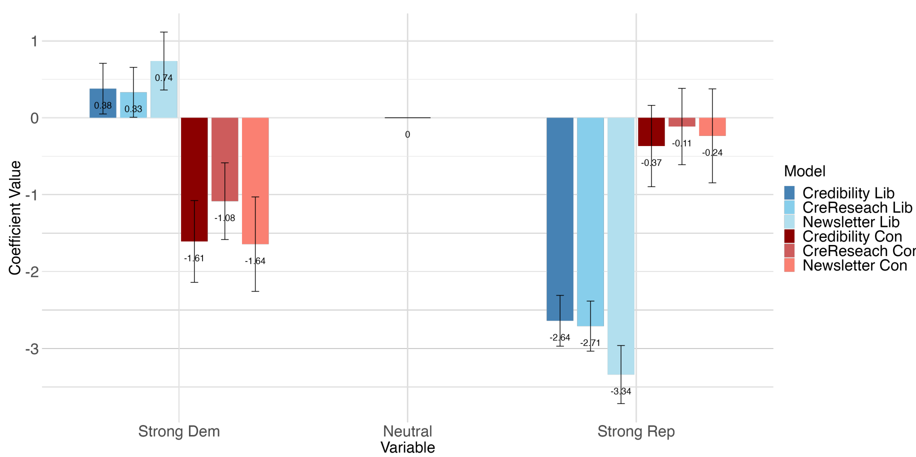
Note: Coefficients are estimated by regressing respondents' perceived credibility and willingness to read opinions from scientists on the scientists' attributes. The x-axis represents different political affiliations of scientists, captured using indicator variables for "Strong Republican" and "Strong Democrat," with "Neutral" as the excluded category. The y-axis displays the estimated coefficients, indicating the impact of political affiliation on credibility and willingness to engage. Respondents were randomly assigned a Neutral scientist profile alongside either a Republican or Democrat profile and were further randomly nudged to rate the latter either higher (blue) or lower (red) relative to the Neutral profile. The results show that credibility and willingness to read peak for Neutral scientists, while left- and right-leaning scientists face credibility penalties, regardless of the demand condition. Standard errors are clustered at the respondent level. (N = 346).

FIGURE A.15.—Impact of scientists' political expression on perceived credibility and willingness to read from the general public. Credibility and public willingness to read peak at neutral, with a penalty for scientists displaying political affiliations to the 'left' and 'right' of neutral (***Journalists***).

A. Base Model

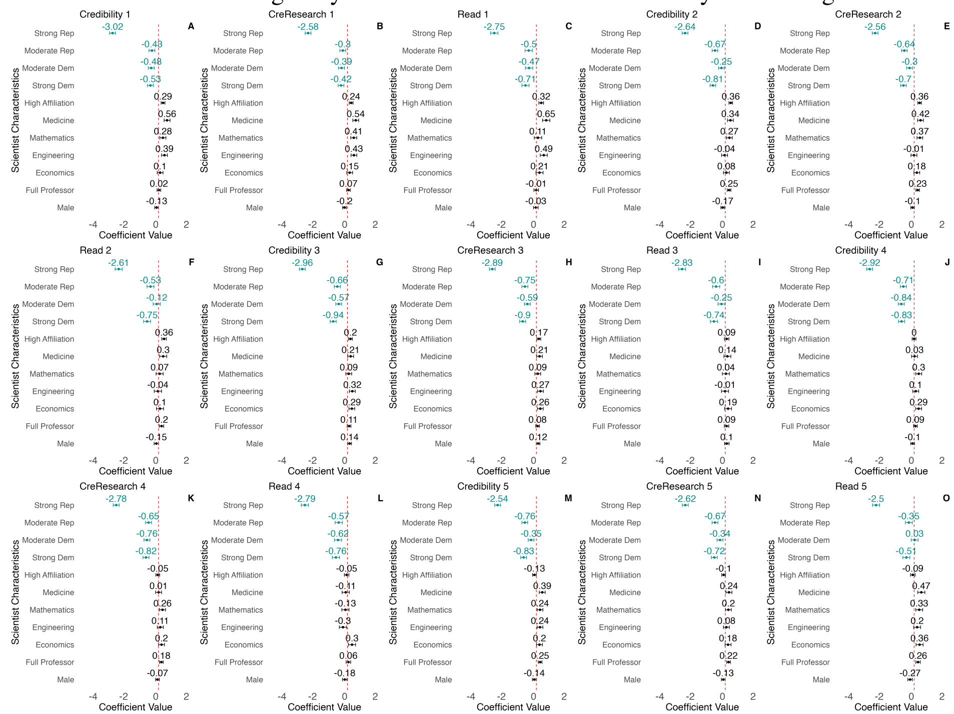


B. Heterogeneity by Journalists' Leaning



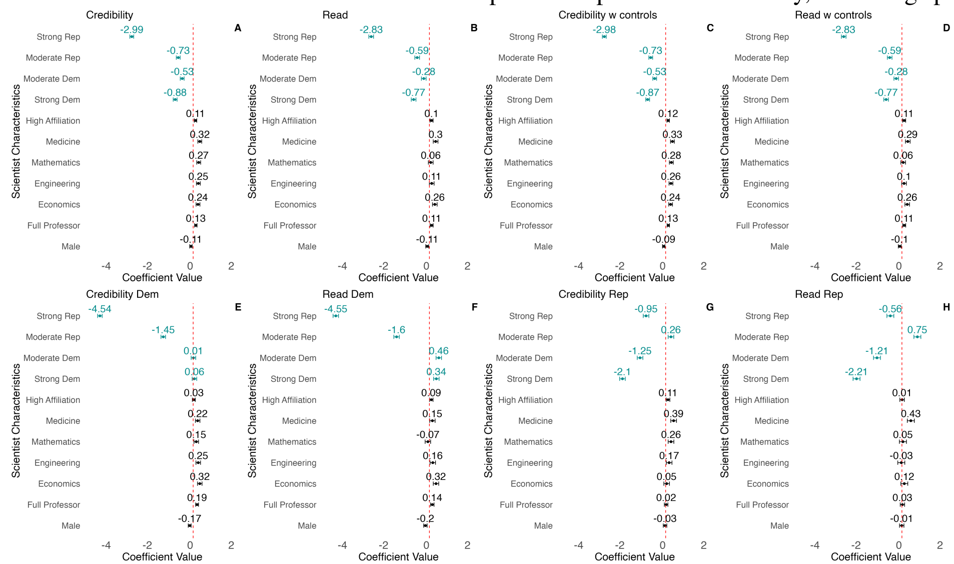
Note: Coefficients are obtained by regressing scientists' characteristics on journalists' perceived credibility or willingness to feature content from scientists. The x-axis represents different political affiliations of scientists, estimated by indicator variables for "Strong Republican" and "Strong Democrat", with "Neutral" as the excluded category. The y-axis shows the coefficient values indicating the impact on credibility and willingness to feature the profile in a newsletter. The data reveals a peak in credibility for neutral scientists, with a decline for both left- and right-leaning scientists. Standard errors are clustered at the individual level. (N = 135, 36 Conservative, 84 Liberal, 15 Moderate leaning.)

FIGURE A.16.—Excluding carryover effects on scientists' credibility and willingness to read.



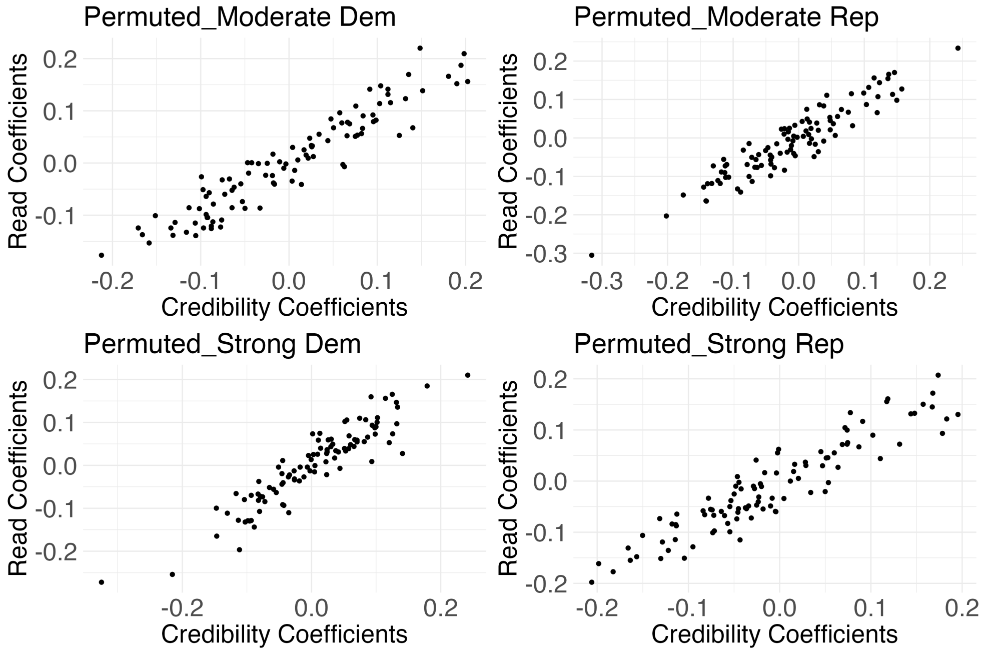
Note: Coefficients were obtained by regressing scientists' characteristics on respondents' perceived credibility or likelihood of reading from similar scientists. We repeat the procedure for each profile the respondents have seen in the study. All the standard errors are clustered at the individual level. Political leaning is indicated by "Strong Republican," "Moderate Republican," "Strong Democrat," or "Moderate Democrat," with "Neutral" as the excluded category. High Affiliation signifies institutions such as Harvard, UC Berkeley, or Chicago, versus Arkansas or Connecticut. Research fields include Medicine, Mathematics, Engineering, and Economics, with Literature excluded. Full professor indicates full professors versus assistant professors. Male is coded as one for male scientists. (N=1704)

FIGURE A.17.—Effect of scientists' attributes on respondents' perceived credibility, excluding speeders.



Note: Coefficients were obtained by regressing scientists' characteristics on respondents' perceived credibility or likelihood of reading from similar scientists. All the standard errors are clustered at the individual level. Political leaning is indicated by "Strong Republican," "Moderate Republican," "Strong Democrat," or "Moderate Democrat," with "Neutral" as the excluded category. High Affiliation signifies institutions such as Harvard, UC Berkeley, or Chicago, versus Arkansas or Connecticut. Research fields include Medicine, Mathematics, Engineering, and Economics, with Literature excluded. Full professor indicates full professors versus assistant professors. Male is coded as one for male scientists. Controls encompass respondents' age, gender, income, ethnicity, education, employment status, religion, region, and political leaning. (N = 1431)

FIGURE A.18.—Permutation Test



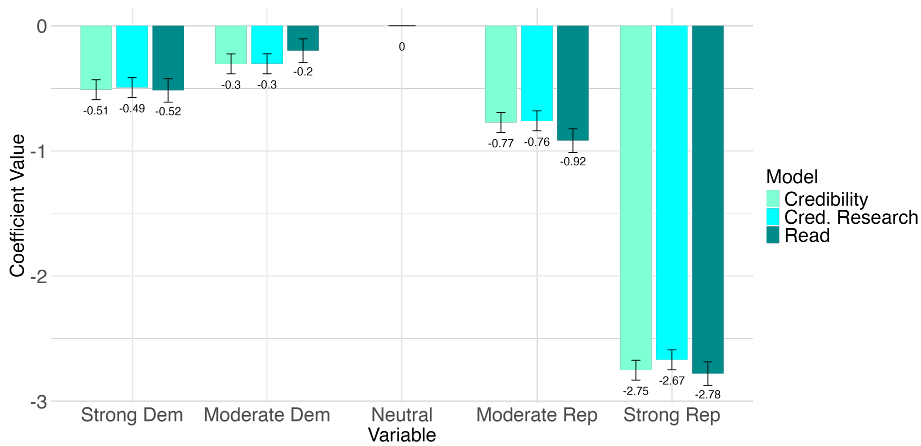
Note: This figure reports the results of a permutation test conducted to assess the robustness of our estimates and ensure that our observed political effects are not due to some unusual feature of the data. To address this, we randomly re-shuffled all political labels across profiles within each respondent, creating a permuted version of the "Political" affiliation of the synthetic scientists' profiles. For each permuted dataset, we ran regressions using these mis-labeled dummy variables to estimate their impact on perceived credibility and willingness to read, repeating the procedure with 100 random permutations. Each scatter plot illustrates the coefficients of the placebo political affiliation of scientists on their perceived credibility (x-axis) and on respondents' willingness to read (y-axis) for the different political profile permutations. The consistent patterns across these plots indicate that the permuted labels do not systematically influence our main effects, as all coefficients remain close to zero and smaller than our estimates, demonstrating that our original findings are not driven by any peculiarities in the data, thereby affirming the robustness of our results.

FIGURE A.19.—Vignettes of economists profiles

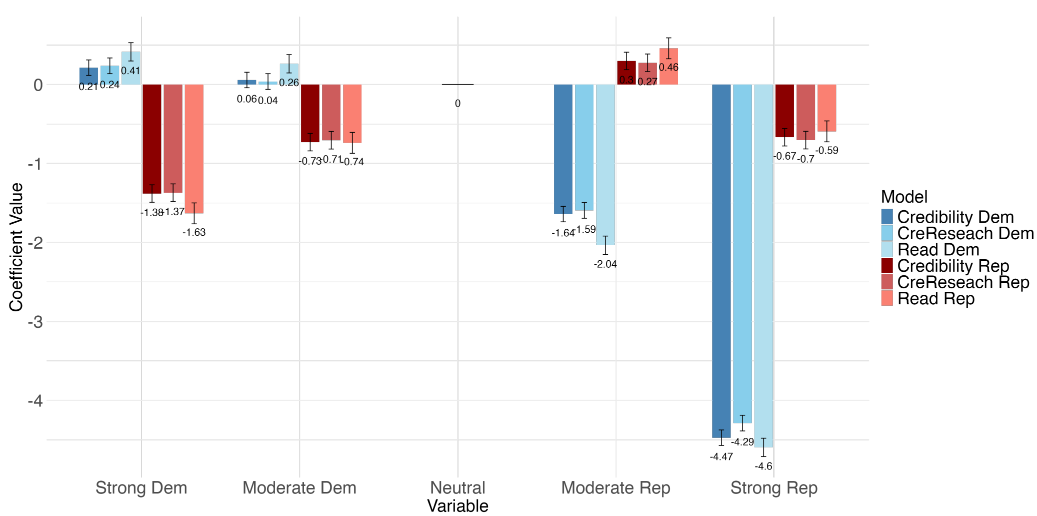
Passive Control 1/6	**Treatment Left (signal)** 1/3
The economist is active on X (formerly known as Twitter). The twitter bio of the economist is: "**Passionate about Research**". This is an example of a tweet: **"In our recent paper, we show that Nash equilibrium uniquely satisfies key axioms across different games, challenging refinement theories. Our findings have implications for zero-sum, potential, and graphical games."**	The economist is active on X (formerly known as Twitter). The twitter bio of the economist is: "**Passionate about Research and Advocate for Equality 🌍**". This is an example of a tweet: **"Our latest study in Lancet Global Health provides evidence on the health impacts of hostile environment policies toward migrants: restrictive entry and integration policies are linked to poorer mental and general health, and a higher risk of death."**
Active Control 1/6	**Treatment Right (signal)** 1/3
The economist is active on X (formerly known as Twitter). The twitter bio of the economist is: "**Passionate about Research**". This is an example of a tweet: **"Our latest study in Lancet Global Health provides evidence on the health impacts of hostile environment policies toward migrants: restrictive entry and integration policies are linked to poorer mental and general health, and a higher risk of death."**	The economist active on X (formerly known as Twitter). The twitter bio of the economist is: "**Passionate about Research and Proud Patriot 🦅**". This is an example of a tweet: **"Our latest study in Lancet Global Health provides evidence on the health impacts of hostile environment policies toward migrants: restrictive entry and integration policies are linked to poorer mental and general health, and a higher risk of death."**

FIGURE A.20.—Impact of scientists' political expression on perceived credibility and willingness to read from Twitter users. Credibility and public willingness to read peak at neutral, with a monotonic penalty for scientists displaying political affiliations to the 'left' and 'right' of neutral.

A. Base Model



B. Heterogeneity by Respondents' Partisanship



Note: Coefficients are obtained by regressing scientists' characteristics on respondents' perceived credibility or willingness to read content from scientists. The x-axis represents different political affiliations of scientists, estimated by indicator variables for "Strong Republican", "Moderate Republican", "Strong Democrat", or "Moderate Democrat", with "Neutral" as the excluded category. The y-axis shows the coefficient values indicating the impact on credibility and willingness to read. The data reveals a peak in credibility for neutral scientists, with a decline for both left- and right-leaning scientists. Standard errors are clustered at the individual level. Additional regressors include indicator variables to control for other scientist characteristics: institutional affiliation (Harvard, UC Berkeley, or Chicago, versus Arkansas or Connecticut), field of research (Medicine, Mathematics, Engineering, Economics, or Literature), seniority of role (Full professor or Assistant Professor), and gender (male or female). (N = 2,000, 1,081 Dem. or Lean Dem., 906 Rep. or Lean Rep., 13 Other leaning or unsure.)

FIGURE A.21.—Wording of Treatment Mechanism Experiment

You are matched with a (migration expert) economist
who posted the following factual tweet on their personal social media account.

Promotion	No-Promotion
Before writing and sharing the post, the economist was informed that we (the researchers) would promote the post to increase its visibility and reach. Promoted posts on X/Twitter are displayed to a broader audience beyond the economist's current followers. Visibility typically extends to users with similar interests (such as other academics, policy professionals, or journalists) based on engagement patterns and topical relevance.	No additional information.

Neutral	Left	Right
Tweet: "In 2022, the U.S. admitted approximately 1 million legal immigrants, maintaining a historical trend of high immigration levels since the 1990s."	Tweet: "In 2022, the U.S. admitted approximately 1 million legal immigrants, maintaining a historical trend of high immigration levels since the 1990s. Immigrants play a key role in the U.S. economy — recent data show that over 20% of all new businesses were started by first-generation immigrants."	Tweet: "In 2022, the U.S. admitted approximately 1 million legal immigrants, maintaining a historical trend of high immigration levels since the 1990s. At the same time, the U.S. recorded over 2.4 million border encounters in 2023 — a record high — raising concerns over illegal immigration, enforcement failures, and border resources."

APPENDIX B: ONLINE TABLES

TABLE B.1

OVERVIEW OF THE EXPERIMENTAL DESIGN

Study	**Participants**	**Description**
Study 1	1,704 respondents from representative sample of U.S. population	Measuring credibility penalty for expressive academics and disentangling impact of signaling political identity vs. posting on salient research
Replication of Study 1	2,000 respondents from representative sample of U.S. population	Replicate Study 1 with modified outcomes to test robustness
Journalist Study	135 international journalists	Replicate a simplified Study 1 with three vignettes among journalists to assess generalizability to a professional media audience
Scientists Study 1	128 international scientists	Measure scientists' perceptions of credibility penalty and perceived norms around political expression in their field
Scientists Study 2	118 international scientists	Measure scientists' perceived professional costs and benefits of political expression on social media
Robustness Experiment	354 U.S. respondents	Simplified Study 1 used to bound experimenter demand effects through treatment-neutral design
Mechanism Study	2,000 U.S. active Twitter users	Replicate Study 1 and test whether beliefs about scientists' strategic motives behind political expression moderate the credibility penalty

Note: *Study 1* is the core experiment used to identify the credibility cost of political expression by academics. The *Replication of Study 1* tests robustness to measurement variation. The *Journalist Study* probes external validity using journalist respondents. *Scientists Studies 1 and 2* measure scientists' beliefs about reputational consequences and norms. The *Robustness Experiment* addresses experimenter demand. *Study 2* explores whether perceived strategic motives attenuate credibility penalties.

TABLE B.2

SUMMARY STATISTICS OF SCIENTIST LEVEL CHARACTERISTICS

Variables	**N**	**% (Filtered)**	**% Politicized (Filtered)**	**% Politicized (Full data)**
Scientists (Full)	97,737	-	-	43.7
Scientists (Filtered)	52,541	100	81.4	-
Male	28,998	55.2	78.3	40.0
Female	22,442	42.7	85.4	49.6
Other	1,101	2.1	79.3	-
Citations: 1-100	19,285	36.7	82.3	41.4
Citations: 101-500	14,097	26.8	80.9	46.0
Citations: 501-1000	5,859	11.1	80.5	44.2
Citations: 1000+	13,299	25.3	80.9	45.0
High Twitter Reach	25,688	50.1	87.8	53.1
Low Twitter Reach	25,583	49.9	74.8	36.3
Field: With Concepts Data	25,719	49.0	81.4	51.7
Field: Humanities	103	0.4	86.4	57.8
Field: STEM	11,819	45.95	79.5	42.5
Field: Social Sciences	6,032	23.5	86.0	64.9
Field: Medicine	7,765	30.19	80.6	38.3

Note: Table shows individual-level summary statistics on key characteristics of scientists. For some key categories relevant to our experiment, we show a breakdown by the number of observations, the proportion of those who tweeted about any of our topics, and among them, the proportion of those who are *politicized* (i.e., whether they have made at least one pro or anti tweet on one of our five topics in the cross-section from 2016 to 2022). The "Filtered" column refers to the subset of scientists who have tweeted about a political topic (pro, anti, or neutral). The "% Politicized" refers to the subset of scientists who have made at least one pro or anti tweet. High and low Twitter reach is defined based on whether the follower count is above or below the median of the filtered dataset (median = 522; 1st quartile = 219; 3rd quartile = 1,302; max = 4,587,745). "With Concepts Data" refers to those for whom we have concepts data. Above 40% of our full sample of academics ever talked about one of the topics of interest during the period of observation.

TABLE B.3

SUMMARY STATISTICS OF TOPIC AND STANCE DETECTION

Topics	**N. Tweets (Full data)**	**% All Tweets (Full data)**	**N. Tweets (Sampled)**	**% All Tweets (Sampled)**	**% Pro**	**% Neutral**	**% Anti**	**% Mention Politician**	**% Mention Trump Biden**	**% Mention Research**
Climate Action	2,423,954	2.09%	97,587	0.08	28	70	2	11.57	3.40	44.50
Immigration	995,558	0.86%	79,892	0.06	20	73	7	21.46	6.57	21.41
Racial Equality	1,738,049	1.50%	79,986	0.07	73	12	15	14.24	3.26	25.99
Abortion Rights	287,346	0.254%	31,351	0.03	37	58	5	21.53	4.07	15.03
Income Redistribution	706,886	0.61%	61,683	0.05	21	74	5	15.34	3.57	25.06
Topical Tweets	6,151,793	5.31%	350,499	0.30	-	-	-	16.01	4.19	28.91
All Tweets	115,744,660	100%	-	-	-	-	-	8.55	1.21	19.22

Note: Table shows tweet-level summary statistics of topic and stance detection steps. The dataset and classification methods are described in detail in Section D. We reproduce here the essential methods for variables used in this paper. The data contains the entirety of these academics' Twitter activities from January 1, 2016, to December 31, 2022. This included original tweets, retweets, quoted retweets, and replies, totaling around 116 million tweets. Topic detection was the primary step in our methodology of stance classification, aiming first to categorize tweets into one of the predefined topics: (1) Abortion Rights, (2) Climate Action, (3) Immigration, (4) Racial Equality, (5) Income Redistribution. This approach is further demonstrated in writefilebrfwritefilebrfwritefilebrfwritefilebrfwritefilebrfwritefilebrfwritefilebrfwritefilebrfwritefilebrfwritefilebrfwritefilebrfGarg and Fetzer (2025). OpenAI's GPT-4 was used to generate dynamic keyword dictionaries to capture the evolving discourse on these subjects. For stance detection, we employed OpenAI's GPT-3.5 Turbo. Tweets were classified into one of four stances: pro, anti, neutral, or unrelated. This was done using the prompt "Classify this tweet's stance towards <topic> as 'pro', 'anti', 'neutral', or 'unrelated'. Tweet: <tweet>.'" A sampling procedure was employed to reduce the total costs of this tweet-by-tweet labeling task. For each year by month, up to three random tweets per author per topic were included in the sample. This ensured we have enough tweets to determine the stance of an author in a given time period. The stance detection results refer to the sampled tweet sample. The final three columns on "% Mention" show results from an additional topic detection step. The "% Mention Politician" column represents the percentage of tweets mentioning any politician or political candidate (including Trump or Biden). The "% Mention Trump/Biden" column represents the percentage of tweets mentioning either Joe Biden or Donald Trump. The "% Mention Research" column represents the percentage of tweets mentioning scientific research papers.

TABLE B.4

EXAMPLE NGRAMS FOR TOPIC DETECTION

Topic	**Example ngrams**
Abortion	abortion, abortion rights, planned parenthood, pro-choice, pro-life
Climate Action	renewable energy, protect the environment, climatehoax, global warming
Immigration	deportation, immigration, undocumented, migrants, ice detention centers
Racial Equality	race relations, black lives matter, xenophobia, affirmative action, #sayhername
Income Redistribution	welfare state, taxation, #ubi, income level, social safety net
Donald Trump	maga, trump administration, trump tower, Russia investigation, #trumptrain
Joe Biden	#buildbackbetter, bidenharris2020, Afghanistan troop withdrawal, biden's first 100 days
Politicians	candidate forum, presidential candidates, vote, swing state, campaign ads
Research	research impact, sample size, researchgate, clinical trials, peer review

Note: Table shows example ngrams used in the topic detection step of our methodology. We used OpenAI's GPT-4 family of models to generate dynamic keyword dictionaries to capture the evolving discourse on these subjects. The prompt used was "Provide a list of <ngrams> related to the topic of <topic> in the year <year>. <twitter fine tuning>. Provide the <ngrams> as a comma-separated list." This process was repeated for each combination of topic, ngram, year, and vernacular type, resulting in 180 prompts. The generated keywords were combined at the topic level and applied to the full corpus of tweets. Tweets containing keywords from a topic's dictionary were labeled as belonging to that topic. These example ngrams are chosen to illustrate the diversity of responses we can obtain.

TABLE B.5

EVALUATION METRICS: GPT 3.5 TURBO, GPT-4, AND TOPIC DETECTION

Task	**Target**	**GPT 3.5 Turbo** (F_{avg})	**GPT 4** (F_{avg})	**Topic Detection** (F_{avg})
A	Feminism	92.44	81.89	67.01
A	Hillary Clinton	89.57	87.53	67.35
A	Abortion	79.52	84.36	74.87
B	Donald Trump	84.18	80.00	71.84

Note: The table presents validation results for stance detection using both GPT-3.5 Turbo and GPT-4 models, comparing their performance on the ACM SemEval-2016 Task 6 dataset. GPT-3.5 Turbo achieved F_{avg} scores ranging from 79.52 to 92.44, with GPT-4 showing slightly better performance on Abortion (84.36) but generally similar results. Topic detection was validated using dictionaries generated from GPT-4, capturing evolving lexical patterns for the same topics. True positives, true negatives, false positives, and false negatives were calculated to measure the accuracy of topic detection, achieving F_{avg} scores of 67.01 to 74.87, indicating high recall and precision in filtering relevant tweets. For further comparisons and details on stance and topic detection validation, see Garg and Fetzer (2025).

TABLE B.6

EXAMPLES OF TWEETS BY STANCE

Stance	**Example Tweet**
Income Redistribution	
Pro	when someone runs an experiment asking" what happens if you give people some money" the answer is, without fail, "their life gets better." No amount of research validating and re-validating this will ever be enough for the politicians who demand suffering as penance for poverty.
Anti	#Civilrights/#prolife colleagues (same thing), FYI. '@daviddaleiden is a national hero for exposing these barbaric practices that #abortion zealots like @JoeBiden want all Americans to approve and fund.'
Neutral	Do corporate tax cuts boost growth? Our paper is out @ European Economic Review. We meta-analyse 441 estimates from 42 studies; results imply: the attention corporate taxation has received as a source of growth has often been exaggerated.
Climate Action	
Pro	Do you remember the famous 97% study - that 97% of climate science supported the consensus on human-caused climate change? Well, we have just published an update for 2012-2021 papers in the same journal, Environmental Research Letters. The figure is now... drumroll please...99.9%!
Anti	The concept of global warming was created by and for the Chinese in order to make U.S. manufacturing non-competitive.
Neutral	The most significant contribution among the highest emitters is from air and land transport, with 41% and 21% among the top 1% of EU households. Air transport is by far the most income-elastic, unequal and carbon-intensive consumption category in our study.
Immigration	
Pro	Our new research in @LancetGH provides evidence of the health effects of hostile environment policies to migrants: restrictive entry and integration policies are associated with worse mental and general health, and an increased risk of death.
Anti	National sovereignty and border security are paramount. Open borders policies invite chaos and undermine the rule of law. A nation must control its borders to protect its citizens and uphold its values.
Neutral	Finally ready to share my paper on individualistic Scandinavian emigrants, and how their departure during the Age of Mass Migration generated lasting cultural change towards collectivism and convergence across migrant-sending districts.
Abortion Rights	
Pro	Texas' latest abortion ban, #SB8, gives people the right to sue those who provide or help others get an abortion after 6 weeks. Bans like these are not based in science and the consequences could potentially be disastrous. Here's what our research says.
Anti	Let's Make Abortion UNTHINKABLE! Who's with me? #prolife #unborn #bhfyp #alllivesmatter #hope #endabortion #prolifegen
Neutral	In the wake of a gene-editing experiment gone wrong, the president of the National Catholic Bioethics Center said that the Church must stand firm against the unborn being "sacrificed on the altar of scientific research."
Racial Equality	
Pro	An article came out in @TheLancet today that is flying under the radar but is absolutely critical to read. It provides rare CAUSAL evidence showing structural racism causes poor health outcomes for Black Americans. Here's the science in a quick thread.
Anti	My study of northern backlash against the Great Migration has no policy prescription, but it has a smoking gun. Police are the only public investment to increase in metro areas w/ more black migration. Good faith pursuit of racial justice starts by questioning this institution.
Neutral	We document the appearance of a new race gap in traffic deaths that emerged after 2014. In fact, this was the first time that the rate of traffic deaths for Black Americans exceeded that of White Americans since at least the early 1970s. Our paper tries to unravel this mystery.

Note: The table presents examples of tweets by stance (pro, anti, neutral) for the five topics: Income Redistribution, Climate Action, Immigration, Abortion Rights, and Racial Equality. These tweets are drawn from the complete timelines of academics, which include (1) original tweets, (2) retweets, (3) quote retweets, and (4) replies. Consequently, some tweets may reflect content shared or quoted by academics rather than authored directly by them.

TABLE B.7

SUMMARY STATISTICS OF SCIENTISTS

	Survey 1	Survey 2
Institute: University	0.63	0.65
Institute: Research Institute (including public agencies)	0.17	0.16
Institute: Private institute	0.18	0.14
Institute: Non profit	0.02	0.02
Institute: Hospital/clinic/facility	0.01	0.02
Seniority: Less than 1 year	0.08	0.11
Seniority: Between 1 year and 3 years	0.19	0.18
Seniority: Between 3 years and 5 years	0.26	0.2
Seniority: More than 5 years	0.48	0.51
Position: Postdoctoral researcher	0.43	0.49
Position: University faculty	0.28	0.32
Position: Industry professional	0.29	0.19
Field: Arts & Humanities	0.05	0.10
Field: Life Sciences & Biomedicine	0.34	0.20
Field: Physical Sciences	0.11	0.09
Field: Social Sciences	0.34	0.43
Field: Technology	0.16	0.17
Employment: Working full time now	0.89	0.81
Employment: Working part time now	0.05	0.08
Employment: Unemployed	0.03	0.02
Employment: Retired	0.01	0.01
Male	0.51	0.45
Female	0.46	0.52
Non-binary	0.03	0.03
Republican	0.02	0.06
Democrat	0.42	0.37
Independent	0.18	0.20
Other	0.28	0.22
Not sure	0.10	0.14

Note: The Scientists samples recruited on Prolific. The characteristics are broken down into different dimensions.

TABLE B.8

SUMMARY OF SCIENTISTS' RESPONSES TO OPEN-ENDED QUESTIONS

Question	**Summary of Responses**	**Example Response**
In your own words, **what qualifies a scientist as 'politicized'** when they actively engage in political discourse? Where do you draw the line between sharing an opinion and promoting an ideology?	Scientists are often seen as "politicized" when their engagement shifts from presenting evidence-based insights to actively promoting specific ideologies or agendas. Intent, evidence basis, and context are key factors in distinguishing opinions from ideology.	*A scientist becomes 'politicized' when they share opinions that are politically laden and fall outside their area of expertise; the key distinction lies in intent.*
In a previous survey, scientists found it broadly acceptable to share political views related to their expertise. In your own words, **what are the boundaries of that expertise** (e.g., own literature, sub-field, discipline)?	The boundaries of expertise include a scientist's research area, sub-field, and relevant literature. While it is acceptable to share views directly related to one's expertise, caution and evidence-based claims are essential to maintain credibility.	*The boundaries of a scientist's expertise encompass their research, broader sub-field, and established knowledge, provided their claims are supported by evidence and consensus.*
Would you like to **share any personal experiences** regarding the costs or benefits of sharing political views online or in public?	Respondents expressed mixed views, highlighting potential professional risks (e.g., backlash, reputation loss, job security) but also noting opportunities for raising awareness and fostering discourse on important issues within their expertise.	*Sharing political views online can help inform the public about important issues, but it can also lead to backlash or harm professional relationships, making caution essential.*

TABLE B.9

SUMMARY STATISTICS MAIN STUDY

	Population	Sample
Income: $<$ 30,000	0.51	0.17
Income: 30-59,999	0.26	0.25
Income: 60-99,999	0.14	0.27
Income: 100-149,999	0.06	0.19
Income: $>$ 149,999	0.04	0.11
Age: 18-34	0.30	0.29
Age: 35-44	0.16	0.18
Age: 45-54	0.16	0.16
Age: 55-64	0.17	0.24
Age: $>$ 64	0.21	0.13
Ethnicity: White	0.7	0.73
Edu: Up to Highschool	0.39	0.26
Edu: Some college	0.22	0.20
Edu: Bachelor or Associate	0.28	0.35
Edu: Masters or above	0.11	0.19
Region: West	0.24	0.17
Region: North-east	0.17	0.22
Region: South	0.38	0.40
Region: Mid-west	0.21	0.21
Male	0.49	0.49
Republican	0.28	0.28
Democrat	0.32	0.31

Outcome	Mean
Credibility	6.35
Credibility Research	6.27
Read	5.63

Note: The population average demographics are computed using the 2022 American Community Survey (ACS) 1-year estimates. The ACS sample includes only individuals above the age of 18. The population share of Republicans is obtained from the average share of people identifying as Republicans across multiple surveys conducted in 2024 by Gallup.

TABLE B.10

SUMMARY STATISTICS MECHANISM STUDY

	Population	Sample
Income: $<$ 30,000	0.51	0.14
Income: 30-59,999	0.26	0.25
Income: 60-99,999	0.14	0.27
Income: 100-149,999	0.06	0.21
Income: $>$ 149,999	0.04	0.13
Age: 18-34	0.30	0.41
Age: 35-44	0.16	0.24
Age: 45-54	0.16	0.18
Age: 55-64	0.17	0.11
Age: $>$ 64	0.21	0.06
Ethnicity: White	0.7	0.69
Edu: Up to Highschool	0.39	0.10
Edu: Some college	0.22	0.17
Edu: Bachelor or Associate	0.28	0.40
Edu: Masters or above	0.11	0.23
Region: West	0.24	0.18
Region: North-east	0.17	0.21
Region: South	0.38	0.40
Region: Mid-west	0.21	0.21
Twitter: Every day		0.37
Twitter: 4-5 days per week		0.17
Twitter: 3 days per week		0.11
Twitter: 1-2 days per week		0.20
Twitter: Not at all		0.15
Male	0.49	0.49
Republican	0.28	0.32
Democrat	0.32	0.36

Outcome	Mean
Credibility	6.46
Credibility Research	6.40
Read	5.93

Note: The population average demographics are computed using the 2022 American Community Survey (ACS) 1-year estimates. The ACS sample includes only individuals above the age of 18. The population share of Republicans is obtained from the average share of people identifying as Republicans across multiple surveys conducted in 2024 by Gallup.

TABLE B.11

CHARACTERISTICS OF THE SCIENTISTS' PROFILES

Attributes	**Categories**	**Options**
Gender	Male, Female	We specify the gender
Research Field	Social Sciences, STEM, Medicine, and Humanities	We mention: Economics, Material Engineering, Mathematics, Medicine, American Literature
Seniority	Senior, Junior	We mention that scientists are: Full Professor or Assistant Professor
University Affiliation	High-ranked, Low-ranked	We use affiliations to Harvard University, Berkeley, University of Chicago, University of Arkansas, University of Connecticut
Twitter Bio and Twitter Post	Strongly Dem, Moderately Dem, Strongly Rep, Moderately Rep, Neutral	Academic. Human rights advocate [rainbow and fist emoji] - "Greta has been arrested for the first time. This signals a moment for more of us to rise and face arrest if necessary, for the future of our planet. Such actions have the power to change the course of events.", Academic. Friend of the environment [wave emoji] - "Researchers at Exxon precisely forecasted the extent of global warming resulting from fossil fuel combustion in studies starting in 1970s, according to a research paper. Despite this, the company cast skepticism on the findings, contributing to a postponement of government climate initiatives. ", Academic. Republican. #biblebelieve [American flag] - "For those advocating for civil rights and pro-life values (which are inherently linked), take note. There are individuals who have courageously highlighted the inhumane procedures that proponents of abortion, such as @JoeBiden, are pushing for nationwide acceptance and funding. This is unequivocally unacceptable", Academic. American. Sharing research, family and community stories [house and handshake emoji] - "I'm not inclined towards the right or the left, but the excessive wokeness of the left has nudged me to the right. Interestingly, when right-wing extremists commit mass shootings against minorities, it doesn't compel me to shift towards the left. Somehow, that's not considered 'too far.'", Academic. Discovering truths of the world [books emoji] - "On December 5, 1932, Albert Einstein received a visa, enabling his journey to the United States. #OnThisDay"
Twitter Bio and Twitter Post (Cross-randomization)	Dem, Rep, Active Control, Pure Control	Passionate about Research and Advocate for Equality [Earth emoji] - "Our latest study in Lancet Global Health provides evidence on the health impacts of hostile environment policies toward migrants: restrictive entry and integration policies are linked to poorer mental and general health, and a higher risk of death.", Passionate about Research and Proud Patriot [Eagle emoji] - "Our latest study in Lancet Global Health provides evidence on the health impacts of hostile environment policies toward migrants: restrictive entry and integration policies are linked to poorer mental and general health, and a higher risk of death.", Passionate about Research - "Our latest study in Lancet Global Health provides evidence on the health impacts of hostile environment policies toward migrants: restrictive entry and integration policies are linked to poorer mental and general health, and a higher risk of death.", Passionate about Research - "In our recent paper, we show that Nash equilibrium uniquely satisfies key axioms across different games, challenging refinement theories. Our findings have implications for zero-sum, potential, and graphical games."

Note: This table provides an overview of the characteristics of the scientists we manipulate in the conjoint experiment and in the last task.

TABLE B.12

SCIENTISTS' PROFILE CREDIBILITY BY SCIENTISTS' POLITICAL AFFILIATION

	Credibility of Scientists by Profile Type:				
	Strong *Rep*	Moderate *Rep*	Neutral	Moderate *Dem*	Strong *Dem*
Male	-0.060	-0.165	-0.014	-0.116	0.021
	(0.150)	(0.117)	(0.097)	(0.110)	(0.129)
Full Professor	-0.024	0.214^{*}	0.313^{***}	-0.045	0.267^{**}
	(0.150)	(0.117)	(0.097)	(0.110)	(0.129)
Economics	0.356	0.288	-0.029	0.410^{**}	-0.105
	(0.234)	(0.187)	(0.154)	(0.171)	(0.201)
Engineering	0.247	0.141	0.075	0.384^{**}	0.168
	(0.230)	(0.189)	(0.151)	(0.172)	(0.212)
Mathematics	0.094	0.362^{**}	0.007	0.549^{***}	0.169
	(0.238)	(0.184)	(0.153)	(0.174)	(0.204)
Medicine	0.084	-0.004	0.134	0.871^{***}	0.389^{*}
	(0.230)	(0.189)	(0.154)	(0.170)	(0.208)
High Affiliation	0.254^{*}	0.274^{**}	0.088	0.337^{***}	-0.229^{*}
	(0.152)	(0.120)	(0.099)	(0.111)	(0.132)
Constant	4.210^{***}	6.294^{***}	7.067^{***}	6.244^{***}	6.396^{***}
	(0.208)	(0.174)	(0.139)	(0.156)	(0.189)
Observations	1,704	1,704	1,704	1,704	1,704

Note: Coefficients were obtained by regressing scientists' characteristics on respondents' perceived credibility. All the standard errors are clustered at the individual level. Each column represents a different scientist based on the political affiliation. High Affiliation signifies institutions such as Harvard, UC Berkeley, or Chicago, versus Arkansas or Connecticut. Research fields include Medicine, Mathematics, Engineering, and Economics, with Literature excluded. Full professor indicates full professors versus assistant professors. Male is coded as one for male scientists. The significance levels are as follows: $^{*}p<0.1$; $^{**}p<0.05$; $^{***}p<0.01$.

TABLE B.13

SCIENTISTS' RESEARCH CREDIBILITY BY SCIENTISTS' POLITICAL AFFILIATION

	Credibility of Scientists Research by Profile Type:				
	Strong *Rep*	Moderate *Rep*	Neutral	Moderate *Dem*	Strong *Dem*
Male	-0.123	-0.154	-0.031	-0.127	0.093
	(0.149)	(0.116)	(0.096)	(0.111)	(0.130)
Full Professor	-0.007	0.288^{**}	0.271^{***}	0.047	0.218^{*}
	(0.149)	(0.116)	(0.096)	(0.111)	(0.129)
Economics	0.331	0.297	0.090	0.326^{*}	-0.147
	(0.233)	(0.186)	(0.153)	(0.173)	(0.202)
Engineering	0.216	0.075	0.002	0.411^{**}	0.156
	(0.230)	(0.188)	(0.150)	(0.174)	(0.213)
Mathematics	0.289	0.341^{*}	0.033	0.539^{***}	0.120
	(0.238)	(0.182)	(0.152)	(0.176)	(0.204)
Medicine	0.175	-0.037	0.179	0.747^{***}	0.289
	(0.230)	(0.187)	(0.152)	(0.172)	(0.209)
High Affiliation	0.169	0.274^{**}	0.109	0.330^{***}	-0.276^{**}
	(0.152)	(0.119)	(0.098)	(0.113)	(0.132)
Constant	4.241^{***}	6.186^{***}	6.933^{***}	6.138^{***}	6.399^{***}
	(0.207)	(0.173)	(0.138)	(0.157)	(0.189)
Observations	1,704	1,704	1,704	1,704	1,704

Note: Coefficients were obtained by regressing scientists' characteristics on respondents' perceived credibility of scientist's own research. All the standard errors are clustered at the individual level. Each column represents a different scientist based on the political affiliation. High Affiliation signifies institutions such as Harvard, UC Berkeley, or Chicago, versus Arkansas or Connecticut. Research fields include Medicine, Mathematics, Engineering, and Economics, with Literature excluded. Full professor indicates full professors versus assistant professors. Male is coded as one for male scientists. The significance levels are as follows: *p$<$0.1; **p$<$0.05; ***p$<$0.01.

TABLE B.14

WILLINGNESS TO READ BY SCIENTISTS' POLITICAL AFFILIATION

	Willingness to Read Opinion of Scientists by Profile Type:				
	Strong *Rep*	Moderate *Rep*	Neutral	Moderate *Dem*	Strong *Dem*
Male	0.073	−0.196	-0.317^{**}	−0.122	0.068
	(0.168)	(0.144)	(0.126)	(0.139)	(0.155)
Full Professor	−0.035	0.213	0.162	0.012	0.270^{*}
	(0.168)	(0.144)	(0.126)	(0.139)	(0.155)
Economics	0.223	0.325	−0.033	0.411^{*}	0.194
	(0.262)	(0.230)	(0.200)	(0.217)	(0.242)
Engineering	0.033	−0.023	−0.001	0.009	0.372
	(0.258)	(0.233)	(0.197)	(0.218)	(0.256)
Mathematics	−0.133	0.169	0.012	0.276	0.094
	(0.268)	(0.226)	(0.199)	(0.220)	(0.245)
Medicine	0.116	0.043	0.061	0.676^{***}	0.531^{**}
	(0.258)	(0.232)	(0.200)	(0.216)	(0.251)
High Affiliation	0.196	0.244^{*}	0.097	0.301^{**}	−0.182
	(0.171)	(0.148)	(0.129)	(0.141)	(0.158)
Constant	3.575^{***}	5.684^{***}	6.485^{***}	5.781^{***}	5.488^{***}
	(0.233)	(0.214)	(0.181)	(0.197)	(0.227)
Observations	1,704	1,704	1,704	1,704	1,704

Note: Coefficients were obtained by regressing scientists' characteristics on respondents' likelihood of reading from similar scientists. All the standard errors are clustered at the individual level. Each column represents a different scientist based on the political affiliation. High Affiliation signifies institutions such as Harvard, UC Berkeley, or Chicago, versus Arkansas or Connecticut. Research fields include Medicine, Mathematics, Engineering, and Economics, with Literature excluded. Full professor indicates full professors versus assistant professors. Male is coded as one for male scientists. The significance levels are as follows: $^{*}p<0.1$; $^{**}p<0.05$; $^{***}p<0.01$.

TABLE B.15

REGRESSION WITH MULTIPLE HYPOTHESIS TESTING CORRECTION

	Dependent variable:					
	Credibility	Cred.Research	Read	Credibility	Cred.Research	Read
	(1)	(2)	(3)	(4)	(5)	(6)
Male	-0.067	-0.068	-0.097	-0.067	-0.068	-0.097
	(0.054)	(0.054)	(0.066)	(0.054)	(0.054)	(0.066)
Full Professor	0.147^{***}	0.165^{***}	0.124^{*}	0.147^{**}	0.165^{***}	0.124^{*}
	(0.054)	(0.054)	(0.066)	(0.055)	(0.055)	(0.066)
Economics	0.185^{**}	0.184^{**}	0.226^{**}	0.185^{**}	0.184^{**}	0.226^{**}
	(0.086)	(0.086)	(0.103)	(0.086)	(0.086)	(0.103)
Engineering	0.202^{**}	0.172^{**}	0.072	0.202^{**}	0.172^{**}	0.072
	(0.086)	(0.086)	(0.104)	(0.088)	(0.087)	(0.105)
Mathematics	0.246^{***}	0.272^{***}	0.092	0.246^{***}	0.272^{***}	0.092
	(0.086)	(0.086)	(0.104)	(0.085)	(0.086)	(0.104)
Medicine	0.299^{***}	0.279^{***}	0.290^{***}	0.299^{***}	0.279^{***}	0.290^{***}
	(0.086)	(0.086)	(0.104)	(0.086)	(0.087)	(0.103)
High Affiliation	0.142^{**}	0.120^{**}	0.128^{*}	0.142^{**}	0.120^{**}	0.128^{*}
	(0.056)	(0.056)	(0.067)	(0.056)	(0.056)	(0.067)
Moderately Dem	-0.505^{***}	-0.483^{***}	-0.293^{***}	-0.505^{***}	-0.483^{***}	-0.293^{***}
	(0.086)	(0.086)	(0.104)	(0.073)	(0.073)	(0.094)
Moderately Rep	-0.660^{***}	-0.617^{***}	-0.521^{***}	-0.660^{***}	-0.617^{***}	-0.521^{***}
	(0.086)	(0.086)	(0.104)	(0.076)	(0.075)	(0.095)
Strong Rep	-2.828^{***}	-2.698^{***}	-2.708^{***}	-2.828^{***}	-2.698^{***}	-2.708^{***}
	(0.086)	(0.086)	(0.104)	(0.089)	(0.088)	(0.105)
Strongly Dem	-0.788^{***}	-0.715^{***}	-0.694^{***}	-0.788^{***}	-0.715^{***}	-0.694^{***}
	(0.086)	(0.086)	(0.104)	(0.081)	(0.081)	(0.100)
Constant	6.994^{***}	6.876^{***}	6.243^{***}	6.994^{***}	6.876^{***}	6.243^{***}
	(0.096)	(0.095)	(0.115)	(0.088)	(0.089)	(0.108)
Observations	8,520	8,520	8,520	8,520	8,520	8,520

Note: Coefficients were obtained by regressing scientists' characteristics on respondents' perceived credibility, scientists' research perceived credibility or likelihood of reading from similar scientists. The p-values in Columns 4, 5 and 6 are corrected for Multiple Hypothesis Testing using FDR procedure. All the standard errors are clustered at the individual level. Political leaning is indicated by "Strongly Republican," "Moderately Republican," "Strongly Democrat," or "Moderately Democrat," with "Neutral" as the excluded category. High Affiliation signifies institutions such as Harvard, UC Berkeley, or Chicago, versus Arkansas or Connecticut. Research fields include Medicine, Mathematics, Engineering, and Economics, with Literature excluded. Full professor indicates full professors versus assistant professors. Male is coded as one for male scientists. The significance levels are as follows: *p$<$0.1; **p$<$0.05; ***p$<$0.01.

TABLE B.16

SUMMARY STATISTICS OF JOURNALISTS

	Sample
Seniority: Less than 1 year	0.10
Seniority: Between 1 year and 3 years	0.23
Seniority: Between 3 years and 5 years	0.14
Seniority: More than 5 years	0.53
Position: Reporter	0.45
Position: Editor	0.33
Position: Opinion Writer	0.14
Position: Columnist	0.08
Job: Daily Newspaper	0.16
Job: Weekly Newspaper	0.04
Job: Freelance	0.28
Job: Online Newspaper	0.35
Job: Blog	0.04
Job: TV	0.12
Political: Conservative	0.27
Political: Liberal	0.62
Political: Moderate	0.11
Employment: Working full time now	0.74
Employment: Working part time now	0.17
Employment: Unemployed	0.02
Employment: Retired	0.03
Country: U.S. and UK	0.59
Country: Other	0.47
Male	0.37
Female	0.59
Non-binary	0.04

Outcome	Mean
Credibility	6.24
Credibility of Research	6.14
Newsletter	5.92

Note: The Journalist sample recruited on Prolific. The characteristics are broken down into different dimensions.

TABLE B.17

JOURNALISTS' BELIEFS

	Sample
Disclosure Leaning: Disagree	0.33
Disclosure Leaning: Neither Disagree nor Agree	0.15
Disclosure Leaning: Agree	0.52
Source Credibility: Disagree	0.16
Source Credibility: Neither Disagree nor Agree	0.14
Source Credibility: Agree	0.70
Readership Reaction: More Backlash	0.39
Readership Reaction: More Engagement	0.18
Readership Reaction: Balanced Mix of Both	0.43
Contact Politicized Scientist: Unlikely	0.21
Contact Politicized Scientist: Neither Unlikely nor Likely	0.27
Contact Politicized Scientist: Likely	0.52
Feature SM Active Scientist: Unlikely	0.21
Feature SM Active Scientist: Neither Unlikely nor Likely	0.24
Feature SM Active Scientist: Likely	0.55

Note: We summarize the journalists' answers to different questions listed below. All the answers were recorder on a 5-item Likert scale. For convenience, we grouped the answers in three categories. We ask them to state the degree of agreement to the following statements: "*A scientist's political leaning should be disclosed when their research is reported*" (Disclosure Leaning) and "*Featuring politically active scientists might affect the newspaper's credibility with its audience*" (Source Credibility). Then, we asked the following questions: "*How do you expect your readership to respond if a scientist's political views are prominently featured in your content?*" (Readership Reaction), "*How likely are you to reach out to a scientist for an interview or expert opinion if their political views are well-known?*" (Contact Politicized Scientist) and "*How likely are you to feature a scientist if they have a politically active social media presence?*" (Feature SM Active Scientist).

TABLE B.18

MECHANISM: SEPARATING THE EFFECT OF COMMUNICATING SALIENT RESEARCH FROM A PURE SCIENTISTS' POLITICAL SIGNAL

	Dependent variable:				
	Credibility	Credible Research	Willing to Read	Yes Newsletter	Trust in Science Idx
Active Control	0.010	-0.120	1.049^{***}	0.089^{**}	0.004
	(0.193)	(0.194)	(0.239)	(0.040)	(0.060)
Treatment Left	-0.096	-0.285^{*}	0.972^{***}	0.062^{*}	0.045
	(0.166)	(0.167)	(0.207)	(0.035)	(0.052)
Treatment Right	-0.121	-0.370^{**}	0.978^{***}	0.039	0.056
	(0.167)	(0.168)	(0.207)	(0.035)	(0.052)
Male	0.048	0.032	-0.123	-0.030	0.021
	(0.111)	(0.112)	(0.138)	(0.023)	(0.034)
Full Professor	0.230^{**}	0.239^{**}	0.375^{***}	0.047^{**}	0.052
	(0.111)	(0.111)	(0.137)	(0.023)	(0.034)
High Affiliation	-0.044	-0.017	0.063	-0.008	-0.090^{**}
	(0.113)	(0.114)	(0.141)	(0.024)	(0.035)
Constant	7.335^{***}	8.082^{***}	5.382^{***}	0.622^{***}	4.067^{***}
	(0.974)	(0.979)	(1.210)	(0.203)	(0.301)
Observations	1,704	1,704	1,704	1,704	1,704
Controls	X	X	X	X	X

Note: Coefficients were obtained by regressing scientists' characteristics on respondents' credibility perceptions, likelihood of reading from similar scientists, willingness to receive a related newsletter, and their general trust in scientists. Each column represents a different outcome variable. High Affiliation signifies institutions such as Harvard, UC Berkeley, or Chicago, versus Arkansas or Connecticut. Research fields include Medicine, Mathematics, Engineering, and Economics, with Literature excluded. Full professor indicates full professors versus assistant professors. Male is coded as one for male scientists. Controls encompass respondents' age, gender, income, ethnicity, education, employment status, religion, region, and political leaning. The significance levels are as follows: *p<0.1; **p<0.05; ***p<0.01.

TABLE B.19

MECHANISM: SEPARATING THE EFFECT OF COMMUNICATING SALIENT RESEARCH FROM A PURE SCIENTISTS' POLITICAL SIGNAL (*Democrat* VS. *Republican* RESPONDENTS)

	Panel A: *Democrats or Leaning Democrat*				
	Credibility	Credible Research	Willing to Read	Yes Newsletter	Trust in Science Idx
Active Control	0.646***	0.489**	1.730***	0.081	−0.056
	(0.232)	(0.231)	(0.304)	(0.055)	(0.074)
Treatment Left	0.773***	0.594***	1.985***	0.110**	0.048
	(0.205)	(0.205)	(0.269)	(0.049)	(0.066)
Treatment Right	0.071	−0.071	1.571***	0.055	0.018
	(0.205)	(0.204)	(0.269)	(0.049)	(0.066)
Male	−0.097	−0.131	−0.233	−0.033	0.063
	(0.136)	(0.136)	(0.178)	(0.033)	(0.043)
Full Professor	0.077	0.097	0.231	0.062*	0.022
	(0.134)	(0.134)	(0.176)	(0.032)	(0.043)
High Affiliation	−0.049	−0.082	−0.140	−0.023	−0.092**
	(0.138)	(0.138)	(0.181)	(0.033)	(0.044)
Constant	7.496***	7.781***	3.577*	0.015	3.285***
	(1.637)	(1.632)	(2.146)	(0.392)	(0.523)
Controls	X	X	X	X	X
Observations	940	940	940	940	940

	Panel B: *Republican or Leaning Republican*				
	Credibility	Credible Research	Willing to Read	Yes Newsletter	Trust in Science Idx
Active Control	−0.818**	−0.879***	0.229	0.084	0.073
	(0.328)	(0.335)	(0.386)	(0.060)	(0.100)
Treatment Left	−1.152***	−1.337***	−0.335	−0.034	0.026
	(0.279)	(0.285)	(0.328)	(0.051)	(0.085)
Treatment Right	−0.479*	−0.825***	0.103	−0.003	0.080
	(0.278)	(0.284)	(0.328)	(0.051)	(0.085)
Male	0.104	0.102	−0.074	−0.038	−0.049
	(0.185)	(0.189)	(0.218)	(0.034)	(0.056)
Full Professor	0.354*	0.350*	0.516**	0.034	0.088
	(0.186)	(0.190)	(0.219)	(0.034)	(0.056)
High Affiliation	−0.138	−0.007	0.201	0.008	−0.098*
	(0.191)	(0.195)	(0.225)	(0.035)	(0.058)
Constant	6.780***	7.484***	6.058***	0.897***	3.711***
	(1.384)	(1.414)	(1.629)	(0.251)	(0.420)
Controls	X	X	X	X	X
Observations	745	745	745	745	745

Note: Coefficients were obtained by regressing scientists' characteristics on respondents' credibility perceptions, likelihood of reading from similar scientists, willingness to receive a related newsletter, and their general trust in scientists. Each column represents a different outcome variable. High Affiliation signifies institutions such as Harvard, UC Berkeley, or Chicago, versus Arkansas or Connecticut. Research fields include Medicine, Mathematics, Engineering, and Economics, with Literature excluded. Full professor indicates full professors versus assistant professors. Male is coded as one for male scientists. The significance levels are as follows: *p<0.1; **p<0.05; ***p<0.01.

TABLE B.20

MECHANISM: SEPARATING THE EFFECT OF POLITICAL EXPRESSION WITH OR WITHOUT INCENTIVES (*Democrat* VS. *Republican* RESPONDENTS)

Panel A: *Democrats or Leaning Democrat*

	Credibility	Credible Research	Willing to Read	Yes Newsletter	Trust in Science Idx	Bias	Credible Info	Other find Not Credible
Control - Incentives	−0.079	−0.238	−0.402	−0.089*	0.122*	0.139	−0.365	−1.157
	(0.206)	(0.205)	(0.270)	(0.051)	(0.069)	(0.182)	(0.230)	(2.014)
Left - No Incentives	0.669***	0.561***	0.771***	0.008	0.106	−1.079***	0.589**	−0.111
	(0.217)	(0.216)	(0.285)	(0.053)	(0.073)	(0.192)	(0.242)	(2.120)
Left - Incentives	0.801***	0.821***	0.838***	−0.007	0.176**	−1.275***	0.685***	−0.161
	(0.218)	(0.217)	(0.286)	(0.054)	(0.073)	(0.193)	(0.243)	(2.132)
Right - No Incentives	−0.431**	−0.477**	−0.586**	−0.055	−0.015	0.595***	−0.588**	−0.910
	(0.209)	(0.208)	(0.275)	(0.051)	(0.070)	(0.185)	(0.233)	(2.046)
Right - Incentives	−0.578***	−0.506**	−0.870***	−0.027	0.083	0.842***	−0.852***	−0.858
	(0.216)	(0.215)	(0.284)	(0.053)	(0.072)	(0.191)	(0.241)	(2.112)
Constant	7.168***	7.031***	6.652***	0.267	4.429***	−1.008	7.559***	24.708***
	(0.919)	(0.916)	(1.208)	(0.226)	(0.308)	(0.814)	(1.026)	(8.996)
Controls	X	X	X	X	X	X	X	X
Observations	1,081	1,081	1,081	1,081	1,081	1,081	1,081	1,081

Panel B: *Republican or Leaning Republican*

	Credibility	Credible Research	Willing to Read	Yes Newsletter	Trust in Science Idx	Bias	Credible Info	Other find Not Credible
Control - Incentives	−0.435*	−0.470*	−0.314	−0.042	0.082	0.154	−0.540*	−1.267
	(0.257)	(0.264)	(0.306)	(0.059)	(0.094)	(0.248)	(0.287)	(2.268)
Left - No Incentives	−1.058***	−1.100***	−0.980***	−0.017	0.060	−0.845***	−1.087***	3.458
	(0.252)	(0.259)	(0.300)	(0.058)	(0.092)	(0.243)	(0.282)	(2.226)
Left - Incentives	−0.322	−0.372	−0.119	0.097*	0.212**	−0.917***	−0.361	−0.628
	(0.253)	(0.261)	(0.301)	(0.058)	(0.093)	(0.244)	(0.283)	(2.237)
Right - No Incentives	0.109	0.060	0.353	0.010	0.004	1.010***	0.218	0.212
	(0.256)	(0.264)	(0.305)	(0.059)	(0.094)	(0.247)	(0.286)	(2.264)
Right - Incentives	0.456*	0.306	0.720**	0.114*	0.228**	0.848***	0.457	0.239
	(0.261)	(0.269)	(0.311)	(0.060)	(0.096)	(0.252)	(0.292)	(2.306)
Constant	6.999***	6.958***	5.651***	0.309	3.228***	−0.137	7.083***	17.165
	(1.240)	(1.275)	(1.476)	(0.283)	(0.455)	(1.197)	(1.386)	(10.951)
Controls	X	X	X	X	X	X	X	X
Observations	906	906	906	906	906	906	906	906

Note: Coefficients were obtained by regressing scientists' characteristics on respondents' credibility perceptions, likelihood of reading from similar scientists, willingness to receive a related newsletter, and their general trust in scientists. Each column represents a different outcome variable. The significance levels are as follows: *p<0.1; **p<0.05; ***p<0.01.

APPENDIX C: ONLINE PRESENCE OF SCIENTIFIC PUBLICATIONS

Using Scopus, we retrieved 114,868 articles (2011–2020) from top journals (*Science*, *Nature*, *PNAS*, *Cell*, *NEJM*, *Lancet*); 107,008 articles with unique DOIs were tracked by Altmetric.[1] Our analysis reveals a consistent upward trend in online presence across diverse media, indicating that scientists increasingly engage broader audiences beyond traditional academia. Figure 1 (Panel A) shows a marked surge in coverage via blogs, newspapers, and Twitter.

Twitter mentions are especially significant: 42,701 articles (40%) were mentioned in blogs, 47,987 (45%) in news articles, and 102,795 (96%) in tweets, underscoring Twitter's pivotal role in scientific communication. The first panel of Figure 1 (Panel A) presents the absolute number of appearances across media; the second, the proportion of articles with any coverage; and the third, the average number of appearances per article.

Overall, all metrics show an upward trend. Blog posts (orange) remain relatively stable in absolute numbers, proportions, and average mentions—possibly reflecting their declining relevance—whereas newspaper coverage (red) increased steadily in the first half of the period, plateaued later, and spiked in 2020. Scientific discourse on Twitter (light blue) has surged since 2013, with nearly all papers mentioned on the platform and an average peak of almost 250 tweets per paper in 2020. Figure 1, Panel B, further illustrates that, from 2011 to 2020, the distribution of Twitter mentions has become less skewed towards zero, with a thicker right tail and more high-mention outliers, highlighting Twitter's emerging prominence as a medium for scientific dissemination and engagement.

APPENDIX D: LLM CLASSIFICATIONS AND VALIDATIONS

Validation of Topic and Stance Detection Methods. Our stance detection method leverages LLMs, with validation detailed in Garg and Fetzer (2025). Comparing GPT-3.5 Turbo's classifications against 40,317 human-coded labels (from 5,000 unique tweets labeled by 137 annotators) for topics like Abortion Rights and Donald Trump, the model achieved F-scores between 79 and 92. Table B.5 reproduces these results. Comparisons with GPT-4, which showed slight improvements on Abortion-related content, confirm our

[1] Altmetric tracks online dissemination of scientific articles across platforms (Alabrese, 2022, Peng et al., 2022); accessed November 10, 2021. See API documentation.

method's robustness. Topic detection was validated using the SemEval-2016 dataset, with GPT-4–generated dictionaries for topics (e.g., Feminist Movement, Hillary Clinton, Legalization of Abortion, Donald Trump) yielding F-scores from 67.01 to 74.87 (see Table B.5).

Gender. We inferred gender using GPT-3.5 Turbo to classify OpenAlex author names as 'Male', 'Female', or 'Unclear', resulting in a distribution of 49% Male, 49% Female, and <1% Unclear. Garg and Fetzer (2025) validated this approach with 147,269 unique names from sources such as the U.S. Social Security Applications (1880–2019), UK Baby Names (2011–2018), British Columbia's 100 Years of Popular Baby Names (1918–2018), and Australian Baby Names (1944–2019), achieving a count-weighted F1 score of 0.9868.

Field. OpenAlex assigns each author's work to 19 hierarchical root-level 'Concepts'. For our analysis, we simplified these into four broad categories: STEM, Medicine, Social Sciences, and Humanities. Each author's primary field is determined by the highest average score across root concepts (based on their 2016–2022 publications). STEM covers Biology, Chemistry, Computer Science, Engineering, Environmental Science, Geography, Geology, Materials Science, Mathematics, and Physics; Medicine is standalone; Social Sciences include Business, Economics, History, Political Science, Psychology, and Sociology; Humanities encompass Art, Philosophy, Literature, Religion, Music, Theater, Dance, and Film. Table B.2 summarizes these classifications, with about half of the authors tweeting on political topics having an identified field and a detailed distribution across these categories.

Fact checking. We classify academics' topical tweets along two dimensions: (i) whether the text contains a checkable factual claim, and (ii) conditional on containing such a claim, whether it is true, false, or unclear. We implement this classification using the GPT-4o mini model and the prompt reported below. The model assigns each tweet to one of four mutually exclusive categories: `TRUE`, `FALSE`, `UNCLEAR`, or `NO FACTUAL CONTENT`. In addition to the label, the model is instructed to provide a brief justification and to extract the underlying factual claim(s), if any. Table D.1 reports representative examples of tweets, assigned labels, and justifications.

```
   You will be shown a tweet and asked to classify it based on
its factual content. Follow the steps below:
1. Choose one of four categories:
  - TRUE: The tweet contains a claim that aligns with well-established
    evidence (e.g., scientific consensus, official statistics,
    historical fact).
  - FALSE: The tweet contains a claim that clearly contradicts
    well-established evidence.
  - UNCLEAR: The tweet implies or contains a claim, but it is
    not possible to verify due to ambiguity, missing context,
    sarcasm, or lack of detail.
  - NO FACTUAL CONTENT: The tweet does not contain any checkable
    factual claim (e.g., pure opinion, emotion, vague statement).
2. Justify your choice: Write one sentence explaining why you
chose that label (e.g., "The tweet makes a claim that contradicts
CDC data").
3. Extract factual claim(s): If the tweet contains one or more
factual claims, list them. Otherwise, leave the list empty.
Tips:
  - Interpret sarcasm or rhetorical language to recover any implied
    claim.
  - Use your judgment based on public data, scientific knowledge,
    or historical facts.
  - Choose NO FACTUAL CONTENT only if there is no checkable claim
    at all.
```

Validation of fact checking. We externally validate this fact-checking pipeline using the PUBHEALTH data as a benchmark. PUBHEALTH contains 11.8k public-health claims hand-labelled by journalists (Kotonya and Toni, 2020) as *true*, *false*, *unproven*, or *mixture*. We restrict our attention to claims labelled *true* or *false* and apply the same prompt used for

TABLE D.1

FACT CHECKING CLASSIFICATION EXAMPLES

Label	Tweet	Justification
False	"Florida has no income tax. Relies heavily on tourism. Tourism is now bust. Florida has no other income. I wonder who officially became Florida man recently!"	While Florida does not have a state income tax and relies on tourism, the claim that it has "no other income" is incorrect, as the state also raises revenue through sales tax, property tax, and other sources.
No factual content	"Please RT important letter from Afghan #women #humanrights defenders at this critical time for their country... 'We believe the only way to achieve durable peace is to ensure equal rights...'"	The tweet expresses a call to action and shares a general statement about human rights without making a checkable factual claim.
True	"UPDATE: More than 50 companies — including @patagonia, @yelp, @stitchfix, @lyft, and @box — have signed a letter opposing Texas' abortion ban."	This accurately reports that over 50 companies signed such a letter, as confirmed by media reports and company statements.
Unclear	"#climate change now accounts for 30 million displaced. Soon to be 2–3× higher. 250k more deaths/yr by 2030. Consider making exports of fossil fuels a capital crime."	The tweet presents large numerical claims about climate impacts, but lacks context or sourcing to verify accuracy, making the claims difficult to assess.

academics' tweets, yielding model outputs in {`TRUE`, `FALSE`, `UNCLEAR`, `NO FACTUAL CONTENT`}. For each claim, we run the model three times and take the modal prediction.

We evaluate performance along two complementary dimensions. First, we assess **veracity classification** conditional on the model issuing a definitive judgment, i.e. dropping cases where the prompt output is `UNCLEAR` or `NO FACTUAL CONTENT`. We therefore compare predicted `TRUE`/`FALSE` labels to the corresponding PUBHEALTH ground truth.

On the resulting 5,844 claims, the model achieves a precision of 0.81, recall of 0.93, F1 score of 0.86, and accuracy of 0.82.

Second, we assess the ability of the prompting strategy to detect **factual content** per se. Since all PUBHEALTH claims are factual statements by construction (i.e., $y = 1$ for all observations), we treat `TRUE`, `FALSE`, and `UNCLEAR` predictions as factual ($\hat{y} = 1$) and `NO FACTUAL CONTENT` as non-factual ($\hat{y} = 0$). Under this definition, the model attains a precision of 1.00, recall of 0.94, F1 score of 0.97, and accuracy of 0.94. Results are nearly identical when defining factual predictions strictly as `TRUE` or `FALSE`. Taken together, these results indicate that the pipeline reliably distinguishes factual from non-factual text and delivers accurate veracity judgments.

APPENDIX E: AUDIENCE COMPOSITION OF ACADEMICS ON TWITTER

We examine the extent to which academics' Twitter audiences consist of other scholars. We focus on academics with primary affiliation in the United States—our main sample—and consider their interactions with other Twitter users. Audience members are classified as *academics* if they appear in the global registry of academic Twitter accounts compiled by Mongeon et al. (2023).

Data. For each U.S. academic account, we link two types of information. First, we observe a cross-sectional snapshot of their social network as of December 2022, including up to 5,000 followers and up to 5,000 followings per account. Second, we observe time-varying interaction data: for each account and year, we record which other users retweet it (2016–2022), reply to it (2015–2022), or mention it (2022 only, due to API constraints). Throughout, each audience member—whether a follower, followed account, or interacting user—is classified as *academic* or *non-academic* based on the Mongeon et al. (2023) global registry.

Audience measures. We construct audience composition measures separately for network links and for interactions. For the cross-sectional snapshot, we compute, for each U.S. academic account, (i) the total number of observed followers, (ii) the number of followers classified as academics, and (iii) the corresponding share of academic followers.

We construct analogous measures for the set of accounts that the academic follows. These follower and following measures are defined once per academic account.

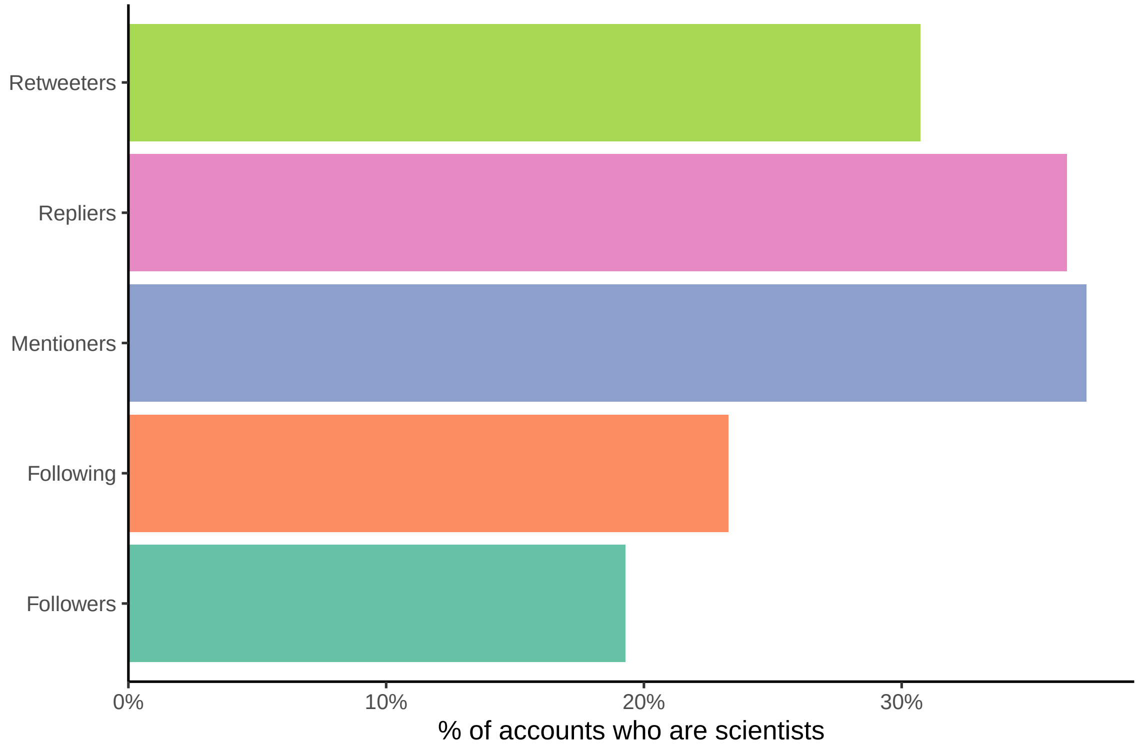
For interactions, we proceed on a year-by-year basis. For each U.S. academic and year, we identify the set of distinct accounts that retweet the academic at least once during that year, count how many of these accounts are academics, and compute the corresponding share. We apply the same procedure to replies, and to mentions in 2022. Interaction measures are defined over distinct pairs (academic account, interacting account) within a given year; i.e. repeated interactions by the same account within a year are counted only once. We then summarise interaction-based audience composition by averaging these within-account shares across U.S. academics in each year.

Sample sizes. Audience coverage is extensive. In the cross-sectional snapshot for U.S. academics, we observe approximately 100 million follower links and 71 million following links. The median academic has 480 observed followers and 487 followings, while the corresponding means are 1,086 followers and 773 followings. These distributions are highly skewed, with some accounts having approximately 35,000 observed followers.

For time-varying interactions, pooling across all years and U.S. academics, we observe approximately 9.0 million distinct retweeter relations, 3.7 million distinct replier relations, and 1.4 million distinct mentioner relations (in 2022). On average, a U.S. academic is retweeted by 98 distinct accounts per year, replied to by 40, and mentioned by 15. The corresponding medians are 27 retweeters, 11 repliers, and 4 mentioners per academic–year, with upper tails extending to tens of thousands of distinct interactions.

Results. Figure E.1 reports cross-sectional averages of the share of audience accounts classified as academics across five audience types: followers, followings, retweeters, repliers, and mentioners. On average, for a U.S. academic, approximately 19 percent of followers are themselves academics, while about 23 percent of followed accounts are classified as academics. The share of academics is higher among interaction-based audiences—roughly 31 percent among retweeters, 36 percent among repliers, and 37 percent among mentioning accounts in 2022. Nevertheless, across all five audience types, the majority of accounts do not appear in the global academic registry, indicating that academics' Twitter audiences—and even their active interlocutors—are frequently non-academic.

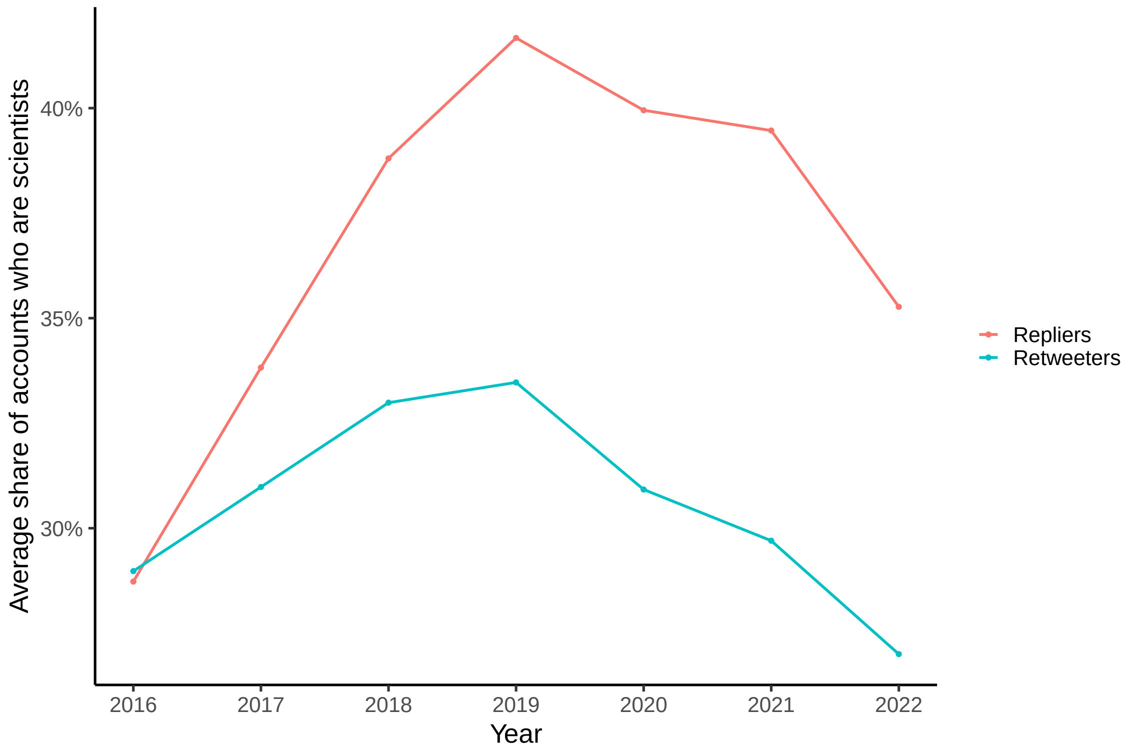
FIGURE E.1.—Share of Academics among Different Types of Audience on Twitter



Note: The figure reports the cross-sectional proportion of audience accounts classified as academics for our main sample of U.S. academics. For each U.S. academic, we compute the fraction of audience accounts classified as academics across five audience types: followers (accounts that follow the academic), followings (accounts the academic follows), retweeters (accounts that retweet the academic at least once between 2016 and 2022), repliers (accounts that reply at least once between 2015 and 2022), and mentioners (accounts that mention the academic at least once in 2022). On average, approximately 19 percent of followers and 23 percent of followings are classified as academics. Among interaction-based audiences, the average academic share is higher—around 31 percent for retweeters, 36 percent for repliers, and 37 percent for mentioners—although in all cases the majority of accounts cannot be identified as academic based on the global registry of Mongeon et al. (2023).

Figure E.2 complements these cross-sectional patterns by documenting the temporal evolution of the average academic share among retweeters and repliers. Between 2016 and 2022, the academic share among retweeters fluctuates around 30 percent, peaking at approximately 33 percent before declining to about 27 percent by 2022. The share among repliers is consistently higher throughout the period, rising from just below 30 percent in 2016 to a peak of roughly 42 percent in 2019, and stabilising in the mid–30 percent range thereafter. Overall, even among the most engaged audiences interacting directly with U.S. academics, the data point to a stable mix of academics and non-academics over time.

FIGURE E.2.—Share of Academics among Different Types of Audience (Retweeters and Repliers) on Twitter over Time



Note: The figure reports the time series of the proportion of audience accounts classified as academics among retweeters and repliers of our main sample of U.S. academics. For each academic and year, we compute the share of distinct retweeters (respectively, repliers) classified as academics based on the global registry of Mongeon et al. (2023), and then take the unweighted mean across academics in each year. The retweeter sample covers 2016–2022, while the replier sample covers 2015–2022; for comparability, the figure reports years 2016–2022. Over this period, the academic share among retweeters fluctuates between approximately 27 and 33 percent. The share among repliers is consistently higher, rising from just below 30 percent in 2016 to a peak of above 40 percent in 2019, and stabilising in the mid–30 percent range thereafter.

Discussion. The global academic registry does not capture all academics active on Twitter; nevertheless, the patterns documented above are informative about the composition of academics' online audiences. Across follower networks and interaction-based audiences, a substantial majority of accounts cannot be identified as academic, indicating that academic accounts are not embedded in a closed professional network but participate in broader online conversations. At the same time, the presence of academics in these audiences is non-trivial: roughly one-fifth of followers and about one-third of retweeters, repliers, and mentioners are themselves academics. Taken together, these findings suggest that academic content on Twitter is plausibly visible to—and circulated within—both a broad public audience and a sizeable community of other academics.

APPENDIX F: REVERSAL EVENTS IN ACADEMICS' POLITICAL STANCES

This section examines whether individual academics change their expressed positions over time. We do so by constructing *reversal events* in academics' political stances on salient topics. We distinguish between three cumulative positions on a topic—*pro*, *anti*, and *neutral*. Intuitively, pro corresponds to a more progressive position and anti to a more conservative one, while neutral indicates no cumulative tilt once all past pro and anti statements are taken into account.

Data. For each U.S.-based academic in our sample, we observe tweets over time classified as *pro*, *anti*, or *neutral* with respect to five political topics: Abortion Rights, Climate Action, Racial Equality, Immigration, and Income Redistribution.

For each academic, topic, and month, we count the number of tweets in each stance category. We then construct a balanced panel spanning from the first month in which an academic tweets about any of the analyzed topics to the last such month. Months without topical tweets are coded as zero for all stance categories. Let $N_{\text{ACADEMICS}}$ denote the total number of unique academics in this resulting panel.

Stock net stance. Our key running measure is an academic's *stock net stance* on a given topic, which cumulates all prior pro and anti statements up to a given month.

Let $\text{CumPro}_{ik\tau}$ and $\text{CumAnti}_{ik\tau}$ denote the cumulative number of pro and anti tweets by academic i on topic k up to month τ (for any month $t \leq \tau$). The stock net stance is defined as:

$$\text{StockNetStance}_{ik\tau} = \begin{cases} \dfrac{\text{CumPro}_{ik\tau} - \text{CumAnti}_{ik\tau}}{\text{CumPro}_{ik\tau} + \text{CumAnti}_{ik\tau}}, & \text{if } \text{CumPro}_{ik\tau} + \text{CumAnti}_{ik\tau} > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Positive values indicate a cumulative pro stance, negative values a cumulative anti stance, and zero indicates neutrality—either because the academic has not yet expressed a stance or because past pro and anti statements exactly offset each other. Months without topical tweets simply carry forward the existing stock stance.

Reversal events. We define a *reversal event* as a change in an academic's cumulative stance on a topic from one month to the next. To characterize these changes, we distinguish between transitions involving neutrality and true sign reversals between pro and anti positions. Specifically, we identify six types s of monthly reversal events:

1. **Neutral → pro**: the academic moves from a neutral cumulative stance to a pro stance:

$$\text{StockNetStance}_{ik\tau} > 0 \quad \text{and} \quad \text{StockNetStance}_{ik,\tau-1} = 0.$$

2. **Neutral → anti**: the academic moves from a neutral cumulative stance to an anti stance:

$$\text{StockNetStance}_{ik\tau} < 0 \quad \text{and} \quad \text{StockNetStance}_{ik,\tau-1} = 0.$$

3. **Pro → neutral**: the academic's cumulative stance returns to neutrality after previously being pro:

$$\text{StockNetStance}_{ik\tau} = 0 \quad \text{and} \quad \text{StockNetStance}_{ik,\tau-1} > 0.$$

4. **Anti → neutral**: the academic's cumulative stance returns to neutrality after previously being anti:

$$\text{StockNetStance}_{ik\tau} = 0 \quad \text{and} \quad \text{StockNetStance}_{ik,\tau-1} < 0.$$

5. **Pro → anti (true sign flip)**: the academic's cumulative stance switches from pro to anti, ignoring any intervening neutral months:

$$\text{StockNetStance}_{ik\tau} < 0 \quad \text{and} \quad \text{StockNetStance}_{ik\tau'} > 0,$$

   where $\tau' < \tau$ denotes the most recent prior month in which $\text{StockNetStance}_{ik\tau'} \neq 0$.
6. **Anti → pro (true sign flip)**: symmetrically,

$$\text{StockNetStance}_{ik\tau} > 0 \quad \text{and} \quad \text{StockNetStance}_{ik\tau'} < 0,$$

   with τ' defined analogously.

In other words, let $\text{StockNetStance}_{ik\tau}$ denote the cumulative stance up to month τ. Transitions involving neutrality are identified by changes between zero and non-zero values of StockNetStance$_{ik\tau}$, while true sign flips capture changes in the sign of the cumulative stance, abstracting from intervening neutral months. True sign flips capture substantive ideological reversals, whereas transitions involving neutrality reflect entry into, exit from, or temporary disengagement from a topic. An academic can experience multiple reversal events over time and across topics. We summarize reversals in two complementary ways.

(i) Ever reversals. For each topic and reversal type, we compute the share of academics who experience at least one such reversal at any point during the sample period:

$$\text{ShareReversers}_k^{(\text{s})} = 100 \times \frac{\#\{\text{academics who ever experience reversal } s \text{ on topic } k\}}{N_{\text{ACADEMICS}}}.$$

Each academic is counted at most once per reversal type and topic.

(ii) Monthly reversal rates. For each month and topic, we compute the fraction of active academics who experience a given reversal in that month:

$$\text{MonthlyReversalRate}_{kt}^{(\text{s})} = 100 \times \frac{\#\{\text{academics experiencing reversal } s \text{ on topic } k \text{ in month } t\}}{\#\{\text{academics active on topic } k \text{ in month } t\}}.$$

An academic is considered active on a topic in a given month t if they post at least one pro, anti, or neutral tweet on that topic. For exposition, we report monthly rates from 2018 onward to reduce noise in the early years.

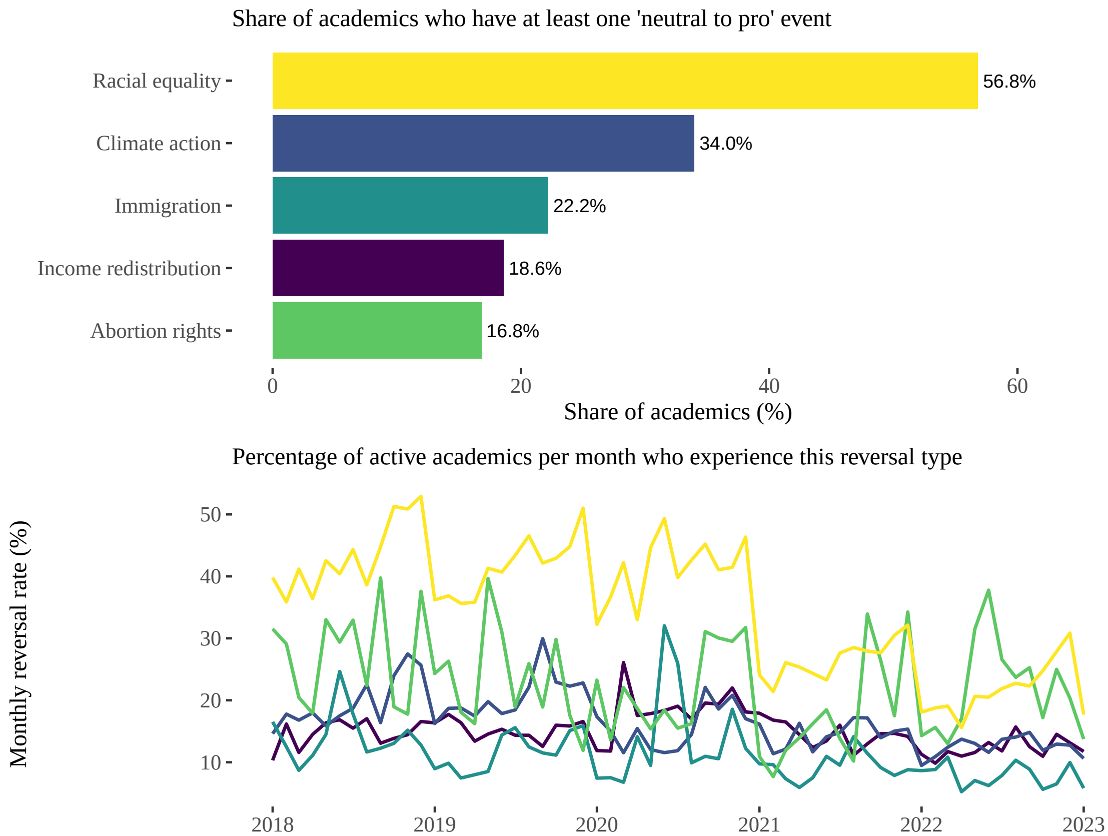
Results Figures F.1–F.6 summarize the empirical patterns for the six types of reversal events. In each figure, the upper panel reports, by topic, the share of academics who experienced at least one reversal of the given type at any point during the sample period. The lower panel plots, for each month from 2018 onward, the percentage of active academics on that topic who experience at least one such reversal in that month.

Two broad patterns emerge. First, entries into *pro* cumulative stances from neutrality are common, whereas entries into *anti* stances are comparatively rare. This asymmetry is particularly pronounced for racial equality: more than 50% of academics ever move from a neutral to a pro stock stance, compared with roughly one-third for climate action and about one-fifth for the remaining topics (Figure F.1). By contrast, neutral-to-anti reversals occur much less frequently. Even on racial equality, only around 12% of academics ever transition from neutrality to an anti cumulative stance, with corresponding shares between roughly 2% and 8% for other topics (Figure F.2).

Second, true sign flips between pro and anti cumulative stances are exceedingly rare, while reversions back to neutrality occupy an intermediate position. Pro-to-anti flips are essentially negligible across all topics, affecting at most about 0.5% of academics (Figure F.3). Anti-to-pro flips are slightly more common but remain uncommon overall, reaching around 2% on racial equality and staying well below 1% elsewhere (Figure F.4). Reversions from pro back to neutrality are observed for up to about 6% of academics on racial equality and less than 1.6% on other topics, with reversions from anti back to neutrality displaying a similar but slightly smaller pattern (Figures F.5 and F.6).

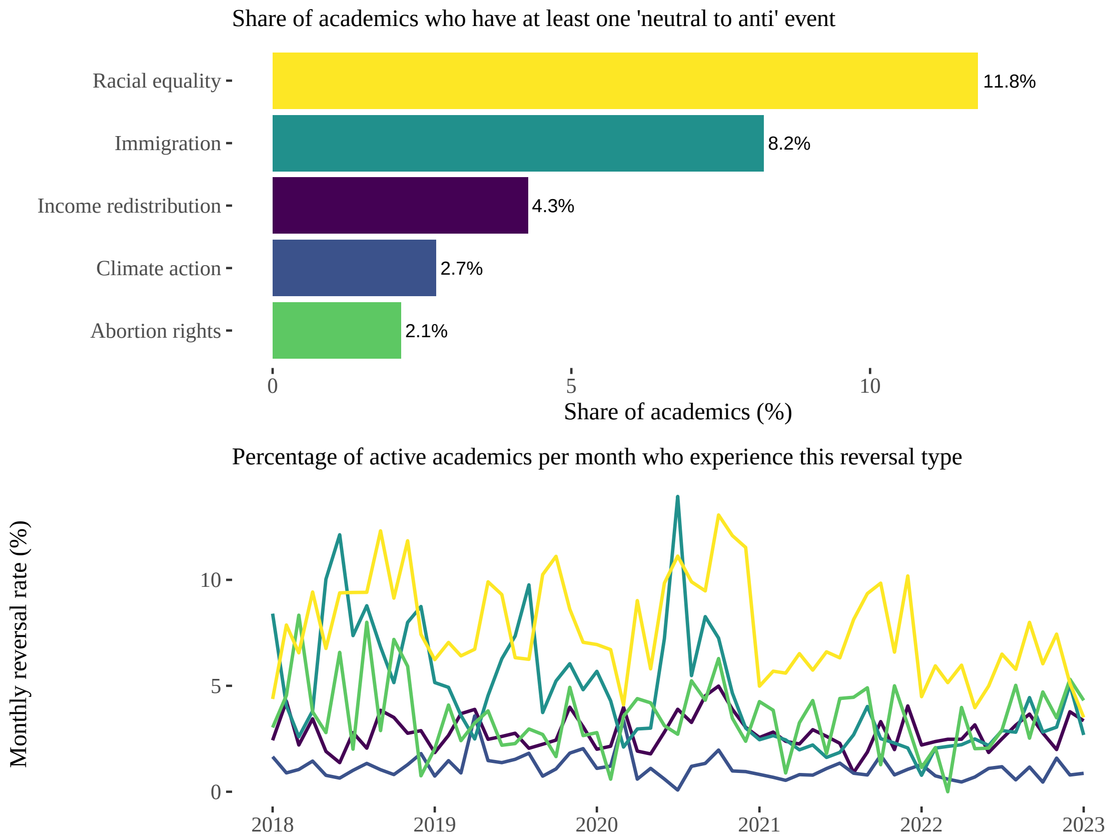
Monthly reversal rates, shown in the lower panels, are generally in the low single digits across all topics and reversal types, with entries into *pro* stances from neutrality being the main exception. This indicates that even when reversals occur, they tend to be dispersed over time rather than concentrated in short periods of abrupt change. Taken together, these patterns suggest that academics' political engagement on Twitter is characterized primarily by entry into, and occasional withdrawal from, cumulative progressive positions rather than by frequent ideological switching between opposing stances.

FIGURE F.1.—Reversal Events: *Neutral* to *Pro*



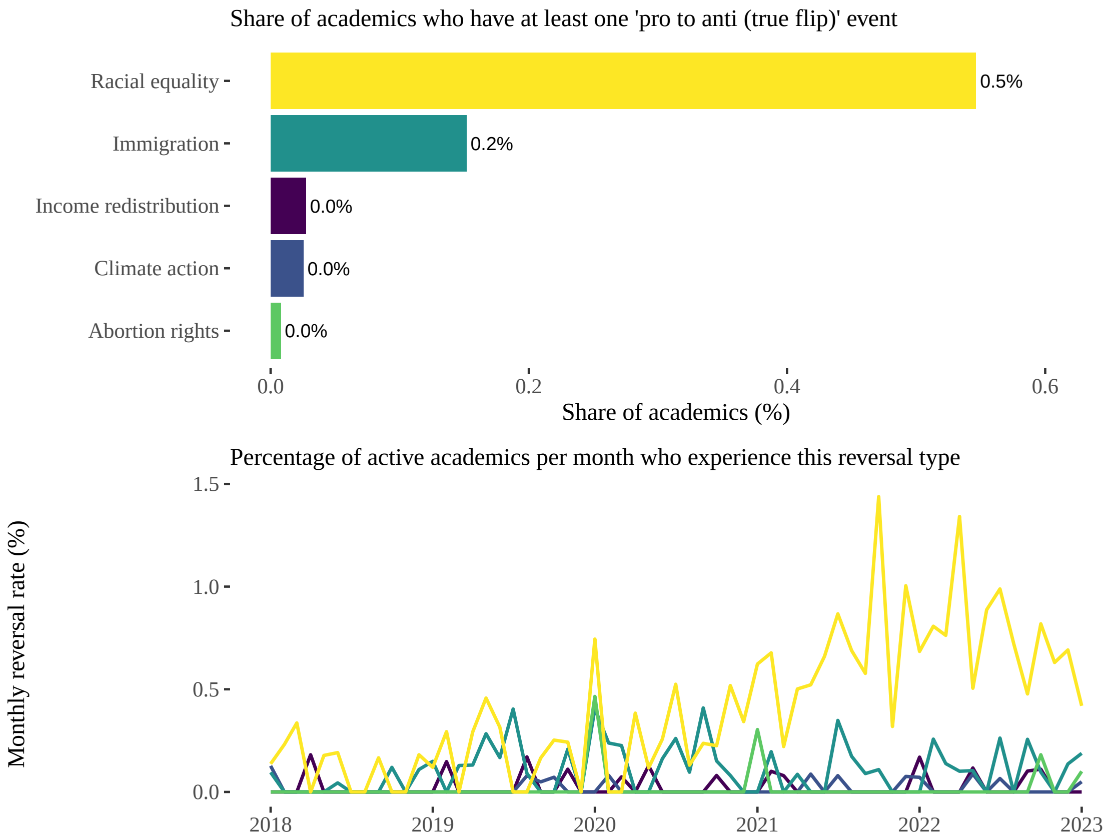
Note: The figure summarises *neutral to pro* reversal events in academics' stock stances on five salient topics. The top panel reports, for each topic, the share of all academics who ever move from a neutral stock stance to a pro stock stance at least once during the sample period. The denominator is the total number of academics $N_{\text{ACADEMICS}}$, so bars are comparable across topics. The share is highest for Racial Equality (above 50%), followed by Climate Action (around one-third), with the remaining topics in the 16–22% range. The bottom panel plots, for each month from 2018 onward, the percentage of active academics on each topic (those with at least one tweet classified as pro, anti, or neutral) who experience a neutral-to-pro reversal in that month, illustrating how frequently academics enter a pro cumulative position from neutrality over time.

FIGURE F.2.—Reversal Events: *Neutral* to *Anti*



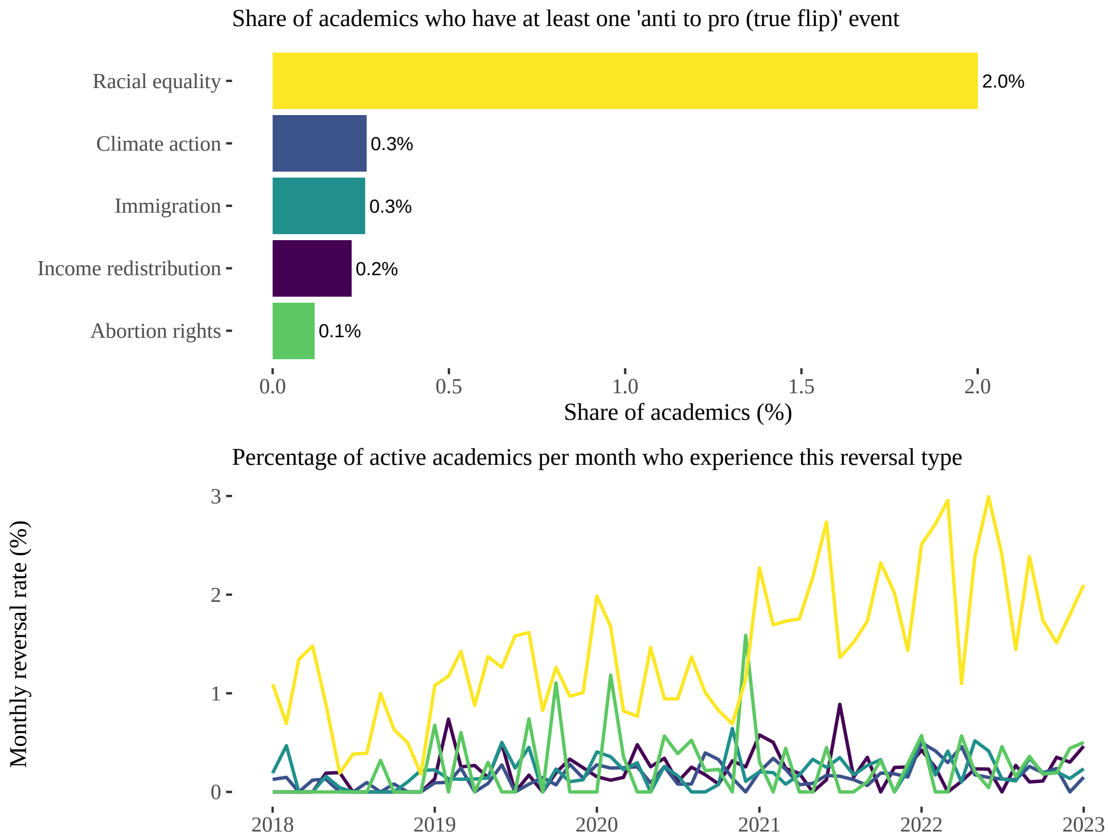
Note: The figure summarises *neutral to anti* reversal events in academics' stock stances on five salient topics. The top panel reports, for each topic, the share of all academics who ever move from a neutral stock stance to an anti stock stance at least once during the sample period. The denominator is the total number of academics $N_{\text{ACADEMICS}}$, so bars are comparable across topics. These shares are markedly smaller than for neutral-to-pro reversals: for Racial Equality, roughly 12% of academics ever go neutral-to-anti, while the corresponding shares for Climate Action, Income Redistribution, Immigration, and Abortion Rights are all below 10%. The bottom panel plots, for each month from 2018 onward, the percentage of active academics on each topic who experience a neutral-to-anti reversal in that month, showing how rarely academics enter an anti cumulative position from neutrality over time. Comparing Figures F.1 and F.2 highlights a strong asymmetry in entry patterns, with academics far more likely to enter debates with a pro rather than an anti cumulative stance.

FIGURE F.3.—Reversal Events: *Pro* to *Anti* (True Sign Flips)



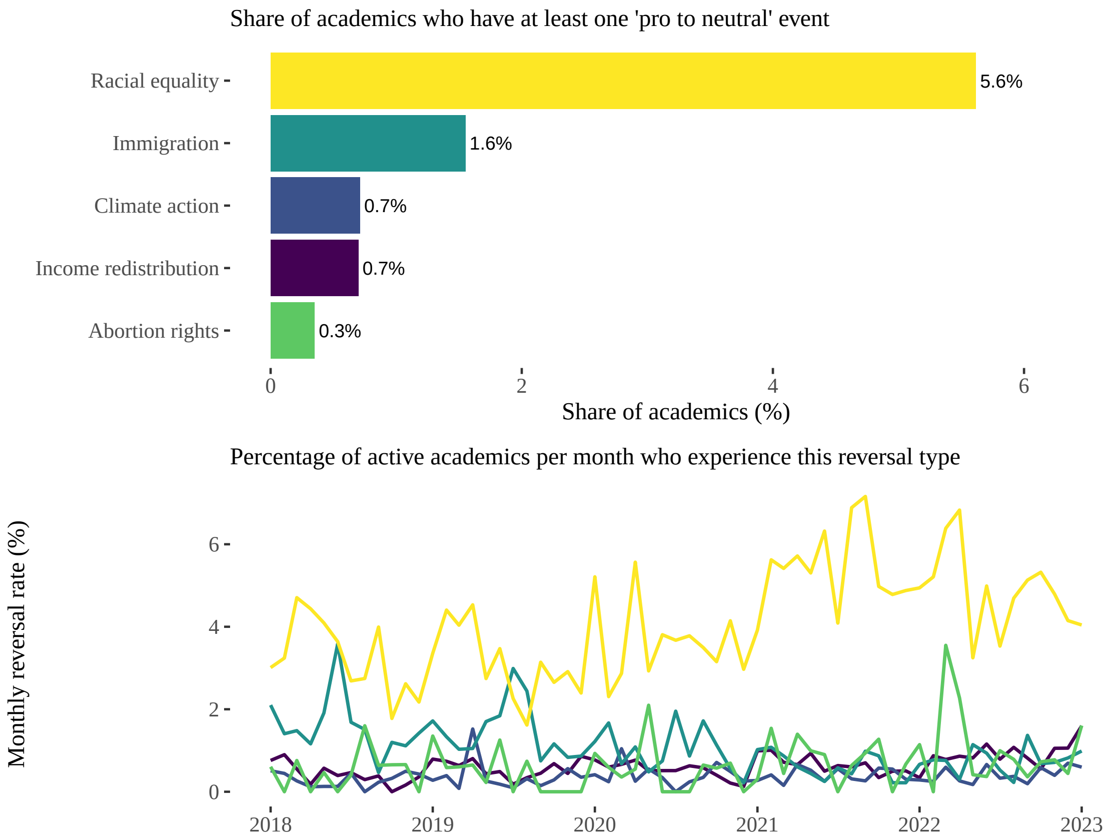
Note: The figure summarises *pro to anti* reversal events that constitute true sign flips in academics' stock stances on five salient topics. The top panel reports, for each topic, the share of all academics who ever move from a pro to an anti cumulative stance at least once during the sample period. The denominator is the total number of academics $N_{\text{ACADEMICS}}$, so bars are comparable across topics. These shares are extremely small, never exceeding about 0.5% of academics on any topic and effectively zero for several topics. The bottom panel plots, for each month from 2018 onward, the percentage of active academics on each topic (those with at least one tweet classified as pro, anti, or neutral) who experience a pro-to-anti reversal in that month. Together, the two panels highlight how rare genuine pro-to-anti ideological reversals are, both over academics' lifetimes and at any point in time.

FIGURE F.4.—Reversal Events: *Anti* to *Pro* (True Sign Flips)



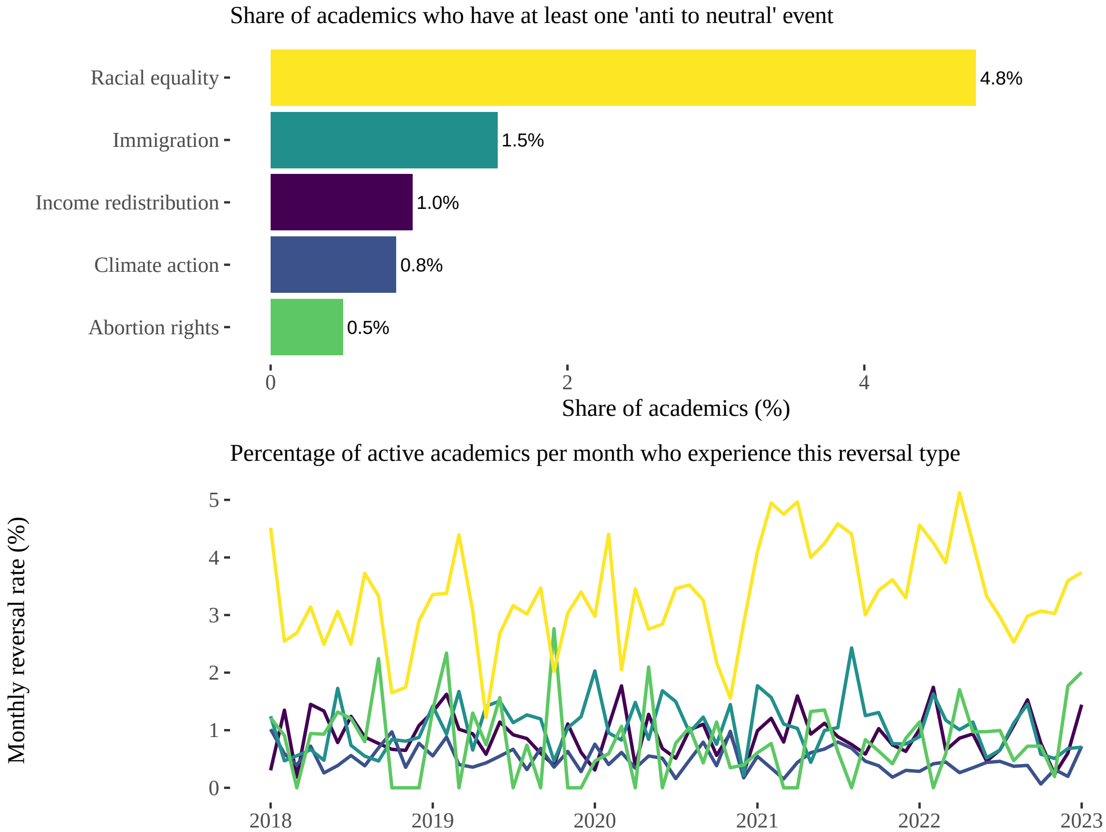
Note: The figure summarises *anti to pro* reversal events that constitute true sign flips in academics' stock stances on five salient topics. The top panel reports, for each topic, the share of all academics who ever move from an anti to a pro cumulative stance at least once during the sample period. The denominator is the total number of academics $N_{\text{ACADEMICS}}$, so bars are comparable across topics. These shares are slightly larger than for pro-to-anti flips but remain small overall, reaching around 2% for Racial Equality and staying well below 1% for the other topics. The bottom panel plots, for each month from 2018 onward, the percentage of active academics on each topic who experience an anti-to-pro reversal in that month. Comparing this figure with Figure F.3 shows that true sign flips, while rare in both directions, occur somewhat more often from anti to pro than from pro to anti.

FIGURE F.5.—Reversal Events: *Pro* to *Neutral*



Note: The figure summarises *pro to neutral* reversal events in academics' stock stances on five salient topics. The top panel reports, for each topic, the share of all academics who ever move from a pro cumulative stance back to neutrality at least once during the sample period. The denominator is the total number of academics $N_{\text{ACADEMICS}}$, so bars are comparable across topics. These shares are highest for Racial Equality (around 6%) and range from roughly 0.3% to 1.6% for the remaining topics. The bottom panel plots, for each month from 2018 onward, the percentage of active academics on each topic who experience a pro-to-neutral reversal in that month. The figure captures how infrequently academics step back from a previously pro cumulative position over time.

FIGURE F.6.—Reversal Events: *Anti* to *Neutral*



Note: The figure summarises *anti to neutral* reversal events in academics' stock stances on five salient topics. The top panel reports, for each topic, the share of all academics who ever move from an anti cumulative stance back to neutrality at least once during the sample period. The denominator is the total number of academics $N_{\text{ACADEMICS}}$, so bars are comparable across topics. These shares are highest for Racial Equality (just under 5%) and fall between roughly 0.5% and 1.5% for the other topics. The bottom panel plots, for each month from 2018 onward, the percentage of active academics on each topic who experience an anti-to-neutral reversal in that month. Taken together with Figure F.5, the figure illustrates how infrequently academics disengage from previously partisan cumulative positions and return to neutrality.

APPENDIX: REFERENCES

ALABRESE, ELEONORA (2022): "Bad Science: Retractions and Media Coverage," *CESifo Working Paper No. 10195*. [5, 10, 42]

GARG, PRASHANT AND THIEMO FETZER (2025): "Political expression of academics on Twitter," *Nature Human Behaviour*. [2, 3, 8, 11, 12, 23, 24, 26, 42, 43]

KOTONYA, NEEMA AND FRANCESCA TONI (2020): "Explainable Automated Fact-Checking for Public Health Claims," *CoRR*, abs/2010.09926. [44]

MONGEON, PHILIPPE, TIMOTHY D. BOWMAN, AND RODRIGO COSTAS (2023): "An Open Data Set of Scholars on Twitter," *Quantitative Science Studies*, 4 (2), 314–324. [10, 11, 46, 48, 49]

PENG, HAO, DANIEL M. ROMERO, AND EMOKE-Á.GNES HORVÁT (2022): "Dynamics of Cross-platform Attention to Retracted Papers," *Proceedings of the National Academy of Sciences*, 119 (25), e2119086119. [10, 42]